

User Manual

ETL & Connectors

This manual provides an overview of the Extraction, Transformation & Loading engine and connectors for data integration.

Version: 5.3.SP36

Wolters Kluwer
CCH® Tagetik
wolterskluwer.com

Table of contents

CCH Tagetik ETL 1

ETL: definition and architecture ... 1

The phases of the process: Extraction, Transformation and Loading ... 2

ETL - basic concepts ... 2

Working with ETL ... 3

ETL Domains 4

ETL Domains page ... 4

ETL Domains definition ... 6

ETL parameters 8

Parameters window ... 8

SQL Models window ... 10

Data source 11

Data Source window ... 11

Data Source Fields Definition window ... 14

Data Source: record types ... 16

Lookup Tables ... 18

Lookup tables window ... 19

Data target: CCH Tagetik ... 23

CCH Tagetik Data Target page ... 24

Fields definition - CCH Tagetik Data Target ... 25

Data Target: Other ... 27

Data Target: Other page ... 28

Fields definition - Data target: Others window ... 30

Routines 33

Routines page ... 34

Commands routine ... 38

Commands routine ... 40

Routine - Extraction ... 42

Routine - Transformation ... 44

Routine - Loading ... 51

Dimensions and dictionaries 55

Dimensions and Dictionaries window ... 55
Dictionary elements ... 59
Completeness checks ... 60

Jobs 61

Job - Definition ... 61
Routines of a job ... 63
Parameters of a job ... 64
Extractions of a job ... 64
Execution jobs ... 66

ETL execution plans 67

Execution plan - Results window ... 67

ETL execution 69

Process cockpit ... 69
Workflow ... 69
ETL - Admin user ... 70
ETL - End user ... 70
Web forms ... 70
ETL execution window ... 71
Purpose of the page ... 71

Logs and audits 77

ETL Log window ... 77

Logs deletion 79

ETL execution data archive 80

Appendix 82

ETL via APIs ... 82
ETL syntax ... 84

ETL connectors 86

Dynamics AX Connector window ... 86
Dynamics NAV Connector window ... 91

SAP Connector Wizard window 94

CCH Tagetik ETL

ETL and Data Integration

Extraction, Transformation and Loading (*ETL*) technology is a fundamental Data Integration tool, with a reliable and efficient process that makes it possible to improve the quality of the data and the accuracy of business analysis.

The ETL module in Tagetik Software S.r.l.

In CCH Tagetik, the ETL engine functions as an interface for loading data, and is able to ensure data security and manage errors; it would not be possible to guarantee this degree of reliability by inserting data directly into an SQL table.
The module is fully configurable, and allows you to extract data from external sources or from a CCH Tagetik database, transform it and load it onto various destinations, both within Tagetik CCH Tagetik (Tagetik databases) and externally (generic databases, Excel files or text files).

ETL Engine features

- **Reliability**: it allows you to transform large volumes of data. It can be used as an alternative method to manually entering and importing/exporting from Excel.
- **High degree of flexibility on targets and sources** : it allows you to extract both data (amounts, statistical data, FX rates) and metadata (elements of dimensions, hierarchies and groupings) from several types of external source simultaneously.

The ETL can either be run on-demand or be automated using scheduling.

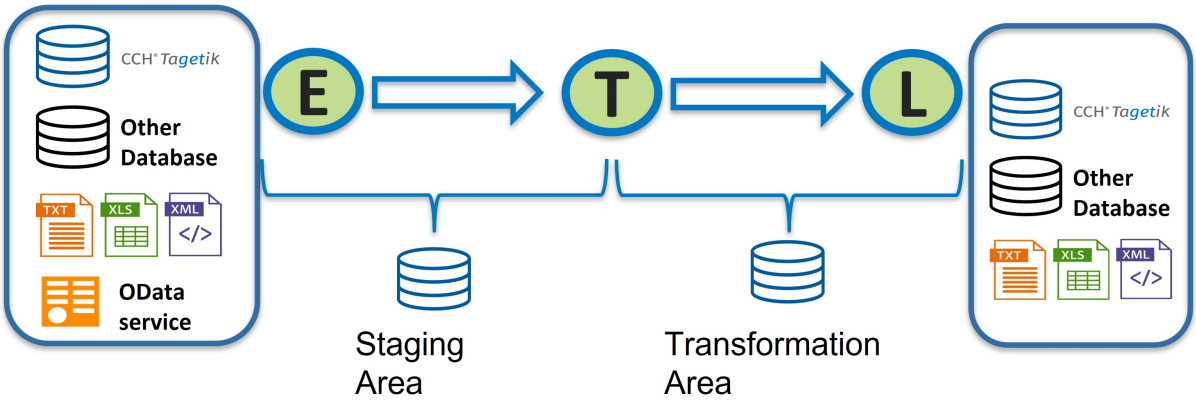
Data loading example

You can load data (values) from an external system, with the transcoding of the source account codes into CCH Tagetik codes and the aggregation of the amounts to align them to the CCH Tagetik standard.

ETL: definition and architecture

Definition

The ETL, an acronym for **"Extraction, Transformation & Loading,** describes the process of collecting financial data and metadata from an unlimited number of sources, and subsequently organizing and centralising them in a single Repository.
In particular, when there is a very high volume of data to be added, ETL is the most flexible and comprehensive data collection tool compared with all other tools (e.g. Quick Data Loader, Data Transformation Package).



The phases of the process: Extraction, Transformation and Loading

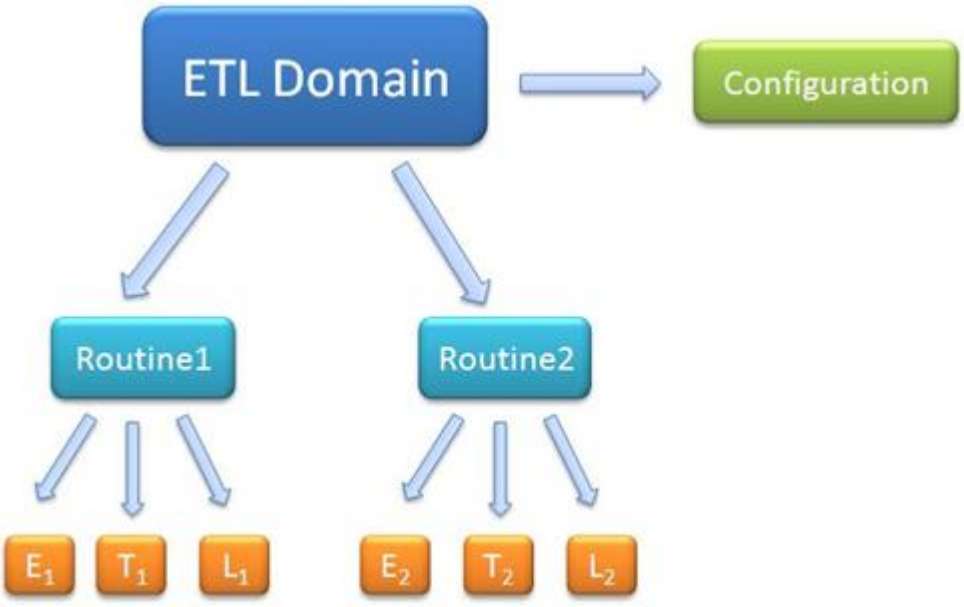
In CCH Tagetik, an ETL process comprises three phases:

Step	Description
1. Extraction	Data can be extracted from any external source: Excel files, text files, XML files, CSVs, TSVs, standard REST APIs and REST APIs with Odata standard, but also data warehouses, ERP systems (SAP, Oracle or Microsoft), or from Tagetik itself. ***IMPORTANT**: you can also extract data simultaneously from several types of external sources: data (amounts, statistical data or FX rates) or metadata (elements of dimensions, hierarchies or groupings)* *If a large number of elements are to be edited/inserted, ETL can be used as an alternative method to manually entering and importing/exporting from Excel.*
2. Transformation	The extracted data can be transformed using SQL rules, lookup tables, transcoding and totalling. For example, you can extract the chart of accounts that exists in Oracle, and transform the code and description of each of its elements (or aggregate several accounts together) to obtain the final chart of accounts to be loaded into Tagetik;
3. Loading	Processed data can be loaded onto various destinations, both within CCH Tagetik Tagetik (Tagetik databases) and externally(custom tables on Tagetik databases, generic databases, Excel files, text files or XML files).

ETL - basic concepts

The structure of an ETL process: Domains, routines and jobs

The structure of the ETL process is organised into several levels:



Structure details

The description of each level is summarised in the following table:

Level	Description
1. ETL domain	An ETL domain is a container macro which includes all the dimensions and dictionaries, SQL models and the parameters related to an ETL rule.
2. Routines and jobs	A **routine** is the minimum data processing unit that can be run in an ETL process and contains all the information for running the routine and the link containing details for extracting, transforming and loading the data. One or several routines can be defined for every ETL domain. An **ETL job** allows you to run, sequentially and in a defined order, several ETL routines (which may also belong to different ETL domains) in a single data process.
3. Details	The details of a routine specify how many and which external sources are available, which data processing/transformation rules to apply to the extracted data and lastly *how and where* the system must load the data that has been previously extracted and transformed.

***IMPORTANT**: All the information related to the configuration, routines and details is specific to an ETL domain. As a result, every management window will ask you to enter the ETL domain to work on as an incoming filter.*

Working with ETL

Set the data source and target

Admin | <application database>

User | <application database>

ETL > ETL Domains > Data Source

1. Define a Data Source. Select the desired source type: specific configuration fields appear.
2. Associate the Data Source with an extraction task of a routine.
3. Select the desired of source data.
4. Click **Run**.

You can only relate several types of source within the same data source if the information contained is represented in the same way.

Configure the dimension mapping

User | <application database>

Admin | <application database>

ETL > ETL Domains > Data Source

1. Access the **data source fields definition**. Relate the source dimensions to the target dimensions by dragging the columns from the data section of the source to the corresponding row of the target dimension.
2. Only with target: Data (Amounts, Intercompany), set the management of amounts without a counterparty.
3. Select the desired previous data deletion policy.
4. Only with submission processes by scenario/period and Data-type target (Amounts, Intercompany), when the **Period length** dimension is set to a non-cumulative value, define the desired previous data carry forward policy.

ETL Domains

Definition

An ETL domain is presented as a container macro containing an entire setup that allows the user to define the following rules:

- extract data from various types of source
- transform them using configurable logics
- load them either to CCH Tagetik or to other target types

ETL domain types: standard and custom

In CCH Tagetik , there are two types of domain:

Element	Description
Standard ETL domain	Domains that are pre-existing by default. These allow you to load both standard data (amounts, intercompany transactions, journals and FX rates) and metadata (elements of dimensions, hierarchies and groupings).
Custom ETL domain	Domains which can be created as fully configurable, and can meet specific business needs.

ETL: Domain setup restrictions

The definition of ETL domains is subject to visibility restrictions.
If the user is subject to restrictions, this page will only show the ETL domains for which they have visibility rights and the user will not be able to add or clone new records.
For more details on the definition of these restrictions, see the section on user rights management

Staging tables

In the execution step, the ETL populates a temporary table ("*Staging* " Table) by extracting the data from one or several external sources, transforming it with a series of rules and then loading it onto CCH Tagetik or other destinations such as files of custom tables. In the window accessible from the *Dimension and Dictionaries* link, you must create alignment between the following structures:

- the structure of the data to be extracted and that of the "*Staging* " table;
- the structure of the "Staging table" and that of the target table or file

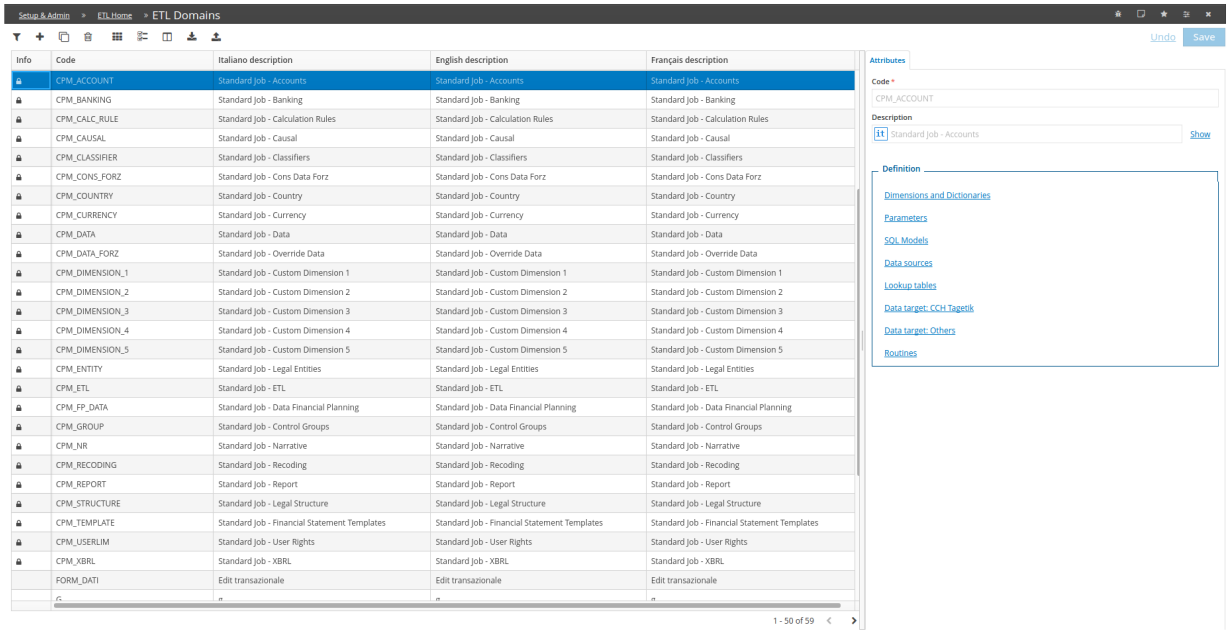
ETL Domains page

Admin | <application database>

ETL > ETL Domains

This page allows you to configure an ETL domain.

What it looks like



Element	Description
A	Action buttons/Tool bar
B	List of ETL Domains
C	Attributes tab

Actions on this page

Element	Description
	This filters the data shown on the page.
	This adds a row in order to create a new domain.
	This duplicates the selected row with all the information inserted, except for the code.
	Deletes the element contained in the selected row.
	Enables/Disables the grid modify function. This allows you to change the characteristics of the selected elements directly in the table, and to hide the configuration tabs.
	Enables multi-row selection to allow mass changes.
	Selects the columns to be shown/hidden in the table and those to be locked when scrolling the table.
	Exports data in Excel format. This file cannot be re-imported into CCH Tagetik.
	Exports the data in Excel format so they can be edited and reimported into CCH Tagetik.

Element	Description
	Imports data from a previously prepared Excel file.
	If only one record is selected, the actions button enables the following utilities: **Reset**: deletes all uses of the ETL domain, its setup and the execution data related to its routines. This window allows you to select the following ETL domain data to be deleted: • ETL Jobs definition • User rights • ETL execution plans (for all processes) • Staging data: extracted • Staging data: transformed • Logs **Copy from**: copies the following (in overwrite mode) onto the selected ETL domain: • the entire configuration (option active by default) • Define Job • User rights • ETL execution plans (for all processes) **Export**: exports the entire configuration of the selected ETL domain and its routines onto a .dat file to replicate it in another existing ETL domain (in the usual application database or another database) through *Import*. **Import**: imports a .dat file previously exported via *Export* to replicate the entire configuration of the exported ETL domain and its routines onto the selected ETL domain. The content of the destination ETL domain will be deleted before the domain is populated with the configuration present in the file.

ETL Domains definition

Admin | <application database>

ETL > ETL Domains

Attributes tab

Every ETL Domain is identified by a **Code** and a number of **Descriptions**.

Field	Description
Code	Unique ID of every ETL Domain
Description	Description in domain language 0 and (optional) in all languages enabled for the domain.

Definition section

This section allows you to access the details of the setup via the following links:

Field	LINK
Dimensions and Dictionaries	see Dimensions and dictionaries
ETL Parameters	see ETL Parameters
SQL Models	see SQL Models
Data Sources	see Data Sources

Field	LINK
Lookup Tables	see Lookup Tables
Data Target: CCH Tagetik	see Data Target: Tagetik
Data target: Others	see Data target: Others
Routines	see Routines

IMPORTANT These links are only enabled if a single ETL domain is selected.

Relationships tab

This tab allows you to create relationships between domains and nodes of the grouping of ETL domains

Field	Description
Nodes	This allows you to view and edit the nodes of the grouping to which the selected ETL domain is related. Every new ETL Domain inserted will be automatically associated with the grouping nodes identified as to be related by default.

ETL parameters

Introduction

Parameters allow you to dynamically select the records that must be processed in the execution step of an ETL.

A **Dimension** is related to each of them

Use of parameters in the ETL process

A summary describing how to use parameters is provided below, based on the following ETL steps:

Steps of the ETL process	Description
Definition of the Routine	filters the storage of extracted and transformed data
Extraction **task**	limits the number of data items to be extracted from the source; filters the data source field related to the Dimension in the **data source fields definition**.
Transformation **of Parameter Choice type**	limits the number of data items to be transformed or used to assign a fixed value to a Dimension. deletes all records that have an invalid value in the field of the staging table related to the Dimension.
Loading	Limits the number of data items to be deleted, added or inserted from or onto the destination. Filters the field of the staging table related to the Dimension in insertion operations and the field of the data destination related to the Dimension in the **data target fields definition** in the deletion step.
Storage	Deletes all records that have an invalid value in the field of the staging table related to the Dimension

Parameters window

Admin | <application database>

ETL > ETL Domains > Parameters

Note: if the ETL domain is locked, the window will open in read-only mode.

Actions on this page

Element	Description
	This filters the data shown on the page.
	This adds a row in order to insert a new parameter.
	This duplicates the selected row with all the information inserted, except for the code.
	Deletes the element contained in the selected row.
	Enables/Disables the grid modify function. This allows you to change the characteristics of the selected elements directly in the table, and to hide the configuration tabs.
	Enables multi-row selection to allow mass changes.
	Selects the columns to be shown/hidden in the table and those to be locked when scrolling the table.

Element	Description
	Exports the data in Excel format so they can be edited and reimported into CCH Tagetik.
	Imports data from a previously prepared Excel file.

Attributes tab

Please find below the most important fields.

Field	Description
Mandatory attributes	
Code	Unique ID of the parameter
Description	Description of the parameter in the languages enabled in the application environment
Type	Parameter type: • *From Dimension dictionary*: range of possible values from among those contained in the dictionary defined for the indicated **dimension**; • *From imported data*: the range of values are those contained in the **dimension** indicated when the parameter is defined. This type of parameter can only be used after the *Extraction* step. • *Free*: The value of the parameter is indicated manually by the user. In cases of **Multiselection** the values must be separated by a *comma*; • *Imported data intersection*: identifies the range obtained from the combinations of values of other **dimensions** defined on other *From imported data* type parameters which have the same Parameter **Code** defined in the **Parent** field. *For example, to be able to filter with the possible Scenario/Period pairs, you can:* *define a "Scenario" parameter and a "Period" whose* ***type*** *is From imported data;* *define a "Scenario/Period" parameter whose* ***type*** *is Imported data intersection;* *associate the "Scenario/Period" parameter with "Scenario" and "Period", parameters as the* ***Parent*** *parameter:* *thus, the values of the "Scenario/Period" parameter are the possible combinations of the values of the "Scenario" and "Period" parameters.* • *From Tagetik*: values contained in a field of a CCH Tagetik table. For example, you can create a Parameter whose values are the entity codes present in the Entity table. In this case, you must define the **Table** attribute as ENTITY and the **Field** as COD_AZIENDA;
Detailed attributes	
Dimension	Mandatory for all parameters, except those whose **type** is ***Imported data intersection***; indicates the date format that the data to be imported will have.
Parent	Only enabled for parameters whose **type** is ***From imported data***; indicates the *Imported data intersection* type parameter shared by other parameters with the same **Parent**.
Table	mandatory for parameters whose **type** is ***From Tagetik***; indicates the table from which to load the selectable values.
Field	mandatory for parameters whose **type** is ***From Tagetik***; indicates the field from which to load the selectable values.
Multiselection	if it is possible to select more than one value for the parameter.

All parameters are used to filter (exclude/delete records depending on where they are used), except for those whose **type** is ***From Tagetik*** and ***Imported data intersection*** with the **Multiselection** option deactivated that are used in a *Parameter choice* type transformation task. In that case, the selected value is used not to delete the records which have a different value, but to override the definition of the field for the dimension related to the parameter for all records present at that time in the staging table.

Example

If the **dimension** related to a parameter defined in the **Identity** field (see Dimensions and Dictionaries window) is one of the *Entity*, *Scenario*, *Period*, *Event scenario*, *Category*, *Custom Dimensions* values, the system will display the values defined in the process, and those for which the current user has visibility, in the range of values.

In the case of a user who only has visibility rights over entities A01, A02 and A03, to be able to select only the entities visible to them, just define a parameter whose **type** is ***From Tagetik*** with the **Table** field defined as the ENTITY value, the **Field** field defined as the COD_AZIENDA value and associate it with a **Dimension** which has its as its **Identity** the *Entity* value.

SQL Models window

Admin | <application database>

ETL > ETL Domains > SQL Models

Note: if the ETL domain is locked, the window will open in read-only mode.

Purpose of window

This window allows you to define SQL syntax that can be reused as text or filters in the tasks of an ETL routine. With this centralised definition, every edit made to an SQL model has an impact on all tasks in which it is used, thus making it easier to maintain setups.

Attributes tab

Please find below the most important fields.

Field	Description
Code	the model's unique ID
Description	description of the model
Text	Contains the SQL syntax. The ETL's dynamic syntax can be used (see ETL syntax

Data source

Contents

This section covers the following topics:

Data Source window 11
Data Source Fields Definition window 14
Data Source: record types 16
Lookup Tables 18
Lookup tables window 19
Data target: CCH Tagetik 23
CCH Tagetik Data Target page 24
Fields definition - CCH Tagetik Data Target 25
Data Target: Other 27
Data Target: Other page 28
Fields definition - Data target: Others window 30

Data Source window

Admin | <application database>

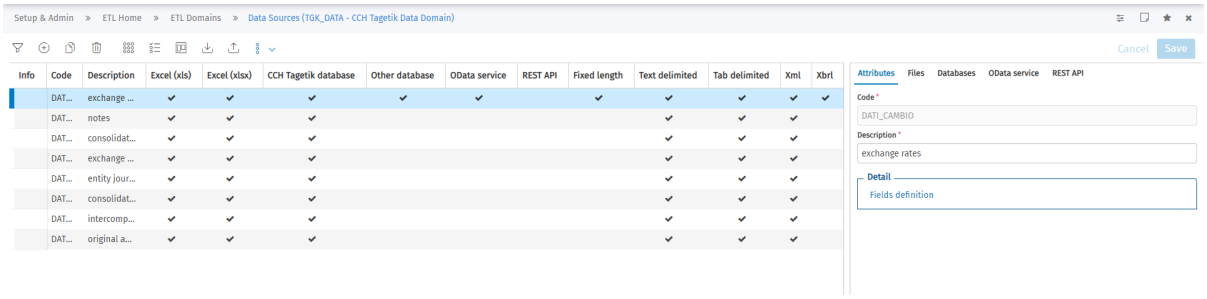
ETL > ETL Domains > Data Source

Note: if the ETL domain is locked, the window will open in read-only mode.

Purpose of window

This window allows you to define the data sources from which to extract the values.

What it looks like



Element	Description
A	Action buttons/Tool bar
B	Data source list
C	Configuration tab

Actions on this page

Element	Description
	This filters the data shown on the page.
	This adds a row in order to insert a new Data Source
	This duplicates the selected row with all the information inserted, except for the code.
	Deletes the element contained in the selected row.
	Enables/Disables the grid modify function. This allows you to change the characteristics of the selected elements directly in the table, and to hide the configuration tabs.
	Enables multi-row selection to allow mass changes.
	Selects the columns to be shown/hidden in the table and those to be locked when scrolling the table.
	Exports the data in Excel format so they can be edited and reimported into CCH Tagetik.
	Imports data from a previously prepared Excel file.
	**Copy from**: this attributes the characteristics of another data source of choice to the selected data source.

Attributes tab

Field	Description
Code	Identification code of the data source to be used in the extraction tasks
Description	Data source description

Link	Description
Fields definition	opens the **Data source fields definition** window, where it is possible to relate the values contained in the data sources defined by the selected data source to the dimensions of the ETL domain.

Files tab

This tab allows you define File type data sources for the selected data source.

***Note**: To complete the activation of certain types of data source, it is necessary to fill out additional fields.*

Data source	Available options
Excel files	**Excel (xls)** enables the possibility of extracting data from an Excel (xls) file. The file must have a defined data area that contains a header record and the records of values. The name of the data area must be reported, paying attention to uppercase and lowercase, in the **Excel (xls) area** field. **Excel (xlsx)** enables the possibility of extracting data from an Excel (xlsx) file. The file must have a defined data area that contains a header record and the records of values. The name of the data area must be reported, paying attention to uppercase and lowercase, in the **Excel (xlsx) area** field.
Text files	**Fixed length**: enables data to be extracted from a text file with a fixed length. Every record must always contain the same number of characters, every item of information must always start from the same position (index of the character within the record) and must have the same number of characters; if the length, in characters, of the value is not equal to the number of characters reserved for the information item, *space* characters must be added (before or after, depending on whether you want to give the value numerical formatting) so that the subsequent information item starts in the pre- established position. In this option, the *Record types* link is active: it allows you to define the **Record types** if the file to be extracted includes records that are not related to the values to be extracted but contain additional information that will automatically define certain fields of the *Staging* table for every record extracted from the file **Text Delimited**: enables the possibility of extracting data from a Text delimited file. In this type of text file, the records may have different lengths, in characters, but they must always be separated by a specific character defined in the **Separator** field. **Tab Delimited** enables data to be extracted from a Text tab delimited file **Charset**: field dedicated to text files, which allows you to indicate the character coding with which the same file was created. This information will be used in the extraction step to correctly interpret the processed characters. If not defined, the system will try to detect the charset automatically.
Other files	**Xml** enables the possibility of extracting data from an Xml (eXtensible Markup Language) file with a pre-established structure. The file must be made up of a *root* and as many *tags*, with the same name, as there are records to be extracted; the tag name must be defined in the **Xml node** field. Each tag must contain as many *sub-tags* as there are items of information to be extracted; the *sub-tags* are all repeated, identically, for each *tag* The names of these tags must then be related to the dimensions in the data source fields definition. **Xbrl** enables data to be extracted from a standard zip file containing information related to the Xbrl. You can tell the system what information is contained in the file by selecting, for the **Xbrl type** field, the *Item*, *Context* and *Instance* values.

Database tab

This tab allows you to define database type data sources for the selected data source.

Data source	Available options/fields
CCH Tagetik	**CCH Tagetik database option**: enables data to be extracted from the CCH Tagetik database. *Field **CCH Tagetik database type**:* indicates the extraction type: ◦ *Object*: from Table or from View as indicated in the **CCH Tagetik database table** field. ◦ *Query*: from a query indicated in the **CCH Tagetik database query** field. ETL's dynamic syntax can be used when writing the query (see ETL syntax).
Other	**Other database** option: enables data to be extracted from another database **Other database connection** field defines the JDBC connection. **Other database type** field: indicates the extraction type: ◦ *Object*: from Table or from View as indicated in the **Other database table** field. ◦ *Query*: from a query indicated in the **Other database query** field. The ETL's dynamic syntax can be used when writing the query (see ETL syntax).

OData service tab

This tab allows you to define Odata type data sources for the selected data source.

Data source	Available options
Data service	This enables the possibility of extracting data from an OData service by defining: ◦ *EntitySet name*: the name of the entity displayed by the OData service ◦ *Query option*: the query options to be specified using the standard OData syntax

***Note**: in the **ETL job extractions** or **execution** definition step, it will be possible to select the OData type **Endpoint** on which the service URL is defined, as well as any authentication method.*

Rest API tab

This tab allows you to define REST API type data sources for the selected data source.

Data source	Available options
REST API	**REST API:** enables data to be extracted via a REST API service; in the *Query option* field, the syntax is defined by the API

***Note**: in the **ETL job extractions** or **execution** definition step, it will be possible to select the REST API type **Endpoint** on which the service URL is defined, as well as any authentication method.*

Data Source Fields Definition window

Admin | <application database>

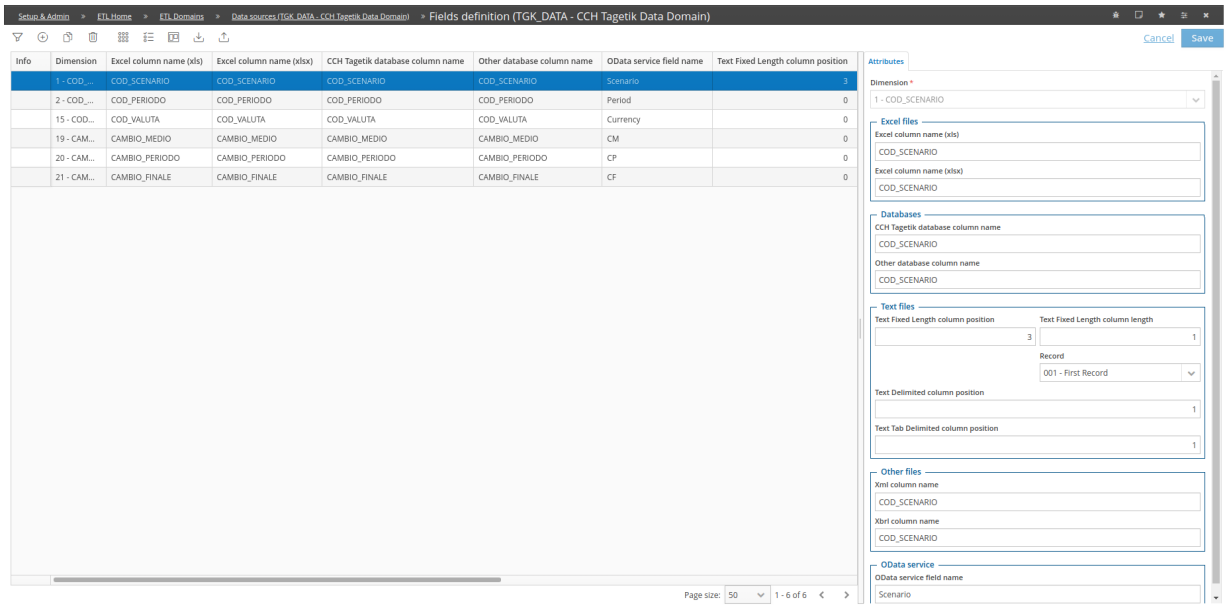
ETL > ETL Domains > Data Sources > Fields Definition link

Note: if the ETL domain is locked, the window will open in read-only mode.

Purpose of window

This window allows you to define alignment between the information present in the data sources to be extracted and the dimensions of the ETL domain, so as to process mapping between the source fields and the fields of the staging table.

What it looks like



Element	Description
A	Action buttons/Tool bar
B	Dimensions list
C	Configuration tab

Actions on this page

Element	Description
	This filters the data shown on the page.
	This adds a row in order to insert a new Data Source
	This duplicates the selected row with all the information inserted, except for the code.
	Deletes the element contained in the selected row.
	Enables/Disables the grid modify function. This allows you to change the characteristics of the selected elements directly in the table, and to hide the configuration tabs.
	Enables multi-row selection to allow mass changes.
	Selects the columns to be shown/hidden in the table and those to be locked when scrolling the table.
	Exports the data in Excel format so they can be edited and reimported into CCH Tagetik.
	Imports data from a previously prepared Excel file.

Attributes tab

Please find below the most important fields.

Field	Description
Dimension	Dimension of the ETL domain whose value, extracted from the source data, will be saved in the relative field of the *Staging* table.
Extraction type	**Field option active**
Depending on the extraction type managed in the **Data source**, the system will show different fields which allow you to define alignment between the Dimension and the values present in the various data sources:	
Excel (xls)	**Excel (xls) column name** indicates the value present in one of the cells of the header record present in the xls file to define the column from which to extract the values.
Excel (xlsx)	**Excel (xlsx) column name** indicates the value present in the xlsx file in one of the cells of the header record to define the column from which to extract the values.
CCH Tagetik database	**CCH Tagetik database column name** indicates the name of the column of the table/view or the query run on the CCH Tagetik database from which to extract the values
Other database	**Other database column name** indicates the name of the column of the table/view or the query run on another database from which to extract the values
Fixed Length	**Text Fixed column position**: indicates the index of the initial character of the value to be extracted from every record of the file **Text Fixed Length column length**: indicates the length in characters of the value to be extracted from every record of the file. **Record**: indicates a Record to define the extraction of the value from a specific record of the file. (see Record types)
Text Delimited	**Text Delimited column position** indicates the number of the column of the file from which to extract the values.
Text Tab Delimited	**Text Tab Delimited column position** indicates the number of the column of the file from which to extract the values.
Xml	**Xml column name**: indicates the name of the tag from which to extract the values
Xbrl	**Xbrl column name**: indicates the name of the information returned by the (*parser*) of the Xbrl file from which to extract the values
OData service	**OData service column name**: indicates the name of the property belonging to the entity displayed by the service from which to extract the values
REST API	**REST API column name**: name of the property of the object displayed by the service from which to extract the values. For compound objects, the name can be indicated with the properties separated by the "." character. E.g. *employee.address.cap*

***IMPORTANT**: If new types of data source are being added to the Data Source, you must access this window to complete the setup; otherwise, an error will be returned during the execution of the extraction task which uses that Data Source.*

The following **Dimensions** cannot be related:

- Dimensions used in the **Dimensions from 1 to 9** fields of **extraction tasks**
- Dimensions which have a set **Identity** whose type is *Num. Record* which automatically defines the field of the Staging table related to the dimension with a progressive number for every extracted record.

Data Source: record types

Admin | <application database>

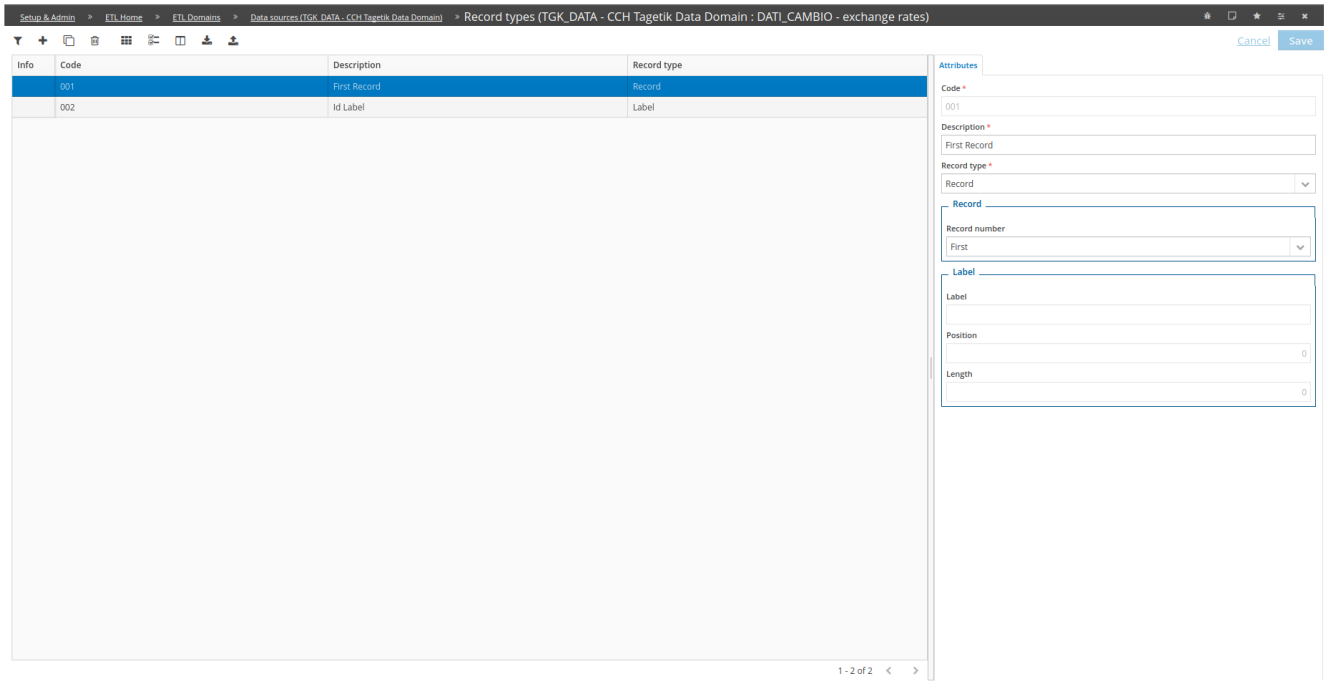
ETL > ETL Domains > Data Sources > Record Types link

Note: if the ETL domain is locked, the window will open in read-only mode.

Purpose of the page

This page, which is only active for data sources which manage extraction from **Text Fixed Length** files, allows you to define certain types of records that do not form part of the data to be extracted but contain information to be used to automatically define certain fields of the Staging table for every record extracted from the file. Record types identify these specific records and, once related to the dimensions managed in the **data source fields definition**, can be used to automatically fill them in with the extracted information.

What it looks like



Element	Description
A	Action buttons/Tool bar
B	Records list
C	Attributes tab

Actions on this page

Element	Description
	This filters the data shown on the page.
	This adds a row in order to insert a new record type.
	This duplicates the selected row with all the information inserted, except for the code.
	Deletes the element contained in the selected row.
	Enables/Disables the grid modify function. This allows you to change the characteristics of the selected elements directly in the table, and to hide the configuration tabs.
	Enables multi-row selection to allow mass changes.
	Selects the columns to be shown/hidden in the table and those to be locked when scrolling the table.
	Exports the data in Excel format so they can be edited and reimported into CCH Tagetik.
	Imports data from a previously prepared Excel file.

Attributes tab

Please find below the most important fields.

Field/section	Description
Code	ID of the record type to be used in the **Record** field of the **data source fields definition**.
Description	description of the record type
Record Type	Describes the record type. You can choose between the following: • ***Record***: to indicate one of the first 5 or one of the last 5 records present in the file. • ***Label***: to indicate a record that contains a specific ID defined in the **Label** field, which starts with the character defined in the **Position** field and has the length defined in the **Length** field.
Record	If the selected **record type** is ***Record***, the record number must be specified: • *F1*: first record • *F2*: second record • *F3*: third record • *F4*: fourth record • *F5*: fifth record • *L5*: fifth to last record • *L4*: fourth to last record • *L3*: third to last record • *L2*: penultimate record • *L1*: last record
Label	If the selected **record type** is *Label*, the following elements must be specified: • **Label**: ID to be sought in every row of the file to identify the record; • **Position**: index of the character which allows you to check whether the **Label** is present, for every row of the file; • **Length**: number of characters in the **Label**. The file records identified by this setup are only discarded if the record types are used in the **data source fields definition**.

Lookup Tables

A lookup table allows you to define a logical structure that creates mapping between one or several input values (input ETL dimensions) and one or several output values (output ETL dimensions). These ETL dimensions relate to fields present in the staging table.

Input1	Input2	...	InputN	Output1	Output2	...	OutputN
In_1	In_2	...	In_N	Out_1	Out_2	...	Out_N
In_3	In_4	...	In_X	Out_3	Out_4	...	Out_X

These tables are useful in **Routine transformation tasks** to transcode the extracted data, making them ready for loading.

Example 1

For amounts present on several general accounting accounts that must be aggregated onto a single account in CCH Tagetik's chart of accounts:

General Ledger Account (in)	Tagetik Account (out)
Revenues 1	Product Revenues
Revenues 2	Product Revenues
Revenues 3	Product Revenues
Revenues 4	Other Revenues
Revenues 5	Other Revenues

Example 2

To transcode a client into another CCH Tagetik client based on the combinations with the cost centres (e.g. Premier or Generic customer). The pair (Customer 2, Sales) is transformed into Premier, while the pair (Customer 2, IT) is transformed into Generic customer:

Customer (in)	Cost Center (in)	Tagetik Customer (out)
Customer 1	Sales	Premiere
Customer 2	Sales	Premiere
Customer 3	Sales	Generic
Customer 1	IT	Premiere
Customer 2	IT	Generic

Lookup tables window

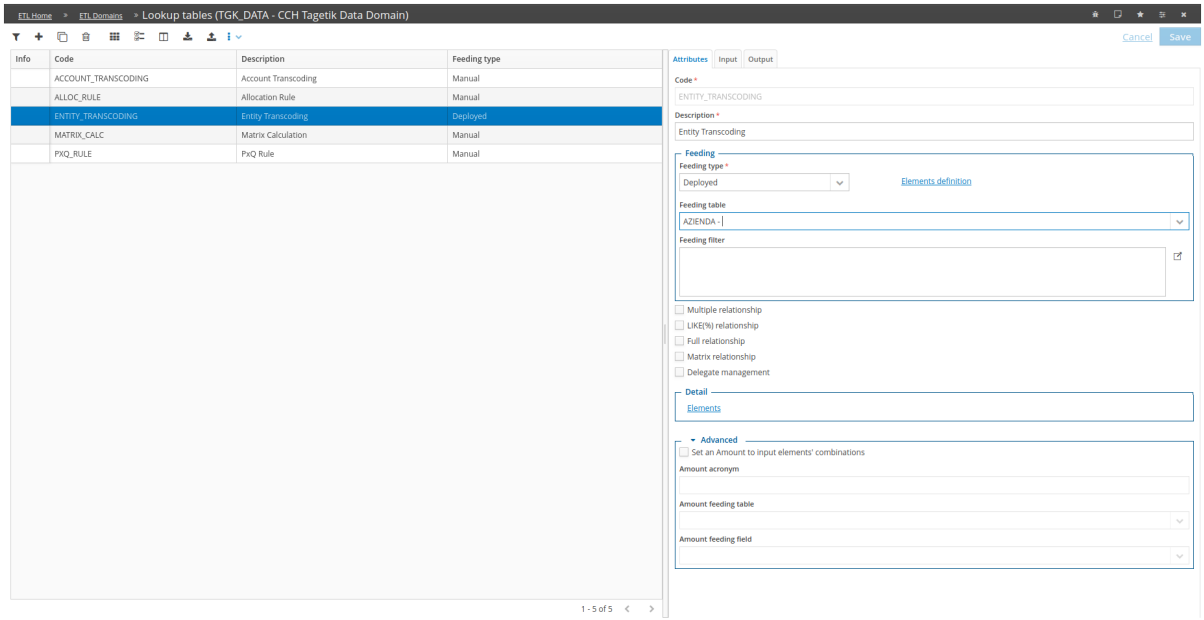
User | <application database>

ETL tile > ETL Domains > Lookup Tables link >

Note: if the ETL domain is locked, the window will open in read-only mode.

Purpose of window

What it looks like



Element	Description
A	Action buttons/Tool bar
B	List of lookup tables
C	Configuration tab

Attributes tab

Element	Description
Code	Code of the lookup table to be used in transformation tasks
Description	Lookup table description
Feeding type	Indicates the method with which the **lookup table elements** must be created. The following types are available: **Manual**: elements defined manually by the user through the relative management **From Tagetik**: elements generated automatically by the system via the **Automatic feeding** data processing based on what is defined in the lookup table. It is possible to subsequently edit automatically created elements, for specific needs, through element management **Deployed**: the elements will be generated automatically by the system through the **Automatic feeding** data processing based on the lookup table definition and the **definition of the lookup table elements** (which synthetically define the rules for creating elements). You can subsequently edit automatically created elements, for specific needs, through element management.
Elements definition	Link only active with **Feeding type**: *Deployed*: allows you to define, synthetically or otherwise (see LIKE(%) relationship in options), the rules for creating elements that will be generated by the **Automatic feeding** data processing
Feeding table	Active if the **Feeding type** is *From Tagetik* or *Deployed*; indicates the CCH Tagetik database table from which to generate, through the **Automatic feeding** data processing, the elements of the lookup table based on how the input and output dimensions have been defined
Feeding filter	Active with **Feeding type** set to *From Tagetik* or *Deployed*: filter used applicable to the lookup table elements generated through the **Automatic feeding** data processing
Options	

Element	Description
Multiple relationship	The same combination of input elements can be associated with different combinations of output elements
LIKE(%) relationship	Each input element can be defined *synthetically* using the % notation *Note*: the **Multiple relationship** and **LIKE(%) relationship** options can be activated simultaneously within the same lookup table.
Full relationship	If active, this allows you to specify that the data present in the staging table when the lookup table is applied through an **Enrichment: lookup table** type transformation task must be registered in the input combinations in the lookup table elements. This is to verify that every data record is actually processed. This check is performed by a **Lookup table diagnostic** type transformation task
Matrix relationship	If active, this allows you to manage additional information for every combination of input elements/output elements. This information is necessary if an **Enrichment: matrix calculation** type task is included in the transformation step of a routine.
Delegate management	If active, this allows the Administrator user, who usually manages lookup tables and their elements independently, to delegate the management of the elements to the contributor user
Elements	This allows you to access the management of **lookup table elements**. It is active for all types allowed by the **Feeding type**

Note: *if the* defined ***transformation tasks of a routine*** *do not include one whose type is* ***Lookup table diagnostic****, the system will check that the defined elements are consistent with the activated options.*

Advanced lookup tables section

This section allows you to define a particular type of lookup table that relates an amount to each combination of input elements, instead of relating a combination of output elements to it.
These rules are used in the **Transfer Price from the rule** and **Allocation** transformation tasks.
Tasks which use these types of rule produce a result using the amount defined in the elements, always with the same input element values present in the fields of the staging table related to the input dimensions.
Normal **Enrichment: lookup table** tasks, however, define the fields of the staging table related to the output dimensions with the output element values with the same values as the input elements present in the fields of the staging table related to the input dimensions.

Element	Description
Assign an amount to the combinations of input elements	enables the rule method detailed above
Amount acronym	allows you to define the name of the column related to the amount that will be shown in the elements of the lookup table instead of the output elements
Amount feeding table	indicates the CCH Tagetik database table from which to generate the amount in the lookup table elements. The behaviour of this field varies based on the value of the **Feeding type** field if it is *Deployed*, the field is visible and takes the value of the **Feeding table** field as its default value if it is *From Tagetik*, the field is not visible and its value will be overridden with that of the **Feeding table** field if it is *Manual*, the field is not visible.
Amount feeding field	indicates the field related to the **Amount feeding table** from which to generate the amount in the lookup table elements. As for the previous field, the behaviour of this field varies based on the value of the **Feeding type** field if it is *Deployed* or *From Tagetik*, the field is visible and allows you to select a field of the table defined on **Amount feeding table** if it is *Manual*, the field is not visible.

Input tab

Element	Description
Input dimension	ETL domain dimension related to a field of the staging table and input element *n* ***IMPORTANT**: it is mandatory to define at least one input dimension; if more than one are defined, they must be sequential.*
Input table	CCH Tagetik database table from which to generate input element *n*. The behaviour of this field varies based on the value of the **Feeding type** field. if it is *Deployed*, the field is visible and takes the value of the **Feeding table** field as its default value if it is *From Tagetik*, the field is not visible and its value will be overridden with that of the **Feeding table** field if it is *Manual*, the field is not visible.
Input field	related to the **Input table** from which to generate input element *n*. Varies based on the value of the **Feeding type** field. if it is *Deployed* or *From Tagetik*, the field is visible and allows you to select a field of the table defined on **Input table** if it is *Manual*, the field is not visible.

Output tab

Element	Description
Output Dimension	ETL domain dimension related to a field of the staging table and output element *n* ***IMPORTANT**: if the **Assign an amount to the combinations of input elements** option is not active, it is mandatory to define at least one output dimension; if more than one are defined, they must be sequential.*
Output table	CCH Tagetik database table from which to generate output element *n*. The behaviour of this field varies based on the value of the **Feeding type** field. if it is *Deployed*, the field is visible and takes the value of the **Feeding table** field as its default value if it is *From Tagetik*, the field is not visible and its value will be overridden with that of the **Feeding table** field if it is *Manual*, the field is not visible.
Output field	related to the **Output table** from which to generate output element *n*. Varies based on the value of the **Feeding type** field. if it is *Deployed* or *From Tagetik*, the field is visible and allows you to select a field of the table defined on **Input table** if it is *Manual*, the field is not visible.

Automatic feeding

The window's actions menu includes the **Automatic feeding** item, which allows you to run the data processing that automatically generates the **Lookup Table elements**.
The menu is only enabled if a single record is selected and if the value of the **Feeding type** field is *From Tagetik* or *Deployed*.

Lookup Table Elements definition

This window can be accessed from the *Elements definition* link in the Lookup Tables window. The link is only enabled if the lookup table's**Feeding type** is *Deployed*.

If the ETL domain is locked, the window will be read-only.

The number of fields shown (**Input 1....20** / **Output 1....9**) will depend on the number of input/output dimensions that have been defined on the rule.

The suffix of each field's name will be the value of the relative dimension's **Acronym** field.

This window allows you to define, synthetically or otherwise, the rules for the creation of the actual elements that will be generated by the **Automatic feeding** data processing.

Lookup Table elements

This window can be accessed from:

- *Elements* link in the Lookup Table window.
- *Lookup table* workflow task belonging to the Data Integration group
- *Lookup Table - elements* tile located in the ETL homepage, accessible from the end user's homepage by selecting an ETL domain and a lookup table.

If the ETL domain is locked, the window will open in read-only mode.

The number of fields shown (**Input 1....20** / **Output 1....9**) will depend on the number of input/output dimensions that have been defined on the rule.

The suffix of each field's name will be the value of the relative dimension's **Acronym** field.

This window allows you to manage records on which combinations between the input elements and output elements of the lookup table are defined.

If the lookup table's **Feeding type** is *Manual*, the elements must be defined manually by the user.

However, if the lookup table's **Feeding type** is *From Tagetik* or *Deployed*, the **Automatic feeding** data processing will pre-populate the lookup table elements depending on how the lookup table has been defined. The user will be able to access and edit the records generated automatically by the data processing in any case.

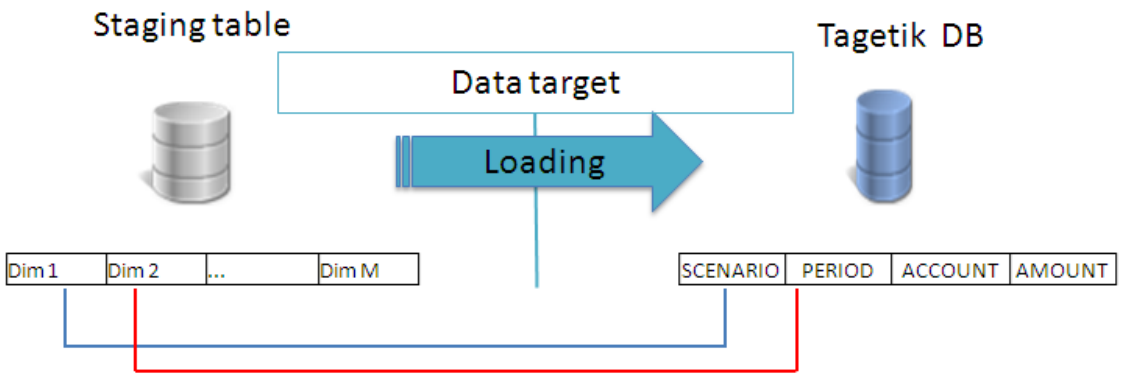
Data target: CCH Tagetik

Data target: CCH Tagetik allows you to define, during the loading step, the CCH Tagetik database tables onto which to load data that have already been extracted and potentially transformed.

To complete the loading step, you must run two operations:

- Define a Data target: CCH Tagetik
- Associate the Data target: CCH Tagetik with a **loading task** of a routine.

***IMPORTANT**: after having defined the tables on which data can be loaded, you must access the **Data target: CCH Tagetik fields definition** window to define the methods for populating every field of each table.*



CCH Tagetik Data Target page

Admin | <application database>

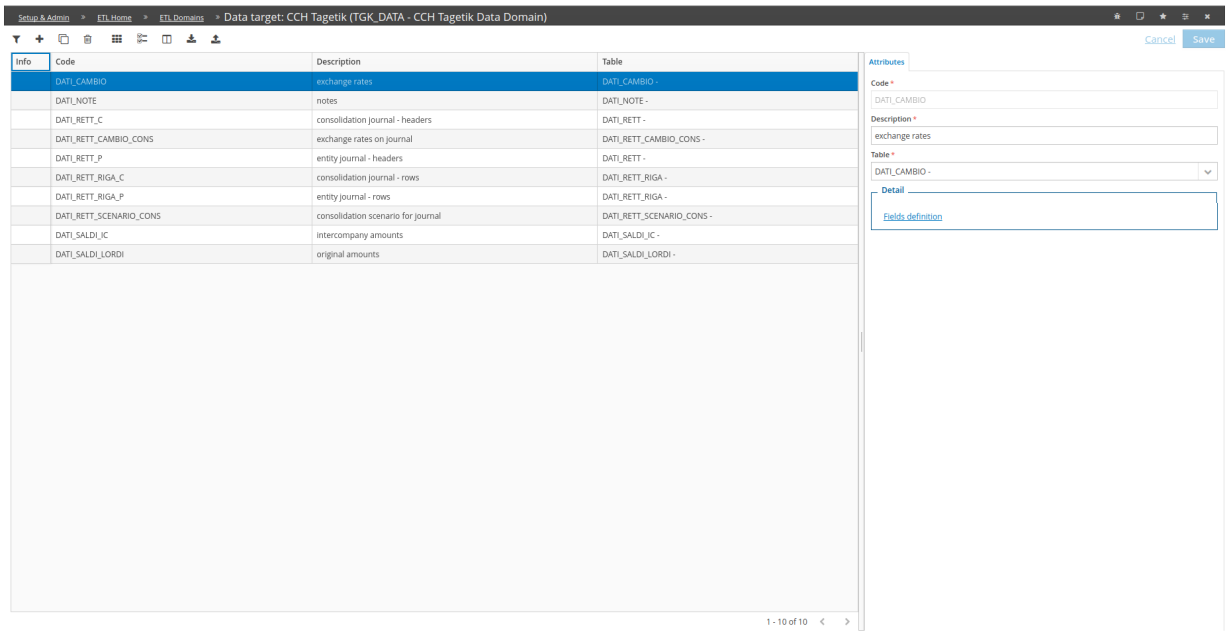
ETL > ETL Domains > CCH Tagetik DataTarget

Note: if the ETL domain is locked, the window will open in read-only mode.

Purpose of window

This window allows you to define the tables onto which data can be loaded

What it looks like



Element	Description
A	Action buttons/Tool bar
B	List of Data Targets
C	Configuration tab

Actions on this page

Element	Description
	This filters the data shown on the page.
	This adds a row in order to insert a new Data Target
	This duplicates the selected row with all the information inserted, except for the code.
	Deletes the element contained in the selected row.
	Enables/Disables the grid modify function. This allows you to change the characteristics of the selected elements directly in the table, and to hide the configuration tabs.
	Enables multi-row selection to allow mass changes.
	Selects the columns to be shown/hidden in the table and those to be locked when scrolling the table.
	Exports the data in Excel format so they can be edited and reimported into CCH Tagetik.
	Imports data from a previously prepared Excel file.

Attributes tab

The following elements can be defined in this tab:

Field	Description
Code	Code of the Data target: CCH Tagetik to be used in the loading task;
Description	Description of the Data target: CCH Tagetik
Table	Name of the CCH Tagetik application database table onto which data must be loaded.
Fields definition	Link from which it is possible to access the **fields definition** window to define how the field must be populated

Fields definition - CCH Tagetik Data Target

Admin | <application database>

ETL > ETL Domains > DataTarget link > FielCCH Tagetikds definition

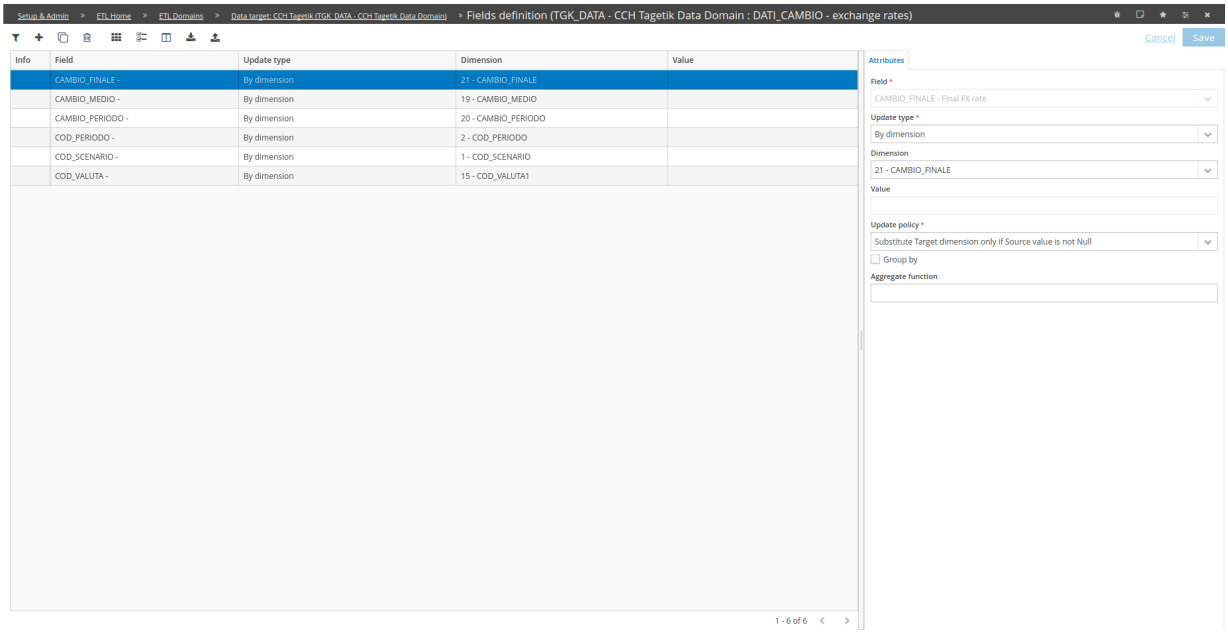
Note: if the ETL domain is locked, the window will open in read-only mode.

Purpose of window

This window allows you to indicate, for every field of the table onto which data will be loaded, whether, during the loading step, it must be populated from the values present in a field of the staging table, related to a dimension of the ETL domain, or from an individual value.

IMPORTANT: if the name of a field of a table in the CCH Tagetik application database is deleted or edited, you must edit the fields definition of every Data target: CCH Tagetik in which the reference to that field is present.

What it looks like



Element	Description
To	Action buttons/Tool bar
B	List of fields in the table
C	Configuration tab

Actions on this page

Element	Description
	This filters the data displayed on the page.
	This adds a row in order to insert a new field
	This duplicates the selected row with all the information inserted, except for the code.
	Deletes the element contained in the selected row.
	Enables/Disables the grid modify function. This allows you to change the characteristics of the selected elements directly in the table, and to hide the configuration tabs.
	Enables multi-row selection to allow mass changes.
	Selects the columns to be shown/hidden in the table and those to be locked when scrolling the table.
	Exports the data in Excel format so they can be edited and reimported into CCH Tagetik.
	Imports data from a previously prepared Excel file.

Attributes tab

The following elements can be defined in this tab:

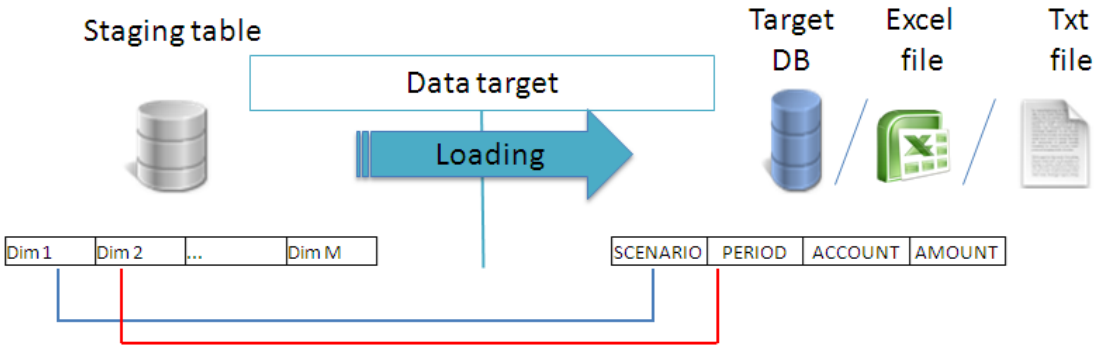
Field	Description
Field	Name of the field to be processed, belonging to the table defined in the Data target: CCH Tagetik. Mandatory
Update type	Field population method. Mandatory. Available options: • ***By dimension***: field populated with values present in the field of the staging table related to a dimension of the ETL domain. • ***By value***: field populated with a fixed value.
Size	If the **Update type *By dimension*** option is active, it indicates the dimension of the ETL domain related to the field of the staging table containing the data to be loaded onto the **Field**
Value	• If the **Update type *By value*** option is active, it indicates the value to be set for the **Field** for all loaded records.
Update policy	indicates how the population of the **Field** on the target table must be managed during an update operation. Mandatory. Possible values: • ***Always substitute Target dimension value with Source value***: the value of the field is always edited, even if its value in the source table is *NULL*; • ***Substitute Target dimension only if Source value is not Null***: the value of the field will only be edited if the new value is not *NULL*; • ***Sum if loading value is not null (only for numeric field)***: if the new value is not NULL, it will be added to the old value present on the field.
Group by	Only active if the **Update type *By dimension*** is enabled, indicates that the staging table data, before being loaded, must be grouped by the field of the staging table related to the dimension of the ETL domain defined in the **Dimension** field;
Aggregate function	Indicates that the field in the *staging* table related to the ETL Environment Dimension defined in the **Dimension** field must have the configured aggregation function applied. Only active in two cases: • the **Update Type** is *From Dimension* • the **Value Type** is *Override current date* ***IMPORTANT**: once the **Group by** option has been activated or a **Aggregate function** has been indicated for a field, it is necessary to set one of the two fields for all the other records whose **Update type** is set to By dimension*

Data Target: Other

Data target: Others allows you to define the custom destinations onto which it is possible to load data that has been previously extracted and potentially transformed, during the loading step. There are the following target types:

- Excel (xls)
- Excel (xlsx)
- Xml
- Fixed length
- Text delimited
- Tab delimited
- Custom table in the CCH Tagetik database
- Custom table in a secondary database

***IMPORTANT**: to complete the loading of data, after a Data target: Others has been defined, it is necessary to associate it with a **loading task** of a routine.*
*Once the tables on which it is possible to load data have been defined, it is necessary to access **Data target: Others fields definition** to manage the population of every field of each custom destination.*



Data Target: Other page

Admin | <application database>

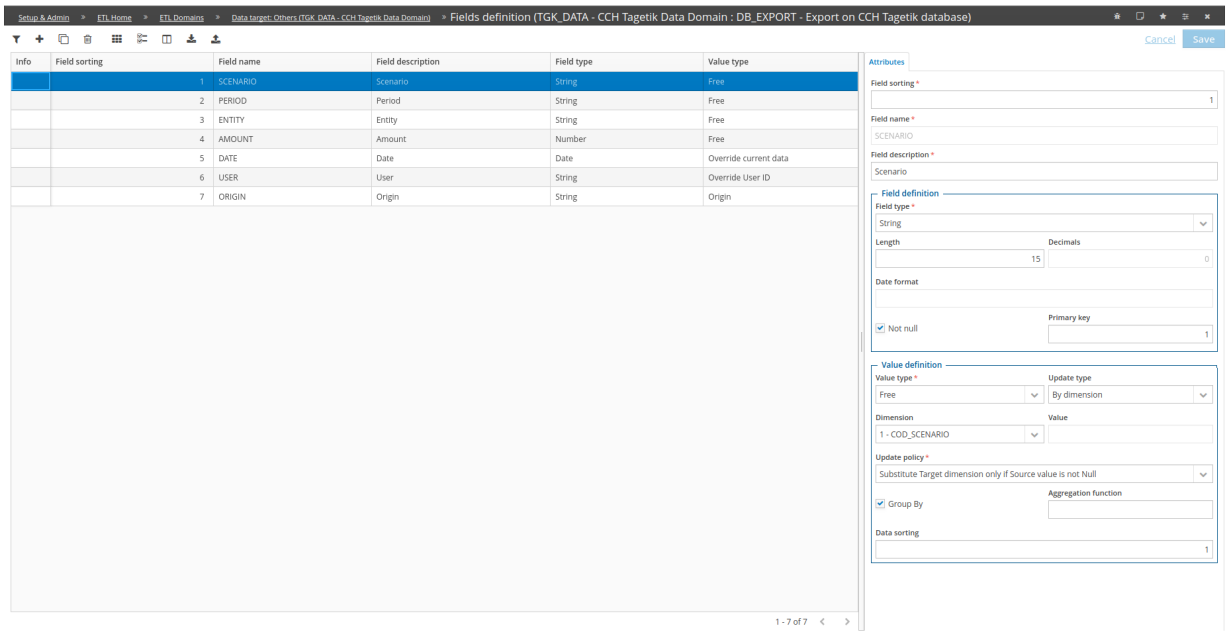
ETL > ETL Domains > Data target: Others link > Fields definition

Note: if the ETL domain is locked, the window will open in read-only mode.

Purpose of the page

This window allows you to define the target destinations on which data can be loaded

What it looks like



Element	Description
A	Action buttons/Tool bar
B	List of Data Targets
C	Configuration tab

Actions on this page

Element	Description
	This filters the data shown on the page.
	This adds a row in order to insert a new field
	This duplicates the selected row with all the information inserted, except for the code.
	Deletes the element contained in the selected row.
	Enables/Disables the grid modify function. This allows you to change the characteristics of the selected elements directly in the table, and to hide the configuration tabs.
	Enables multi-row selection to allow mass changes.
	Selects the columns to be shown/hidden in the table and those to be locked when scrolling the table.
	Exports the data in Excel format so they can be edited and reimported into CCH Tagetik.
	Imports data from a previously prepared Excel file.

Attributes tab

Please find below the most important fields.

Field	Description
Code	*Data target: Others* code to be used in the loading task. Mandatory
Description	*Data target: Others* description. Mandatory
Type	type of *Data target: Others* onto which data must be loaded. Mandatory
Table name / File name	mandatory. Depending on the selected type, this may indicate one of the following options: • custom table name if the type relates to a database • file name if the type relates to a file. If the type relates to a system file, in the file creation step, it will process the name defined both with reference to a loading task and with the default extension. This is all to prevent file overwriting problems if the same routine contains two loading tasks that use the same Data target: Others. For example, if the defined name is “FILE” for a Data target: Others whose type is ***Text Delimited***, and the loading task to which the Data target: Others is associated has 10 as its sort position, the file created will be named “FILE10.txt”. ◦ to tell the system to remove the sort position, simply add the # character to the end of the name. For example, if the defined name is “FILE#”, the created file will be named “FILE.txt”. ◦ to tell the system to remove the sort position and not apply the default extension, simply add the * character to the end of the name. For example, if the defined name is “FILE.csv*”, the created file will be named “FILE.csv”. **Note**: The ETL’s dynamic syntax can be used to define a parametric name (see ETL syntax).

Field	Description
Detail	includes the *Fields definition* link which opens the **fields definition** window where you can indicate, for each field of the custom destination, whether, during the loading step, it must be populated from the values present in a field of the staging table related to a dimension of the ETL domain or from an individual value (defined manually or calculated automatically by the system).
Databases	Active if the selected type relates to a database: • **Create table**: if active, this option tells the system to create the table defined using the **fields definition** on first execution (if it does not already exist); • **Other database connection**: indicates the JDBC connection to connect to a secondary database (selected in the type field) • ***Delete table*** link: allows you to delete, directly from the application, the custom table defined in the **Table name/File name** field;
Files	• a section that can be used if the selected type relates to a file, which contains: ◦ **Excel area name/Xml tag name**: if the type relates to an *Excel (xls)*, *Excel (xlsx)* or *Xml* file, this allows you to indicate: ▪ for Excel files, the name to be given to the sheet and the data area; ▪ for xml files, the name to be given to the tag related to the individual record. if the value is not specified, the value defined in the **Table name/File name** field will be used. ◦ **Add header**: if active, this option tells the system to add a record containing the column headers as the first row of the created file; ◦ **Separator**: if the type relates to a *Text delimited* file, this allows you to specify the character used to separate information items contained in the file columns; ◦ **Charset**: if defined, this indicates the character coding with which the file must be created. If it is not defined, the system will use the default; ◦ **Export file in zip format**: if active, this option tells the system to create the file in zip format. This option is particularly useful to avoid “Out of Memory” problems if there is a lot of data to be loaded;
Use Trim function on data	if active, this tells the system to remove all spaces present in the text or at the end of the value to be loaded.
Use quote in the fields name	if active, this tells the system to delimit the names of the tables and fields with the " character, both in reading data to be loaded and in creating and populating the custom table (if the type is related to a database). The purpose is to preserve the *case* of the defined names and to allow you to use names containing characters that would otherwise be prohibited, like spaces.

Fields definition - Data target: Others window

Admin | <application database>

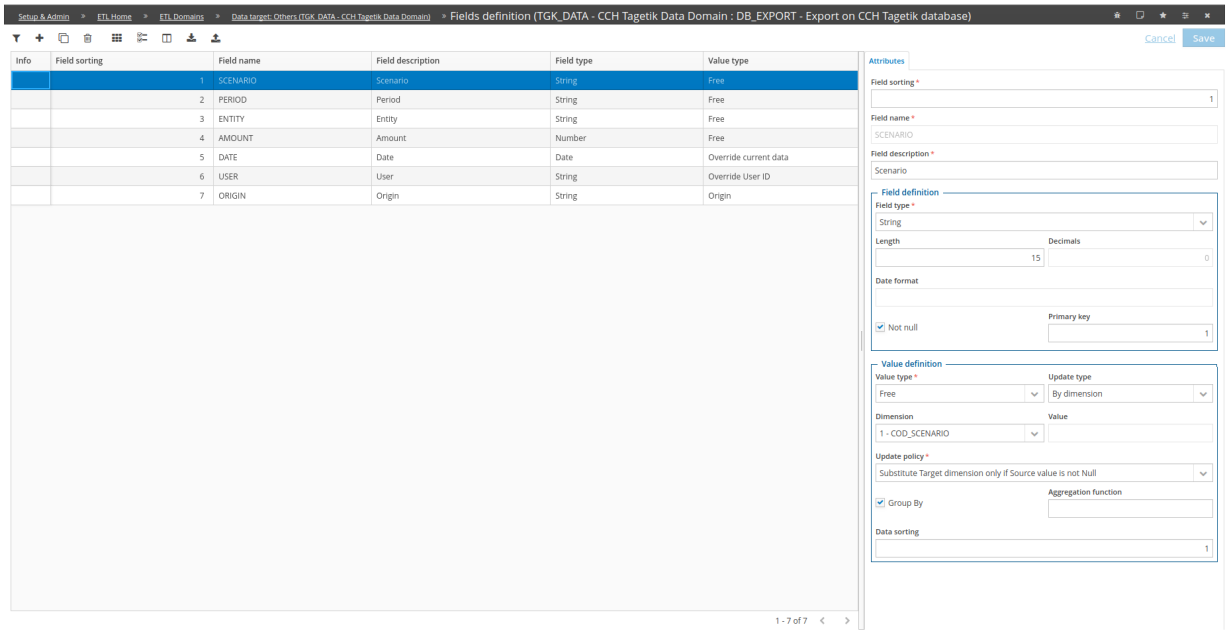
ETL > ETL Domains > CCH Tagetik DataTarget

Note: if the ETL domain is locked, the window will open in read-only mode.

Purpose of window

This window allows you to define the fields onto which data can be loaded

What it looks like



Element	Description
A	Action buttons/Tool bar
B	List of related fields
C	Configuration tab

Actions on this page

Element	Description
	This filters the data shown on the page.
	This adds a row in order to insert a new Data Target
	This duplicates the selected row with all the information inserted, except for the code.
	Deletes the element contained in the selected row.
	Enables/Disables the grid modify function. This allows you to change the characteristics of the selected elements directly in the table, and to hide the configuration tabs.
	Enables multi-row selection to allow mass changes.
	Selects the columns to be shown/hidden in the table and those to be locked when scrolling the table.
	Exports the data in Excel format so they can be edited and reimported into CCH Tagetik.
	Imports data from a previously prepared Excel file.

Attributes tab

The following elements can be defined in this tab:

Field	Description
Code	Data target field code to be used in the loading task;
Description	Description of the data target field

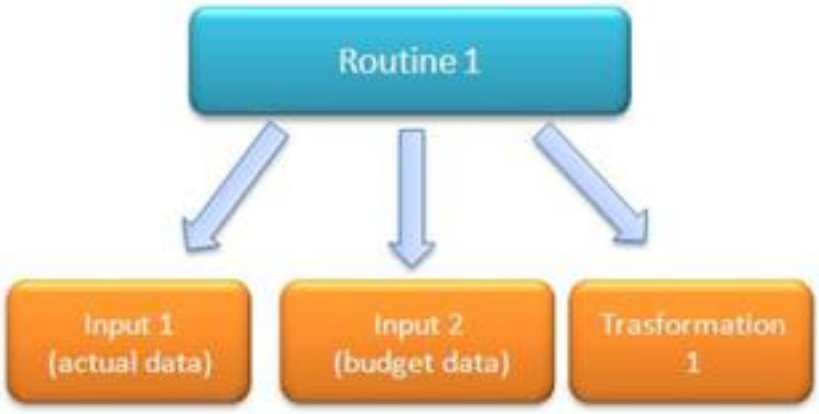
Field	Description
Table	Name of the CCH Tagetik application database table onto which data must be loaded.
Value	If the **Update type** *By value* option is active, it indicates the value to be set for the **Field** for all loaded records.
Fields definition	Link you can use to access the **Data target: CCH Tagetik fields definition** window

Routines

A routine is the minimum data processing unit that can be run in an ETL process and contains all the information for extracting, transforming and loading the data. It is made up of:

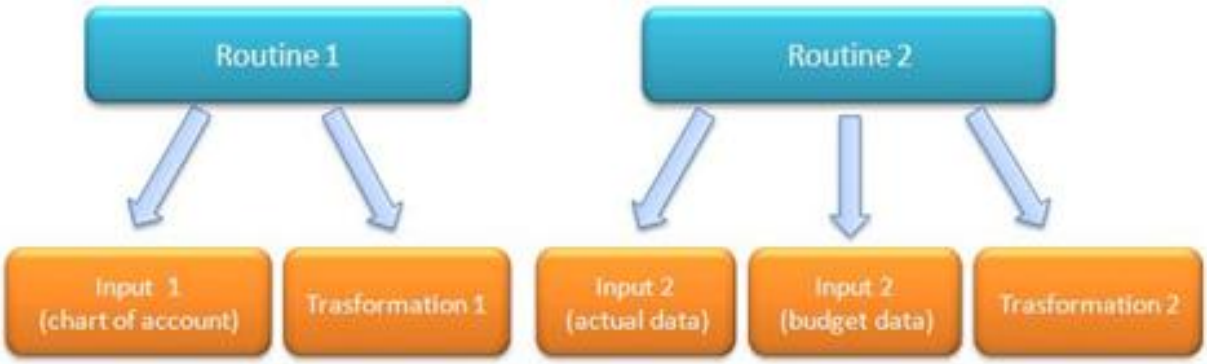
- one or several extraction tasks which allow you to extract data from one or several types of source;
- one or several transformation tasks that allow you to manipulate and convert extracted data;
- one or several loading tasks that allow you to load previously extracted and/or transformed data onto one or several target types.

***IMPORTANT**: as a best practice, when the data to be extracted and processed are "uniform" (e.g. they are either all amounts of various nature or all structural elements) it is recommended to create a single routine.*



Thus, the data will all be extracted together and transformed according to the same logics (Transformation 1).

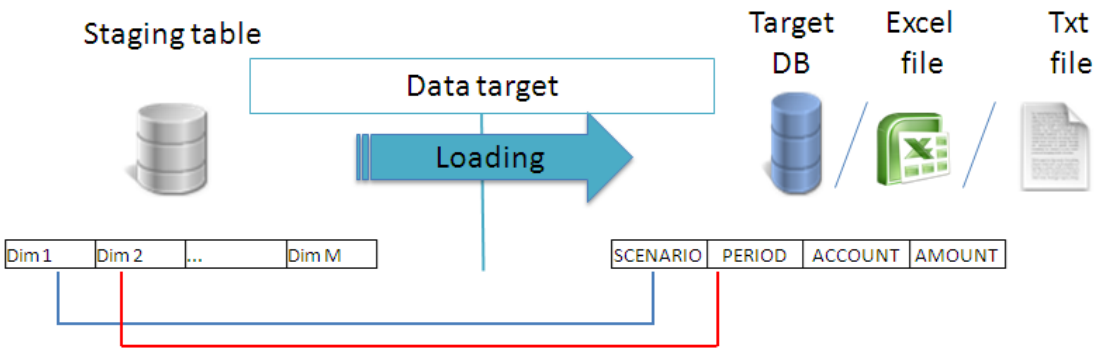
If, however, the data are "non-uniform" (e.g. amounts and structural data), it is recommended to define a routine for each type of data to be extracted.



In this case, the data extracted by Routine 1 will be processed by Transformation 1, whereas the data extracted by Routine 2 (Input 2 and Input 3) will be processed by Transformation 2 in the example.

In CCH Tagetik, the **standard routines** belonging to the **standard ETL domains**exist by default. Using these,you can load both "structural" data (elements of dimensions, hierarchies and groupings) and "values" (amounts, intercompany values, adjustments and FX rates). These routines cannot be edited or deleted by the user.

New **custom routines**, which are fully configurable, can be created (including as copies of standard routines), in order to meet specific business needs.



Routines page

Admin | <application database>

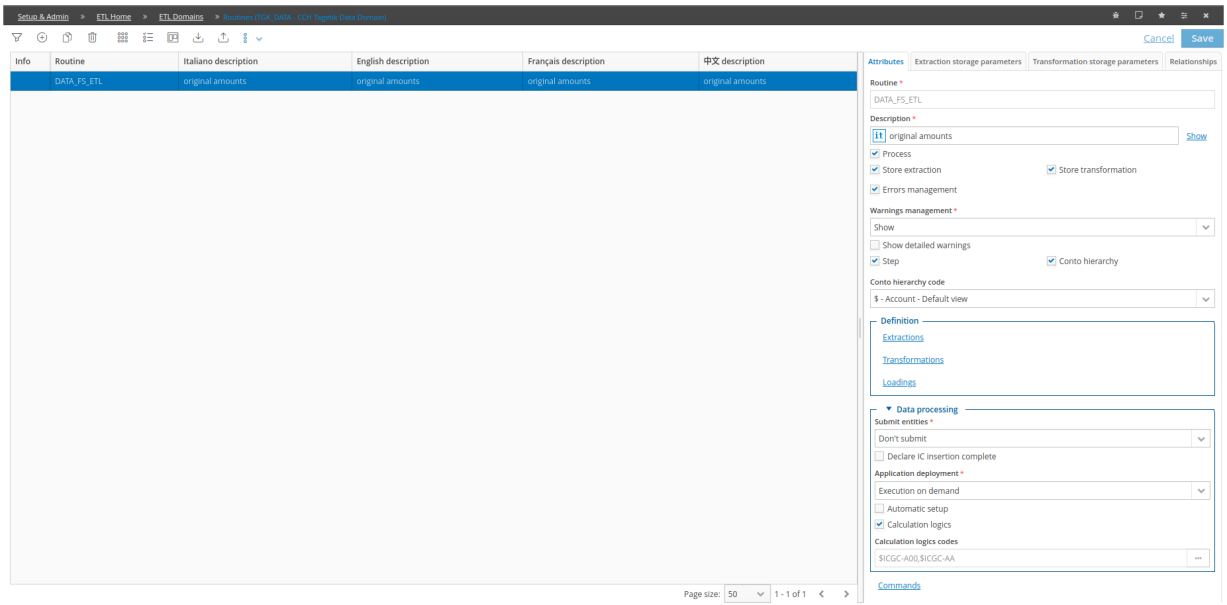
ETL > ETL Domains > Routines link

Note: if the ETL domain is locked, the window will open in read-only mode.

Purpose of the page

This window allows you to define the routines of ETL domains

What it looks like



Element	Description
A	Action buttons/Tool bar
B	List of Routines
C	Configuration tab

Actions on this page

Element	Description
	This filters the data shown on the page.
	This adds a row in order to insert a new routine
	This duplicates the selected row with all the information inserted, except for the code.
	Deletes the element contained in the selected row.
	Enables/Disables the grid modify function. This allows you to change the characteristics of the selected elements directly in the table, and to hide the configuration tabs.
	Enables multi-row selection to allow mass changes.
	Selects the columns to be shown/hidden in the table and those to be locked when scrolling the table.
	Exports the data in Excel format so they can be edited and reimported into CCH Tagetik.
	Imports data from a previously prepared Excel file.

Attributes tab

Please find below the most important fields.

Field	Description
Code	*Routine* code to be used in the loading task. Mandatory
Description	*Routine* description. Mandatory
Process	If activated, this tells the routine that it must request a filter on the process in the execution step. It must be activated for routines that manage “value” data and have a CCH Tagetik table as the loading destination. Thus, the system can run all the necessary editability checks to understand whether or not a data item can be loaded;
Store extraction	If activated, this tells the routine that, at the end of the extraction step, it must save the data contained in the staging table on the extracted data storage table. The data are saved by ETL domain/routine/user and potentially by the extraction storage parameters defined in the routine. This is useful: • to be able to subsequently run queries on the extracted data and have full traceability, over time, of what data have been extracted, by whom, and when; • to be able to run a routine in partial mode, starting with the transformation phase or considering data saved on a the storage table of a previous execution as the data to be extracted

Field	Description
Store transformation	If activated, this tells the routine that, at the end of the transformation step, it must save the data contained in the staging table on the transformed data storage table. The data are saved by ETL domain/routine and potentially by the transformation storage parameters defined in the routine. This is useful: • to subsequently run queries on the transformed data and have full traceability, over time, of what data have been transformed, by whom, and when; • to run a drill-through query within a form to understand in detail how a data item from ETL originated. For example, suppose we extract data from the payments and salaries subsystem, that in this subsystem the data are detailed by sector, and that we load them onto CCH Tagetik, but on a single Salaries account. After the Routine has been executed on the Salaries account, the amount will be, for example, 1,000 but, as we have stored the extraction data, by carrying out a drill-through query within a form, we can see how the number 1,000 originated, and each sector's contribution to the total of 1000; • to be able to run a routine in partial mode, starting with the loading step or considering data saved on a the storage table of a previous execution as the data to be transformed;
Error management	If activated, this tells the routine that, where there are errors, the system must intercept them, continue with the execution, and only at the end allow them to be displayed using the indicators (coloured traffic lights) associated with each step of the routine. If deactivated, however, where there are errors in the routine execution step, user will be presented with the stack trace of the error as soon as it is detected and the execution will be stopped;
Warnings management	Allows you to set how the routine must behave if warnings (yellow traffic light) are returned in the execution step. The possible behaviours are as follows: • **None**: the routine is not stopped; • **Show**: the routine is stopped and the user can decide whether to continue with its execution; • **Lock**: the routine is stopped and does not allow the user to continue with its execution
Show detailed warnings	If activated, this tells the routine to show, in detail (non-grouped pre-loading records), any warnings generated in the loading onto CCH Tagetik step for the insertion and updating of grouped data operations.
Step	in the loading onto CCH Tagetik tables step related to Amounts, Intercompany values and Adjustments, you can partialise the data to be inserted /deleted based on the Account. If activated, this tells the routine to load only data which make reference to accounts that are present in the steps (which will be specified in the execution step) of the process being run;
Account hierarchy	in the loading onto CCH Tagetik tables step related to Amounts, Intercompany values and Adjustments, you can partialise the data to be inserted/deleted based on the Account. If activated, this tells the routine to load only data which make reference to accounts that are related to nodes (which will be specified in the execution step) of the **Account hierarchy** indicated in the field of the same name (this value can also be specified in the execution step).
Declaration	This contains the links to access the management of the following tasks: Extractions Transformations Loading

Field	Description
Submit entities	If activated, this tells the routine that the data are submitted automatically at the end of the loading step. In particular, if this option is set to: • **Don't submit**, the system will not run data submission at the end of the loading step; • **Submit all steps**. the system will automatically submit all steps of the process run for all scenarios, period and entities proceed as parameters; • **Submit all previous steps to the IC matching step**. the system will automatically submit all steps strictly lower than the step in which the first intercompany matching cockpit is present, for all scenarios, period and entities processed as parameters ***Note**: Once activated, in order for the option to work correctly, in the execution step a process must be indicated and the routine must use parameters related to Dimensions which have Scenario, Period and Entity as their "Identity". Thus, the scenarios, periods and entities to be submitted are indicated.*
Declare IC insertion complete	If activated, this tells the routine that, after the data loading step, the insertion of intercompany entries is automatically declared as finished for all the intercompany matching cockpits belonging to the process run for all scenarios, periods and entities processed as parameters; ***Note**: Once activated, in order for the option to work correctly, in the execution step a process must be indicated and the routine must use parameters related to Dimensions which have Scenario, Period and Entity as their "Identity". Thus, the scenarios, periods and entities to be submitted are indicated.*
Application deployment	If activated, this tells the routine that, before every loading task on CCH Tagetik "values" tables, it must run the necessary deployments deriving from other loading tasks run by the same routine on CCH Tagetik "structure" tables. If the routine does not contain any loading task on "structure" tables, the deployment is never run regardless of how this option has been set. The possible values are: • **Never**. Deployments must be run manually after the routine has been executed; • **Execution on demand**. The user will tell the system whether it must run the necessary deployments when the routine is executed; • **Always**. The system will automatically run the necessary deployments.
Automatic setup	If active, this tells the routine to automatically create the user rights related to the loaded "structural" data (entities, custom dimensions, categories, etc.). This is useful if the routine requires loading onto "value" tables using those elements, thus avoiding errors related to editability checks;
Calculation logics	If active, this tells the routine that the data processing is run at the end of the loading step • "Entity data processing - Standard formulas" • "Entity data processing: Formulas". on **Calculation logics** defined in the field of the same name. ***Note**: the data processing is run for the original scenarios, periods and entities selected in the routine execution step via ETL parameters that have related dimensions whose identity is Scenario, Period and Entity.*

Extraction storage parameters tab

Please find below the most important fields.

Field	Description
Extraction - Parameter 1....9	If the routine manages the storage of extracted data, up to nine **parameters** can be defined for each operation. The parameters, together with the routine ID, filter the deletion of the storage table and the subsequent insertion of data from the staging table.

Transformation storage parameters tab

Please find below the most important fields.

Field	Description
Transformation - Parameter 1....9	If the routine manages the storage of transformed data, up to nine **parameters** can be defined for each operation. The parameters, together with the routine ID, filter the deletion of the storage table and the subsequent insertion of data from the staging table.

Relationships tab

Please find below the most important fields.

Field	Description
Nodes	This tab allows you to view and edit the grouping nodes to which the selected routine is related. Every new routine inserted will automatically be associated with the grouping nodes indicated as to be related by default.

Commands routine

Admin | <application database>

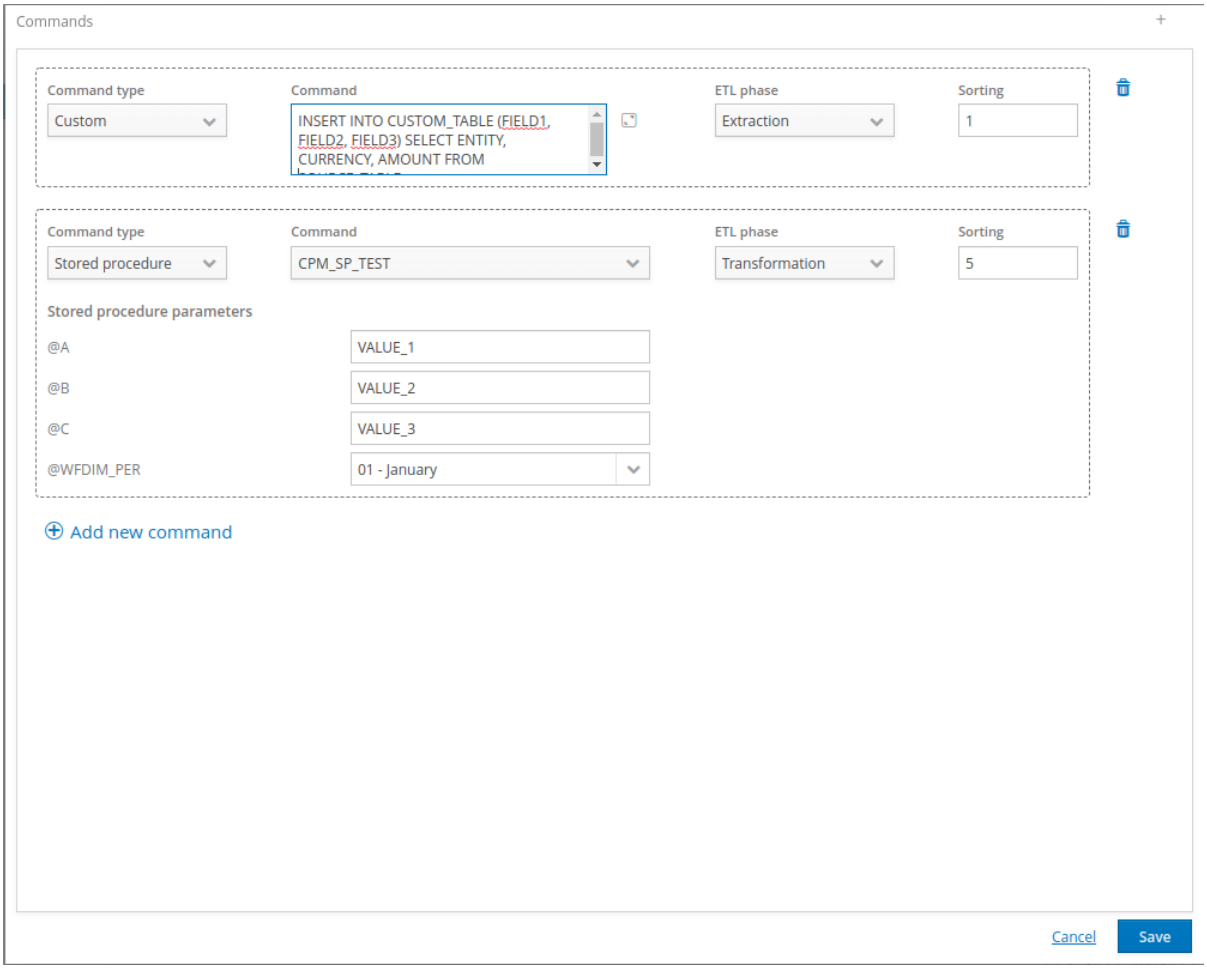
ETL > Domains > Routines link in the Definition section > Commands link

Note: this link is visible only if the 0003874 function is included in the user's role and is enabled. If the ETL domain is locked, the link will not be selectable.

Purpose of the page

This page allows you to define *custom statements* and *stored procedures* that can be run during the execution of an ETL routine.

What it looks like



Element	Description
A	Commands Management section
B	Link to add new commands

Buttons on the page

Element	Description
	Allows you to add a command.
	Allows you to remove a command.

Commands

Please find below the most important fields.

Field	Description
Command type	Allows you to define the type of command to be run during the execution of a routine: • **Custom** allows you to manually define a generic SQL statement. • **Stored procedure** allows you to define a stored procedure.
Command	If the type is **Custom**, it will contain the SQL text to be run; the ETL's dynamic syntax can be used (see Filter Syntax or SQL Query) If the type is **Stored procedure**, it will allow you to select one of the stored procedures present in the CCH Tagetik database whose name has the prefix *CPM_SP*.
ETL phase	Indicates the step of the ETL routine (Extract, Transform, Load) in which to run the command
Sorting	Indicates the order in which to run the command in respect of the Extraction, Transformation or Loading tasks involved in the step of the ETL routine; it the sorting is the same, the task is run first.
Parameters	If the selected stored procedure requires the selection of parameters, these will be shown in the **Stored procedure parameters** panel below. Depending on the parameter name, different editors will be shown to select the value (see Stored procedure manual). The ETL's dynamic syntax can be used (see Filter Syntax or SQL Query) as the value of a parameter.

Commands routine

Admin | <application database>

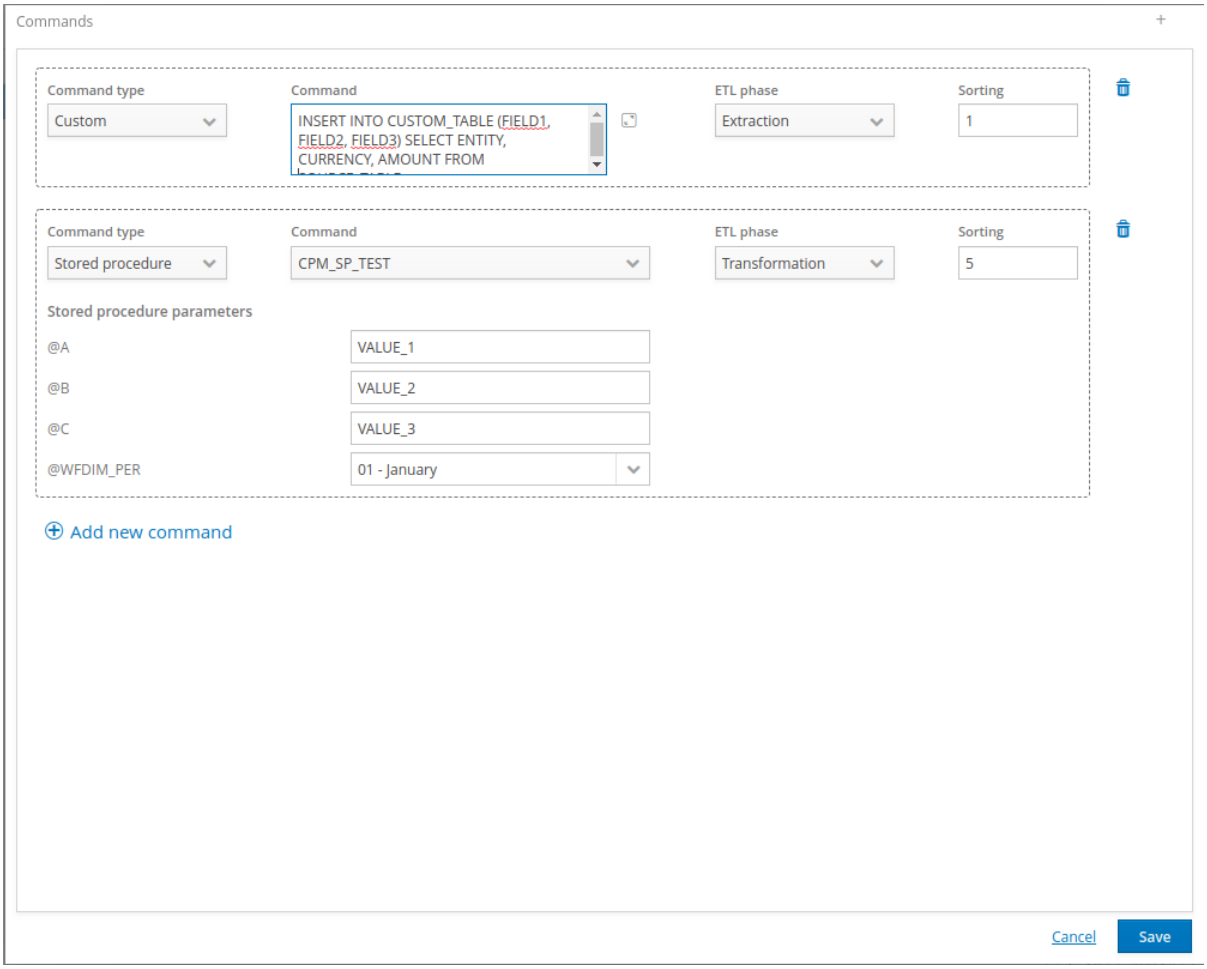
ETL > Domains > Routines link in the Definition section > Commands link

Note: this link is visible only if the 0003874 function is included in the user's role and is enabled. If the ETL domain is locked, the link will not be selectable.

Purpose of the page

This page allows you to define *custom statements* and *stored procedures* that can be run during the execution of an ETL routine.

What it looks like



Element	Description
A	Commands Management section
B	Link to add new commands

Buttons on the page

Element	Description
	Allows you to add a command.
	Allows you to remove a command.

Commands

Please find below the most important fields.

Field	Description
Command type	Allows you to define the type of command to be run during the execution of a routine: • **Custom** allows you to manually define a generic SQL statement. • **Stored procedure** allows you to define a stored procedure.
Command	If the type is **Custom**, it will contain the SQL text to be run; the ETL's dynamic syntax can be used (see Filter Syntax or SQL Query) If the type is **Stored procedure**, it will allow you to select one of the stored procedures present in the CCH Tagetik database whose name has the prefix *CPM_SP*.
ETL phase	Indicates the step of the ETL routine (Extract, Transform, Load) in which to run the command
Sorting	Indicates the order in which to run the command in respect of the Extraction, Transformation or Loading tasks involved in the step of the ETL routine; it the sorting is the same, the task is run first.
Parameters	If the selected stored procedure requires the selection of parameters, these will be shown in the **Stored procedure parameters** panel below. Depending on the parameter name, different editors will be shown to select the value (see Stored procedure manual). The ETL's dynamic syntax can be used (see Filter Syntax or SQL Query) as the value of a parameter.

Routine - Extraction

Admin | <application database>

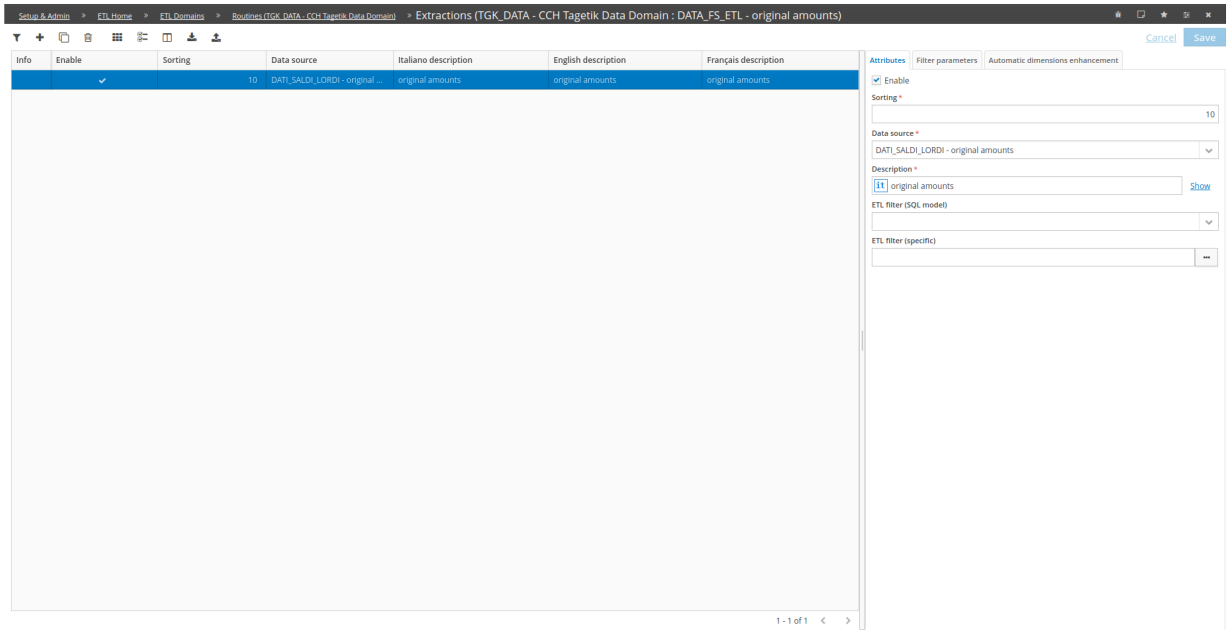
ETL > Domains > select the desired domain > Routines > Extractions link

Note: if the ETL domain is locked, the link will not be selectable.

Purpose of the page

On this page, you can manage the tasks of a Routine that allow you to extract data from sources defined in the **Data sources** and save it in the ETL's staging table

What it looks like



Element	Description
A	List of extraction tasks
B	Extraction task attributes

Attributes tab

If the **Active** option is enabled, the task will be run (default).

Field	Description
Sorting	indicates the order in which the task must be run. You cannot set the value 0 or a sorting that already exists
Data source	indicates, through a **data source**, the possible data extraction methods to be selected later in the execution step (mandatory)
Descriptions	The description of the routine in the languages enabled in the environment
ETL filter (SQL Model)	allows you to select a filter that has been defined previously in the **SQL models** to limit the number of data items to be extracted.
ETL filter (specific)	allows you to indicate a filter with the SQL syntax to limit the number of data items to be extracted. The ETL's dynamic syntax can be used (see ETL syntax) ***IMPORTANT**: If the user specifies both the **ETL filter (SQL Model)** and the **ETL filter (specific)**, the system will concatenate the conditions with the logic operator AND.*

Filter parameters tab

This tab allows you to set up to nine filter **parameters** to dynamically limit the number of data items to be extracted.

Field	Description
Filter parameter 1, 2...9	In each field, you can select one of the parameters created in the ETL domain

***IMPORTANT**: only From CCH Tagetik, From Dimension Dictionary and Free type parameters can be used.*

Note: If the user has also specified one of the two ETL filters, the system will concatenate the conditions with the logic operator *AND*.

Automatic dimensions enhancement tab

This tab allows you to define up to nine dimensions, automatically, in the execution step, by setting the **Dimension 1...9** / **Value 1...9** pair as indicated in the following example:

Field I	Field II
Dimension 1	**Value 1**
Dimension 2	**Value 2**

Note: These dimensions cannot also be specified in the **data source fields definition**.
Data extraction will only take place using one of the methods defined by the **data source** selected in the execution step.

***IMPORTANT**: If several extraction tasks are required, for each task the system will insert the extracted data into the staging table in addition to those extracted from any previous loading tasks.*

Routine - Transformation

Admin | <application database>

ETL > Domains > select the desired domain > Routines > Transformations link

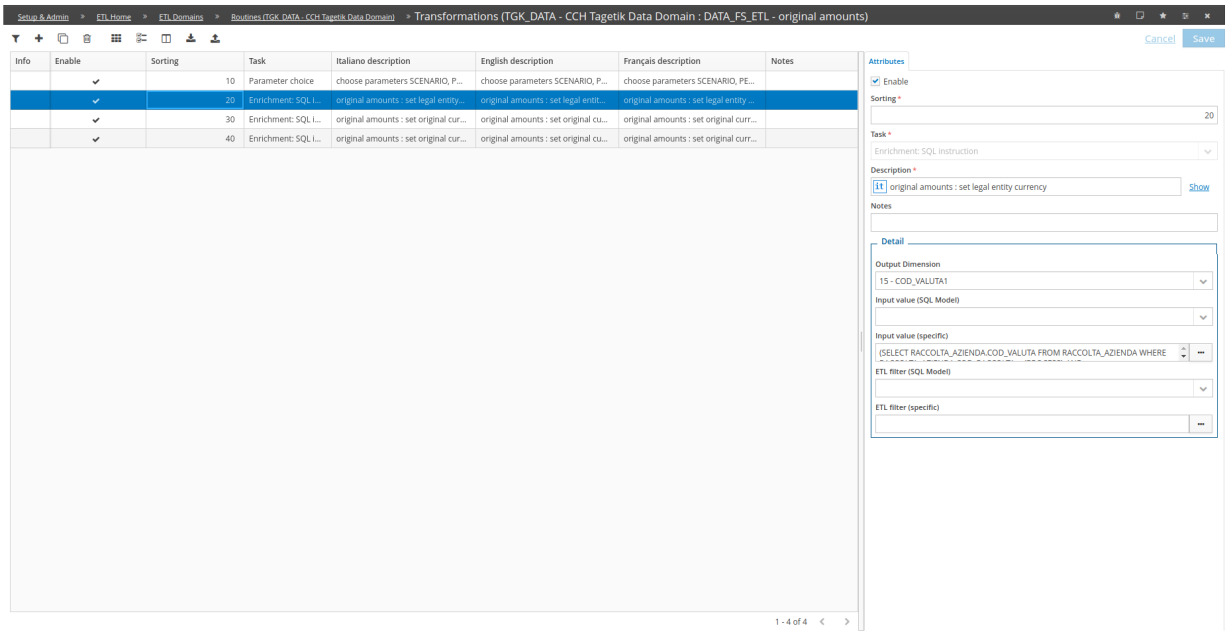
Note: if the ETL domain is locked, the link will not be selectable.

Purpose of the page

This page allows you to manage Transformations, i.e. tasks of a routine which, in the execution step, allow the following activities on data in order to make them ready for loading:

- filtering
- check
- enrichment

What it looks like



Element	Description
A	List of transformation tasks
B	Transformation task attributes section

Attributes tab

Field	Description
Enable	If enabled, the task will be run. Selected by default.
Sorting	indicates the order in which the task must be run. You cannot set the value 0 or a sorting that already exists
Task	indicates the activities that can be run, subdivided into the following macro types: • Filter task • Check task • Enrichment task ***IMPORTANT**: Transformation tasks are run sequentially following the **Sorting** defined in the setup step. Every task acts on the staging table and processes the data produced by the previous task (except for the first, whose input data are those generated by the extraction step).* **Note**: Each **task** may require further information to be defined. A specific details panel will appear in the window depending on the selected **task**. The details of each panel are explained in the section on task type.
Descriptions	Descriptions of the tasks in the languages enabled in the environment.
Notes	indicates any notes
ETL filter (SQL Model)	allows you to indicate a filter that has been defined previously in the **SQL models** to limit the number of data items to be extracted;
ETL filter (specific)	allows you to indicate a filter with the SQL syntax to limit the number of data items to be extracted. The ETL's dynamic syntax can be used (see ETL syntax) **Note**: If the user specifies both the **ETL filter (SQL Model)** and the **ETL filter (specific)**, the system will concatenate the conditions with the logic operator *AND*

Filter task

Task	Description
Parameter choice	allows you to define up to 9 **Parameters** and, based on the definition type, enable the following options: • define a dimension for all records to be transformed (if the parameter type is *Free*, *From Tagetik* or *From dimension dictionary* and it does not have the **Multiselection** option active); • filter the data to be transformed by deleting from the staging table all records for which the dimension related to the parameter does not contain a certain value selected by you for the parameter (if the parameter has a different configuration from that described in the previous point). **Note**: The values to be assigned to a parameter are selected on execution and depend on the definition of the parameter used (see **Parameters**).
Delete by SQL	allows you to delete from the staging table all records that do not satisfy the filter defined in the **ETL filter (SQL Model)** and/or **ETL filter (specific)** fields.

Check task

Task	Description
Invalid characters check	allows you to check whether the values of a given **Dimension** defined in the **Dimension to check** field contain characters that are not allowed in CCH Tagetik codes.
Dictionary check	allows you to check, for the ETL domain dimensions of the routine with which a **Dictionary** is associated, that the values present in the processed data belong to the set of values allowed for the defined dictionary. By default, the system runs the check on all dimensions present in the routine's ETL domain for which a dictionary has been defined. If the **Data Source** field is defined, you can limit the check to the ETL domain dimensions used in the **data source fields definition.** By activating the **Signal empty values in the check** option in the dimension's dictionary definition, you can make sure that this task also includes the absence of a value for the dimension as one of the check's fail conditions.
Completeness check	allows you to check, for the ETL domain dimensions of the routine with which a **Completeness Check** is associated, that the values present in the processed data satisfy the completeness rules related to the record. This means that for every record of the staging table and for every value of a dimension included in the check, all other dimensions defined as mandatory must contain a value, whereas those defined as prohibited cannot contain any value (*NULL* value). This check is useful for verifying that: • data extraction has taken place correctly; • enrichment tasks following extraction have been run correctly. By default, the system runs the check on all dimensions present in the routine's ETL domain for which a completeness check has been defined. If the **Data Source** field is defined, you can limit the check to the ETL domain dimensions used in the **data source fields definition**. **Note**: These tasks generate user warnings, for the staging table records on which they are run that do not pass the checks.
Completeness and dictionary check	This task allows you to run both the *Dictionary Check* task and the *Completeness check* task.
Check with SQL	This task allows you to identify all the records that do not satisfy the filter defined in the **ETL filter (SQL Model)** and/or **ETL filter (specific)** fields. It is usually used to search for data containing logically incorrect information. This task may be useful, for example, if run before a *Delete by SQL* task in order to allow the user to see what the system will delete in the next task and thus allow them to make any corrections.
Lookup Table Diagnostic	allows you to check that the **lookup table** defined in the **Lookup Table** field has been set up correctly. The following checks are performed: • if the lookup table has the **Full relationship** option active, the system checks that the staging table contains all the values that male up the input combinations defined in the lookup table elements; • if the lookup table does not have the **Multiple relationship** option active, the system checks that the combination of input values defined in the lookup table elements is unique; • if the lookup table has the **Multiple relationship** option active, the system checks that the combination of input and output values defined in the lookup table elements is unique.

Enrichment task

Task	Description
Lookup table feeding	Only valid for lookup tables that have their feeding type defined as *Tagetik* or *Deployed*: allows you to feed the **lookup table** elements defined in the **Lookup Table** field. The system will delete all existing elements (even if edited) and insert the new elements from the data source defined by the lookup table.
Data grouping	allows you to aggregate the data obtained in the extraction step or the data enriched by a previous transformation task. In the **Dimensions to aggregate (SQL Model)** / **Dimensions to aggregate (specific)** fields, you must specify the list of fields (separated by commas) of the *staging* table that must be aggregated, including the aggregation function to be used. You need not use upper case when writing the text *E.g.: "MAX(CAMPO1), MIN(CAMPO2), SUM([conversion function in number format](CAMPO3))").* When a numerical aggregation function is used, you must set, for the *staging* table field, the conversion function in number format required by the database type. Alternatively, you can have the system manage any conversion in number format by replacing the *CAMPOx* syntax with the value of the **Acronym** field between *{}* of the dimension related to the *staging table field*. Any filters set on the task will only perform the aggregation of the portion of the data that satisfies that filter. The non-aggregated portion of the data will support the new aggregated data and will be available for the next transformation task.
Enrichment: lookup table	this allows you to apply the transcoding rule defined in the **lookup table** specified in the **Lookup Table** field. In the staging table, the output dimensions are defined with the values of the output elements for the records where the values of the input dimensions are aligned with the values defined in the input elements. If the **Multiple relationship** option is active for the rule, the system will open as many records as there are combinations of values defined in the output elements, with the same combination of input values, for every combination of values found in the input dimensions.
Enrichment: matrix calculation	Function used in the context of the Italian banking system. Allows you to manage specific additional information that is useful for relating data of various origin and nature (loans, current accounts, etc.), with the consolidated chart of accounts.

<table>
<tr><th>Task</th><th>Description</th></tr>
<tr><td>Enrichment: SQL instruction</td><td>allows you to define the staging table field related to the dimension defined in the Output Dimension field using an SQL syntax defined in the Input text (SQL model) or Input text (specific) fields, where it is also possible to use the ETL's dynamic syntax (see Filter Syntax or SQL Query). You can indicate a query or a fixed value delimited by single inverted commas.

Example 1: Suppose we want to define the field related to the custom dimension (6 - Custom dimension) with the last three characters of the value of the field related to the account (5 - Account) only for accounts starting with IC.
To do this, we must define:

Output Dimension: 6 - Custom dimension

Input text: RIGHT(CAMPO5, 3)

ETL filter (specific): LEFT(CAMPO5, 2) = 'IC'

Thus, the following result is obtained:

<table>
<tr><th>5 - Account</th><th>6 - Custom dimension</th></tr>
<tr><td>IC_001</td><td>001</td></tr>
<tr><td>IC_002</td><td>002</td></tr>
<tr><td>003</td><td></td></tr>
<tr><td>004</td><td></td></tr>
<tr><td>IC_005</td><td>005</td></tr>
</table>

Example 2: Suppose we want to define the field related to the entity currency (14 - Entity currency) with the currency defined for the entity (defined on dimension 3 - Entity) on process $PROCESS. To do this, we must define:

Output Dimension: 14 - Entity currency

Input text: (SELECT COD_VALUTA FROM RACCOLTA_AZIENDA WHERE COD_RACCOLTA = '$PROCESS' AND COD_AZIENDA = CAMPO3)

Thus, the following result is obtained:

<table>
<tr><th>3 - Entity</th><th>14 - Entity currency</th></tr>
<tr><td>ENTITY_001</td><td>EUR</td></tr>
<tr><td>NOT_AN_ENTITY</td><td></td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Task</th><th>Description</th></tr>
<tr><td>Enrichment: formula</td><td>allows you to define the staging table field related to the dimension defined in the Output Dimension field by applying a numerical formula previously saved as the value of the field related to the dimension defined in the Input dimension field that uses, in its syntax, references to other fields made through the ID of the dimension number between braces (EXAMPLE: "({2}*{3} / 100)").
All the dimensions used both in the definition of the task and in the syntax of the formula must be numerical type and contain a valid number.
Example: Suppose that the field related to the amount to be loaded onto CCH Tagetik (4 - Result) is represented by numerical operations defined as a formula in the field related to dimension "1 - Formula" from among the amounts contained in the fields related to dimensions "2 - Amount1" and "3 - Amount2".
To do this, we must define:
Input dimension: 1 - Formula
Output Dimension: 4 - Result
Thus, the following result is obtained:
<table>
<tr><th>1 - Formula</th><th>2 - Amount1</th><th>3 - Amount2</th><th>4 - Result</th></tr>
<tr><td>({2} * {3}) / 100</td><td>7000</td><td>10</td><td>700</td></tr>
<tr><td>{3} - {2}</td><td>200</td><td>300</td><td>100</td></tr>
</table>
</td></tr>
<tr><td>Enrichment: amounts total</td><td>allows you to define the staging table field related to the dimension defined in the Output Dimension field with the sum of the amounts contained in the field related to the dimension defined in the Input dimension field, by summing up, with the same values contained in other fields of the staging table listed in the Dimensions to aggregate (SQL model) / Dimensions to aggregate (specific) fields (e.g. "CAMPO1, CAMPO2, CAMPO3").
Note: The input and output dimensions must be numerical type and the relative fields must contain numerical values.
Example: Suppose we want to sum up the amount contained in the field related to dimension "4 - Amount" for the fields related to dimensions "1 - Scenario", "2 - Period", "3 - Entity" and save it in the field related to dimension "5 - Total amount".
To do this, we must define:
Input dimension: 4 - Amount
Output Dimension: 5 - Total amount
Dimensions to aggregate (specific): CAMPO1,CAMPO2,CAMPO3
Thus, the following result is obtained:
<table>
<tr><th>1 - Scenario</th><th>2 - Period</th><th>3 - Entity</th><th>4 - Amount</th><th>5 - Total amount</th></tr>
<tr><td>2017ACT</td><td>01</td><td>001</td><td>100</td><td>300</td></tr>
<tr><td>2017ACT</td><td>01</td><td>001</td><td>200</td><td>300</td></tr>
<tr><td>2017ACT</td><td>01</td><td>002</td><td>200</td><td>500</td></tr>
<tr><td>2017ACT</td><td>01</td><td>002</td><td>300</td><td>500</td></tr>
<tr><td>2017ACT</td><td>02</td><td>001</td><td>100</td><td>150</td></tr>
<tr><td>2017ACT</td><td>02</td><td>001</td><td>50</td><td>150</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Task</th><th>Description</th></tr>
<tr><td>Enrichment: ranking</td><td>This task allows you to define the staging table field related to the dimension defined in the Output Dimension field with a cumulative number with the same values contained in other fields of the staging table.
The method by which the cumulative number is assigned is indicated in the Ranking Type field:
• Differentiate sorting (value priority): the cumulative number is increased for every record contained in the range of records with the same values in the fields on which to run the ranking. However it is reset for every new range of records;
• Keep sorting (value priority): the same cumulative number is assigned to the records contained in the same range of records with the same values in the fields on which to run the ranking. It is increased for every new range of records.
The fields of the staging table on which to run the ranking and for which the sorting takes place and the cumulative numbering is reset are defined in the Dimensions to aggregate (SQL model) / Dimensions to aggregate (specific) fields (e.g.: GROUP BY CAMPO1, CAMPO2, ORDER BY CAMPO1, CAMPO2); these fields can contain:
• a GROUP BY clause which indicates the staging table fields which, with the same values, are inserted in the same range of records on which the ranking is applied, as defined in the Ranking Type field;
• an ORDER BY clause which indicates the staging table fields by which to sort before assigning the range to each record.
Example: Suppose we want to sort the amounts contained in the field related to dimension "2 - Amount” with the same value as the code of the account contained in the field related to dimension "1 - Account”, sorting by amount.
To do this, we must define:
Ranking Type: Differentiate sorting (same values)
Output Dimension: 3 - Output ranking
Dimensions to aggregate: GROUP BY CAMPO1 ORDER BY CAMPO2 DESC
Thus, the following result is obtained:
<table>
<tr><th>1 - Account</th><th>2 - Amount</th><th>3 - Ranking output</th></tr>
<tr><td>001</td><td>100</td><td>1</td></tr>
<tr><td>001</td><td>200</td><td>2</td></tr>
<tr><td>001</td><td>300</td><td>3</td></tr>
<tr><td>002</td><td>50</td><td>1</td></tr>
<tr><td>002</td><td>150</td><td>2</td></tr>
<tr><td>002</td><td>250</td><td>3</td></tr>
</table>
</td></tr>
<tr><td>Year-to-date calculation</td><td>Transforms the loaded amounts, by period, into cumulative values.
Based on Period length dimension.
This calculation is performed in order to standardise the imported data with the data managed by CCH Tagetik and is only applied to Profit & Loss and Other Variations accounts or to Variation accounts. The calculation depends on the process type.</td></tr>
<tr><td>Verticalization</td><td>This task allows you to define, in Tagetik, details related to the dimension defined in the Output Dimension field, which are subdivided among various fields in the staging table.
The calculation multiplies the rows, combining the fields for all the combinations required for the Output Dimension</td></tr>
<tr><td>Counterparty zone unbundling</td><td>Task used by Italian banking entities for loading their disclosures to the Bank of Italy</td></tr>
<tr><td>Transfer Price from the Rule</td><td>The amount to be defined is calculated as “Price X Quantity”
The formula is applied by indicating the field that expresses the quantity and the field on which to write the result of the formula</td></tr>
<tr><td>Allocation</td><td>Task used to repair the data based on the percentage obtained via a driver (which is defined through a staging table field: Weight dimension)</td></tr>
</table>

Routine - Loading

Admin | <application database>

application database > Routines > Loading link

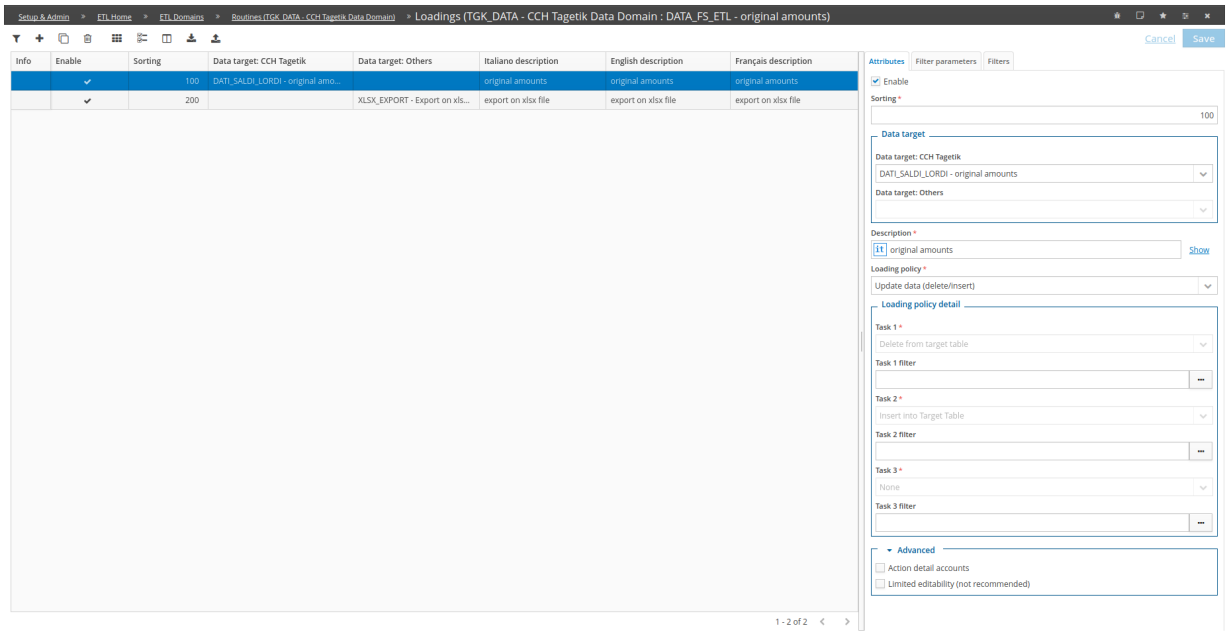
Note: if the ETL domain is locked, the link will not be selectable.

Purpose of the page

On this page, you can manage the loading, or the tasks of a routine that allow you to load data from the staging table onto destinations defined in **Data target: CCH Tagetik** and in **Data target: Others**.

Note: Loading carried out on CCH Tagetik requires editability checks to verify that the information to be inserted or deleted is correct. Any incorrect records will be shown to the user via warnings at the end of ETL execution.

What it looks like



Attributes tab

Field/section	Description
Enable	If enabled, the task will be run. Selected by default.
Sorting	indicates the order in which the task must be run. You cannot set the value 0 or a sorting that already exists

Field/section	Description
Data Target	indicates the loading policy for the respective targets: • **Data target: CCH Tagetik**: load onto CCH Tagetik database • **Data target: Other**: load onto another database or onto one of the managed types of file. ***IMPORTANT**: Transformation tasks are run sequentially following the **Sorting** defined in the setup step. Every task acts on the staging table and processes the data produced by the previous task (except for the first, whose input data are those generated by the extraction step).* ***Note**: Each **task** may require further information to be defined. A specific details panel will appear in the window depending on the selected **task**. The details of each panel are explained in the section on task type.*
Descriptions	Descriptions of the tasks in the languages enabled in the environment.
Loading policy	Indicates the policy according to which the system must load data into the destination: *Table (update/insert)*: can only be used if the data loading takes place on a table that has a primary key defined. Mainly used to enter *structural* data or data regarding tables on which no deletions are required. The following detail tasks are run: • **Task1**: Update target table; • **Task2**: Append into target table; • **Task3**: None. *Update data (delete/insert)*: can only be used if the data loading takes place on a table that has a primary key defined. Typically used to update *numerical* data. The detail tasks are: • **Task1**: Delete from target table; • **Task2**: Insert into Target Table; • **Task3**: None. *Append data (insert)*: mainly used to insert new *structural* or *numerical* data. The detail tasks are: • **Task1**: Insert into Target Table; • **Task2**: None; • **Task3**: None *Specify*: indicates that the detail tasks to be run must be defined manually by the user.

Field/section	Description
Loading policy detail	• a **Task1** which indicates the first detail task to be run; • a **Task1 filter** which allows you to indicate a filter to be applied to the operation run by **Task1**; the ETL's dynamic syntax can be used (see Filter Syntax or SQL Query); • a **Task2** which indicates the second detail task to be run; • a **Task2 filter** which allows you to indicate a filter to be applied to the operation run by **Task2**; the ETL's dynamic syntax can be used (see **Filter Syntax or SQL Query**); • a **Task3** which indicates the third detail task to be run; • a **Task2 filter** which allows you to indicate a filter to be applied to the operation run by **Task3**; the ETL's dynamic syntax can be used (see **Filter Syntax or SQL Query**); Allows you to specify the policy for loading data into the destination, through detail tasks: in the **Task1**,**Task2** and **Task3** fields, one of the following types can be selected: ◦ *Delete from target table*: deletes the data present in the target table defined in the data target; ◦ *Delete from target table*; deletes data present in the target table defined in the data target that are also present in the staging table (existence check for primary key); ◦ *Delete non-ETL data from target table*: deletes data present in the target table defined in the data target that are not also present in the staging table (existence check for primary key); ◦ *Insert into Target Table*: inserts all data present in the staging table onto the data target. ◦ *Append into target table*: allows you to insert, into the target table defined in the data target, non-existent data that is present in the staging table (existence check for primary key); ◦ *Update target table*: allows you to update the data present in the target table with the values present in the staging table (existence check for primary key)
Advanced options	Only for **Data target: CCH Tagetik**, the following options can be activated: **Action detail accounts**: for loading onto the CCH Tagetik Gross Amounts and Intercompany tables with **Loading policy** set to *Update data (delete/insert)*. This activates the loading policy via the window that details, for every account, which amounts are present in the staging table, which are present in the target table with origin related to the current routine, and which are present in the target table with other origins. From this window, you can select which amounts must be inserted and which must be deleted. • **Restricted editability (not recommended)**: for loading onto the CCH Tagetik tables of values, where editability checks are used to verify that the inserted information is correct; it disables the checks related to: ◦ Account restrictions - Custom dimensions ◦ Entity restrictions - Custom dimensions ◦ Counterparty rules

Filter parameters tab

This tab allows you to set up to nine filter **parameters** to dynamically limit the number of data items to be loaded.

Field	Description
Filter parameter 1, 2...9	You can select one element in each field

Note: If the user has also specified one of the other filters, the system will concatenate the conditions with the logic operator *AND*.

Filters tab

This tab allows you to limit the number of records processed by each detail loading task using the following filters:

Section	Description
ETL filters	• **ETL filter (SQL Model)** allows you to indicate a filter that has been defined previously in the **SQL Models** This filter is applied to the staging table during insert and update operations. • **ETL filter (specific)** allows you to indicate a filter manually using the SQL syntax ETL's dynamic syntax can be used (see **Filter Syntax or SQL Query**). This filter is applied to the staging table during insert and update operations. **Note**: If the user specifies both filters, the system will concatenate the conditions with the logic operator *AND*.
Target filter	• a **Target filter (SQL Model)** which allows you to indicate a filter that has been defined previously in the **SQL Models**. This filter is applied to the staging table during delete and update operations. • a **Target filter (specific)** which allows you to indicate a filter manually using the SQL syntax. ETL's dynamic syntax can be used (see **Filter Syntax or SQL Query**), This filter is applied to the target table during delete and update operations.
Origin	• a **Filter on origin** which allows you to indicate the deletion policy to be applied to the field of the target table identified as the data origin. For CCH Tagetik tables, there is a predefined field (ORIGIN) whereas for custom tables both from the CCH Tagetik database and from other databases, the field identified as the origin is indicated via the **Value type** attribute in the Data target: Others fields definition. The available options are: ◦ *Do not delete*: no data in the target table will be deleted. This option is also used to let you indicate a different value in the execution step if the **Deletion management** option is active for the selected routine; ◦ *Delete all*: does not apply the filter to the origin field; ◦ *Delete manual origin records*: deletes data in the target table whose origin is*MAP**, *INPUT_DEFORM*, *INPUT_WEB* and *QDL*; ◦ *Delete manual and other origin records*: deletes data in the target table whose origin is *MAP**, *INPUT_DEFORM*, *INPUT_WEB*, *QDL* and all other origins defined in the **Other origins** field; ◦ *Delete manual and calculation origin records*: deletes data in the target table whose origin is *MAP**, *INPUT_DEFORM*, *INPUT_WEB*, *QDL*, *CONS_CALCULATED* and *PROC_CALCULATED*; ◦ *Delete origin records from current routine*: default value, deletes from the target table data whose origin is the calculated origin of the current routine (MAP_'ETL_Domain_code'_'Routine_code'). • a **Suffix** which allows you to differentiate, for each Loading task, in the insert and deletion phase, the origins generated by the current routine. If specified, the generated origin will be MAP_'ETL_Domain_code'_'Routine_code'_'Suffix'; • an **Other Origins** which allows you to indicate a set of other origins to be deleted, if the value of the **Filter on origin** is *Delete manual and other origin records*.

Dimensions and dictionaries

In the execution step, the ETL populates a temporary table (“*Staging* “ Table) by extracting the data from one or several external sources, transforming it with a series of rules and then loading it onto CCH Tagetik or other destinations such as files of custom tables.

In this window, it is possible to set alignment between the following elements, as necessary for the correct execution of an ETL:

- the structure of the data to be extracted and that of the “*Staging* ” table;
- the structure of the “*Staging* ” table and that of the target table or file.

The dimensions represent the columns of the “*Staging*” table (the field names are formed of "CAMPO" + dimension **Code**). Up to 200 dimensions can be managed.

> ***Note**: It is good practice to define the **Description** and the **Acronym** so that they clearly identify the content of the dimension.*

The definition of the Dimensions and Dictionaries can be accessed from the *Dimension and Dictionaries* link in the **ETL Domains** window.

Dimensions and Dictionaries window

Admin | <application database>

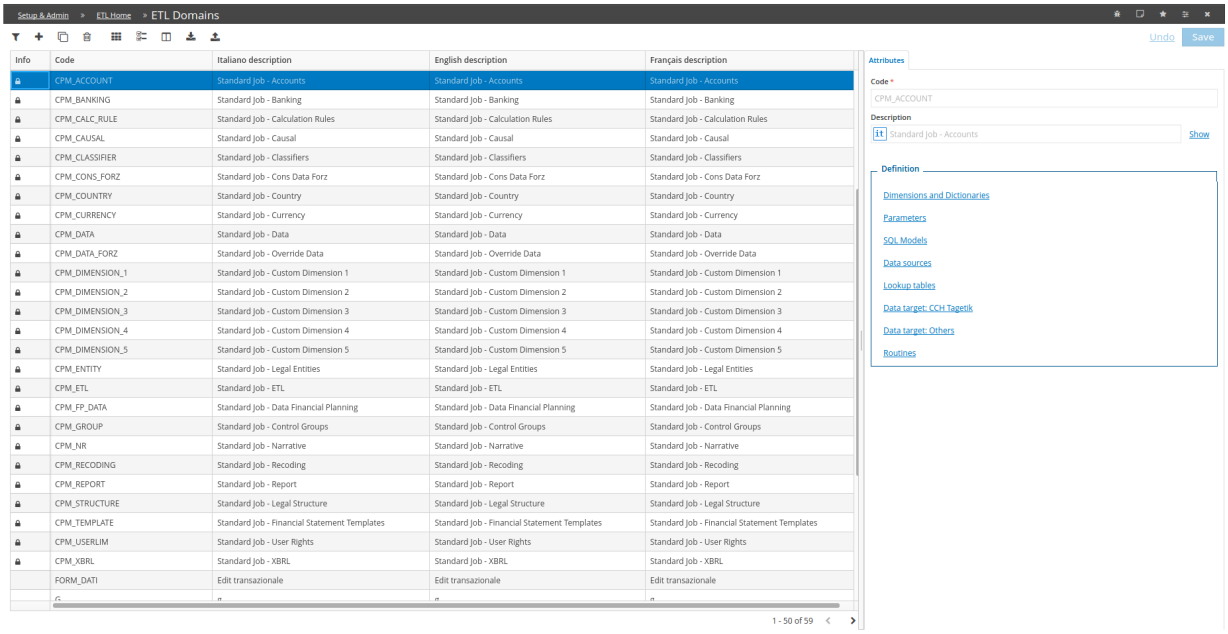
ETL > ETL Domains > Dimensions and Dictionaries

***IMPORTANT**: if the ETL domain is locked, the window will open in read-only mode.*

Purpose of window

This window allows you to define the staging table fields and a set of the permitted values for the dimension.

What it looks like



Element	Description
A	Action buttons/Tool bar
B	List of ETL dimensions
C	Configuration tab

Actions on this page

Element	Description
	This filters the data shown on the page.
	This adds a row in order to insert a dimension.
	This duplicates the selected row with all the information inserted, except for the code.
	Deletes the element contained in the selected row.
	Enables/Disables the grid modify function. This allows you to change the characteristics of the selected elements directly in the table, and to hide the configuration tabs.
	Enables multi-row selection to allow mass changes.
	Selects the columns to be shown/hidden in the table and those to be locked when scrolling the table.
	Exports the data in Excel format so they can be edited and reimported into CCH Tagetik.
	Imports data from a previously prepared Excel file.

Attributes tab

This tab allows you to define dimensions with the following values:

Field	Description
Code	Numerical value which identifies one of the columns of the staging table. ◦ dimensions 1 to 89 are *"varchar"* type, with a length of 50 characters; ◦ dimensions 90 to 99 are *"varchar"* type, with a length of 255 characters; ◦ dimensions 100 to 130 are *"varchar"* type, with the length defined in the **Text length** field; ◦ dimensions 131 to 200 are *"clob"* type **Note**: up to 200 dimensions can be managed. See Staging Tables Topic
Description	Contains the description of the dimension (numerical, text-based, etc.).
Acronym	Brief description which can be used to refer to the dimension in the dynamic syntax of texts and the ETL's SQL filters.
Type	Indicates the type of data that populates the dimension: ◦ ***String***: only for dimensions whose code is between 1 and 99; ◦ ***Number***: only for dimensions whose code is between 1 and 99; ◦ ***Date***: only for dimensions whose code is between 1 and 99; ◦ ***Text***: only for dimensions whose code is between 100 and 130; ◦ ***Clob***: only for dimensions whose code is between 131 and 200. ***IMPORTANT**: Clob type dimensions, which allow you to manage large volumes of text-based content, can be used in all setups and have some specific features:* • they cannot be stored in extracted data or transformed data • for ETL domains that use those dimensions, all steps of the routine must be run. ***IMPORTANT**: Since all columns in the staging table are "varchar" type, the system only uses the**type** indicated for every dimension in the following cases* • to format the data to be displayed following a query on the staging table or log table. Numerical data are formatted based on the user's nationality; • to perform an automated conversion of the value of the element of a dimension every time the **Acronym** is inserted within a dynamic syntax in an ETL filter, an SQL model or an SQL query. *For example, suppose we have a number type ETL dimension (e.g.CAMPO10) whose acronym is "IMP" and we want to multiply the value by -1, just write {IMP} * -1 in the query in order for the system to automatically convert the value to a number before multiplying it by -1.*
Date format	Indicates the date format that the data to be imported will have. Mandatory for dimensions whose **type** *is* ***"Date"***.
Text length	Indicates the number of characters that the dimension can contain. Mandatory for dimensions whose **type** *is* ***"Text"***.

Field	Description
Identity	Allows you to relate the dimension that will contain the values to a CCH Tagetik logical dimension. Identities are useful for restricting the possible values in selecting parameters linked to these dimensions and in certain transformation or loading tasks. The possible identities are as follows: • ***None*** • ***Entity*** • ***Scenario*** • ***Period*** • ***Event scenario*** • ***Account*** • ***Custom dimension*** • ***Classifier*** • ***Counterparty zone*** • ***Counter*** • ***Category*** • ***Currency*** • ***Original currency*** • ***Record num.*** ***IMPORTANT**: you cannot specify an identity for **Clob type** dimensions.*
Identity code	Mandatory for the following **identities**: • **Custom dimension** indicates one of the managed custom destinations • **Classifier**: indicates a classification variable **Note**: The value of this field must be unique for the ETL domain.
Encode	If active, this option indicates that the values entered in this dimension must be encoded in the extraction step.
	Checks section
Completeness checks	Link via which the **Dimension completeness checks** window is accessed ***IMPORTANT**: if the ETL domain is locked, the window will open in read-only mode.*

Dictionary tab

This tab allows you to set dictionaries for every *"String"* or *"Text"* type dimension to indicate a set of permitted values for the dimension.

***IMPORTANT**: The use of a dictionary does not prevent the insertion of values that are not present in that dictionary for a dimension.*
However, the insertion of a Dictionary check type transformation task in an ETL routine will mark data processed with values that not present in the dictionary with warnings during the execution step.
*By activating the **Signal empty values in the check** option, you can tell the check to signal any undefined fields with warnings.*

Field	Description
Dictionary	The following options can be defined for the selected dimension: **None**: no dictionary defined. **Manual**: activates the **Dictionary elements** definition link **Note**: if the ETL domain is locked, the window will open in read-only mode. **From Tagetik**: the dictionary elements are represented by the values of a field read from a CCH Tagetik table.
Tagetik table	Only for **From Tagetik**. Allows you to select a table of the CCH Tagetik DB from which to extrapolate the dictionary elements.
Dictionary code field	Only for **From Tagetik**. Specifies the field of the selected table from which the dictionary item codes will be extracted.

Field	Description
Dictionary description	Only for **From Tagetik**. Specifies the field of the selected table from which the dictionary item descriptions will be extracted.
Dictionary filter	Allows you to restrict the dictionary elements. *Example: even if all the entity codes present in the ENTITY table have been entered in the dictionary, it is possible to consider only those whose code starts with A0 (in this case the SQL condition to enter is COD_AZIENDA LIKE 'A0%')*
Signal empty values in the check	If active, the option marks undefined fields with warnings.
Delegate management	Only for **Manual**. If active, this option allows you to delegate the management of the dictionary elements to the End User (this is generally done by the Admin).

Example - From CCH Tagetik

To use the entity codes present on the "ENTITY" table as dictionary elements, you must define the **Tagetik table** element as ENTITY, the **Dictionary code field** as COD_AZIENDA and the **Dictionary description field** as DESC_ AZIENDA0.
The dictionary elements can be restricted by inserting a filter condition in the **Dictionary filter** attribute. For example, even if all the entity codes present in the ENTITY table have been entered in the dictionary, it is possible to consider only those whose code starts with A0 (in this case the SQL condition to enter is COD_AZIENDA LIKE 'A0%').

Dictionary elements

Admin | <application database>

ETL > ETL HOME > ETL Domains > Dimensions and Dictionaries > Dictionary tab > Dictionary elements

User | <application database>

ETL > ETL HOME > Dictionary - elements

Workflow task ETL Dictionary

Note: if the ETL domain is locked, the window will open in read-only mode.

***IMPORTANT**: Generally, it is the administrator user who manages the Dictionary Elements window that is present where **Manual dictionaries** are used. To edit this setting and delegate management to the end user (making the window editable both from the end user's homepage and from the workflow task), you must enable the **Delegate management** option in the Dictionary tab. Tor more details, see Dimensions and Dictionaries*

Attributes tab

Field	Description
Code	Uniquely identifies the dictionary element
Description	Describes the dictionary element code

Completeness checks

Admin | <application database>

ETL > Domains > Dimensions and Dictionaries > Completeness Checks

Purpose of window

This window allows you to check whether, in line with the indicated values of the dimension in question, the values of other dimensions are mandatory, optional or prohibited.

Attributes tab

Please find below the most important fields.

Field	Description
Element code	Value which the specific dimension can take during the execution step
Lists of options for other dimensions in the ETL domain	If the value of the specific dimension is the value defined in the Element code field, the options allow you to define the value of other dimension as: ◦ **Mandatory**: the dimension related to the option must be defined; ◦ **Optional**: the dimension related to the option is optional; ◦ **Prohibited**: the dimension related to the option must be left empty.

***Note**: In order for the checks to be run, you must define a "Completeness check" transformation task within an ETL routine. Thus, in the execution step, processed data that do not satisfy the rule will be reported to the user as warnings.*

Example

Suppose you enable the management of two custom dimensions, "Function" and "Segment", and you have account "1000 - Salaries" which must be detailed for Function but not for Segment. To verify the consistency of the extracted data, you can define a rule that specifies that for the "Account" dimension, in relation to element 1000, the "Function" dimension must be mandatory whereas the "Segment" dimension must be empty.

Jobs

An ETL job allows you to run, sequentially and in a defined order, several **ETL routines** (which may also belong to different **ETL domains**) in a single data process.

There are two types of job in CCH Tagetik:

Element	Description
Standard ETL jobs	Default. These allow you, through the **standard ETL routines** related to them, to load both standard data (amounts, intercompany transactions, journals and FX rates) and metadata (elements of dimensions, hierarchies and groupings).
Custom ETL jobs	Fully configurable jobs which can be created to respond to specific business needs.

Job - Definition

Admin

application database > ETL > Job >

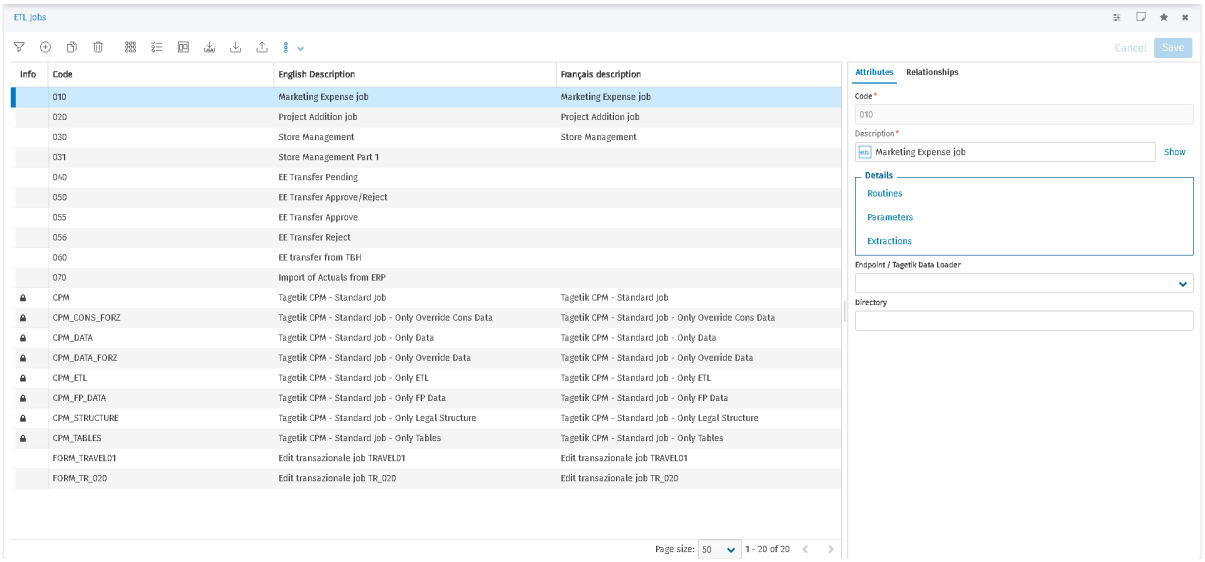
The definition of standard jobs is locked. For these records, the icon is visible in the first column of the table.

The job can be executed:

Purpose of the page

This page allows you to configure **custom ETL jobs** and view **standard ETL jobs**

What it looks like



Element	Description
A	Action buttons/Tool bar
B	Job list
C	Configuration tab

Buttons on the page

Element	Description
	This filters the data shown on the page.
	Adds a row in order to insert a new element
	This duplicates the selected row with all the information inserted, except for the code.
	Deletes the element contained in the selected row.
	Enables/Disables the grid modify function. This allows you to change the characteristics of the selected elements directly in the table, and to hide the configuration tabs.
	Enables multi-row selection to allow mass changes.
	Selects the columns to be shown/hidden in the table and those to be locked when scrolling the table.
	Exports the data in Excel format so they can be edited and reimported into CCH Tagetik.
	Imports data from a previously prepared Excel file.
	If only one record is selected, the actions button enables the following utilities: **Reset**: deletes the job data related to one of these options: • all user restrictions • ETL execution plans for all processes **Note**: To run the operation, at least one element to be deleted must be selected. **Copy from**: copies (in overwrite mode) the data related to the following job uses onto the selected job, in addition to the entire active default configuration: • user rights, • the ETL execution plans of all processes To run the operation, it is necessary to indicate the source job from which to copy. **Connectors**: allows you to automatically generate a comprehensive setup of ETL domains and routines that allow data to be loaded onto CCH Tagetik from one of the relative sources: • SAP© Connector Wizard • AX 2012 Connector Wizard • AX 2012 R2 Connector Wizard • NAV 2013 R2/2015/2016 Connector Wizard

Attributes tab

Field/box	Description
Code	Identifies the job code
Description	description of the job. Mandatory
Details	allows you to define the job details via the following links: • **Routines** • **Parameters** • **Extractions** **Note**: Links only active if a single job is selected.

Field/box	Description
Endpoint/Tagetik Data Loader	indicates a remote resource on which the files created by any loading task that uses a **Data Target: Other** must be saved. You can indicate either a **Endpoint** or a **Tagetik Data Loader**.
Directory	indicates the directory in which the files created by any loading task that uses a **Data Target: Other** must be saved. If an **Endpoint/Tagetik Data Loader** is also specified, the specified directory will be shown as a sub-directory of the selected remote resource; however, if the user only defines the **Directory** field, then it will mean the absolute or relative path visible by the Application Server that must be indicated in the permitted directories To meet CCH Tagetik's security requirements, the aforementioned directory must match or be a sub-folder of the one defined as the *writable folder* (set in system property "it.grupposervizi.easy.impexp.writableFolder") or if that property is not defined, it must match the one defined in the *java tmp path*. If, in the knowledge that it will compromise the software's security, you want to use different directories, you must activate the **Allow ETL to read or write to directories other than the one that is configured as secure** option present in the *Advanced* section of the *data model*'s general configuration.

Relationships tab

This tab allows you to view and edit the nodes of the grouping to which the selected job is related.

Field	Description
Nodes	Every new job inserted will be automatically associated with the grouping nodes identified as to be related by default.

Routines of a job

Admin | <application database>

application database > ETL > Jobs > Routines link

Note: if the job is locked, the routines window will be read only

Purpose of the page

This page allows you to define the routines of the job.
For every job, you must specify the routines to be run and their sorting.

***Note**: The same routine can be specified more than once.*

Attributes tab

If the **Enable** option is active, the routine will be run by the job (default).

Field	Description
Sorting	indicates the order of the routine within the job. **Note**: the values of the **Sorting** field must be unique and different from 0.
Enable	indicates whether, by default, the routine must be run by the job;
ETL domain	indicates the routine's ETL domain. **Note**: You can define routines of several ETL domains.
Routine	indicates the routine code

***IMPORTANT: OPENING JOB PARAMETERS** After the insertion of a job, in the save, edit or delete step, the system runs an automatic update procedure on the **job parameters** with all the parameters used, both from the defined routines and from all their active extraction, transformation and loading tasks.*

Parameters of a job

Admin | <application database>

application database > ETL > Jobs > Parameters link

Note: if the job is locked, the Parameters window will be read only

Purpose of the page

This window allows you to manage all the **parameters** used by the **job routines** and their extraction, transformation and loading tasks.

Attributes tab

This tab only allows you to edit the **Parameter value** and **Define fields automatically** (insertions and deletions are not permitted).

Field	Description
ETL domain	indicates the ETL domain to which the parameter belongs
Parameter	indicates the parameter code ***IMPORTANT**: If the same parameter is used several times, the system will only consider it once.*
Parameter type	indicates the parameter type, which defines the fields to be completed • *From Tagetik* and *From dimension dictionary*: it is possible to populate the **Parameter value** field or activate the **Define automatically** field • *Free*: it is only possible to populate the **Parameter value** field; • *From imported data* or *Imported data intersection*: you can only activate the **Define automatically** option.
Parameter value	allows you to assign a default value to the parameter
Define automatically	allows you to define the parameter automatically during the job's execution.

Extractions of a job

Admin | <application database>

application database > ETL > Jobs > Extractions link

Note: if the job is locked, the extractions window will be read only

Purpose of the page

This window allows you to detect and define certain necessary information to include the execution of the active **extraction tasks** belonging to the defined **job routines**.

***IMPORTANT**: the window does not allow new records to be inserted manually.*
To register and edit the extraction tasks, you must run the Update extractions utility present in the window's actions menu. The utility will automatically insert as many records as there are active extraction tasks for that routine, for every routine of the active job.
However, you can delete records if, for that particular job, you do not want to run certain extraction tasks that are active by default in the job routine, and have been automatically inserted into the window by the Update extractions utility.

Attributes tab

***Note**: in the job execution step, you can define or edit the attributes set in this window only in relation to that specific execution.*

Field	Description
Sorting	indicates the order of the routine within the job. **Note**: the values of the **Sorting** field are unique and different from 0.
ETL domain	indicates the ETL domain to which the extraction task belongs
Routine	indicates the routine to which the extraction task belongs.
Extraction task description	indicates the description of the extraction task.
Data Source	indicates the data source defined for the extraction task.
Data source type	allows you to select the input type from those configured in the extraction task's data source.
Extraction type	allows you to select the extraction type from among the following: ◦ *Application server*: loading is carried out either from queries or from files present on directories visible by the application server indicated in the Permitted dictionaries; ◦ *Client*: loading is carried out via the uploading of files by the client; ◦ *Endpoint/Tagetik Data Loader*: loading is carried out using an Endpoint that also allows you to use the Tagetik Data Loader. ***Note**: If the type of external source from which to extract the data is a database, the Application server and Client options will be the same and tell the system that the query must be run on the database configured in the data source.*
Endpoint/Tagetik Data Loader	allows you to select an endpoint that will be used if the **Extraction type** is *Endpoint/Tagetik Data Loader*;
Directory	allows you to define a directory that will be used if the **Data source type** is from file and the **Extraction type** is not *Client*;
File names	allows you to list the names of the files, separated by a comma (comma separated). which will be used when the **Data source type** is from file and the **Extraction type** is not from *Client*. **Note**: If it is necessary to indicate several files of the same type, just separate the various names by a semicolon. In addition to indicating several files, you can also use the wildcard "*" to extract only files that respect the indicated filter. When the input File names includes a wildcard (*), the file name must be enclosed in square brackets [ ]. For example, you can write: • [*.fbk] to indicate all files that end with the extension ".fbk"; • [name*] to indicate all files that start with "name"; • [name*.fbk] to indicate all files that start with "name" and end with the extension ".fbk"; • [name{PARAM_AZIENDA}*] to indicate all files that start with "name" + entity parameter value. The file name can also be defined using SQL strings by utilising the parameters defined in the ETL syntax If no file name is specified, the system will extract the data of all files of the type related to the value of the **Data source type** field that exist in the specified directory.

Field	Description
File management type	Only for loaded files whose**Data source type** is from file and **Extraction type** is not from *Client*. Allows you to select from the following options: • *None*: no action will be run on the file • *Delete*: the file will be deleted if loading is completed successfully • *Archive*: the file will be moved to an “archive” file if loading is completed successfully.
File ID	Only for loaded files whose**Data source type** is from file and **Extraction type** is from *Client*. This indicates the ID of the file to be loaded via upload, to classify the extraction tasks with the same data source (in order to limit the number of uploads of files containing data to extract). **Note**: if the same **File ID** is attributed to two or more tasks, the system will load a single file, thus preventing it from requesting the uploading of superfluous files that have already been loaded.

Execution jobs

A job can be run:

- from the dedicated window (see **ETL Execution** for the execution methods and entry points).
- through Scheduling;
- on saving a data entry form;
- from the related task located within an application job;
- through Web Service.

ETL execution plans

Admin | <application database>

Process > Select process > > Process definition > Data Processing section > ETL >

Purpose of window

This window allows you to define a process, the jobs and/or ETL routines to be run for every entity.
Note: The restrictions defined in the execution plan are applied in the **ETL execution** window only if opened from the *Process cockpit* and *Workflow* entry points described in the relevant manual.

Attributes tab

Field	Description
Sorting	Mandatory. This indicates the order in which the deploy data processing must process the defined records in order to generate the execution plan correctly;
Action	Mandatory. Indicates whether the jobs or routines defined in the record must be *Included* in or *Excluded* from the execution plan;
Entity	Mandatory. Allows you to select an individual entity or node of the hierarchy of entities;
The **Job**, **All Jobs**, **ETL Domain**, **Routine** and **All Routines** fields can only be defined as alternatives.	
Job	allows you to select a single ETL job or a node of the grouping of ETL jobs;
All Jobs	allows you to select all ETL jobs in the table automatically;
ETL domain	allows you to select a single ETL domain or a node of the grouping of ETL domains;
Routine	allows you to select a single ETL routine or a node of the grouping of ETL routines
All Routines	allows you to select all ETL routines in the table automatically.
Extraction task description	description of the extraction task.
Data Source	Data source defined for the extraction task.

***Note**: in the job execution step, you can define or edit the attributes set in this window for that specific execution.*

***Note**: the fields that allow you to indicate a single entity or node of the relative hierarchy/grouping allow you to tell the deploy data processing that either the individual entity or all the entities related to the selected node must be taken into consideration for the record.*

***IMPORTANT**: The ETL execution plan only generates restrictions on jobs and/or ETL routines. For this, the selection of an **ETL domain** or a node of the grouping of ETL domains indicates that all the **routines** of that ETL domain or the ETL domains belonging to the node must be taken into consideration.*

Execution plan - Results window

User | <application database>

Process > Select process > > Process definition > Data Processing section > ETL > > Definition results

Purpose of window

After every save, the *deploy* data processing is performed, automatically generating the result records based on the records inserted in the definition. These records will be visible in the *Definition results* window.

Every result record is identified by:

- **Entity**: indicates the entity of the process on which the restriction regarding the execution of a job or ETL routine was defined;
- **Job**: if defined, this indicates the ETL job that can be run for the entity;
- **ETL Domain** / **Routine**: if defined, this indicates the ETL routine that can be run for the entity.

***Note**: The **Job** and **ETLDomain**/**Routine** fields are defined as alternatives, and a restriction either on a job or on an ETL routine is indicated in every record (entity of the process).*

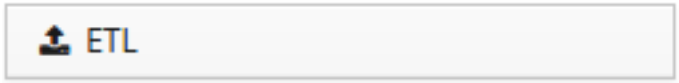
***IMPORTANT**: the restrictions regarding the ETL routines are only valid for the execution of the individual routine and not for restricting routines belonging to an ETL job. If the execution plan states that it is possible to run an ETL job, you will be able to run all the routines related to it, even if they are not present in the execution plan.*

ETL execution

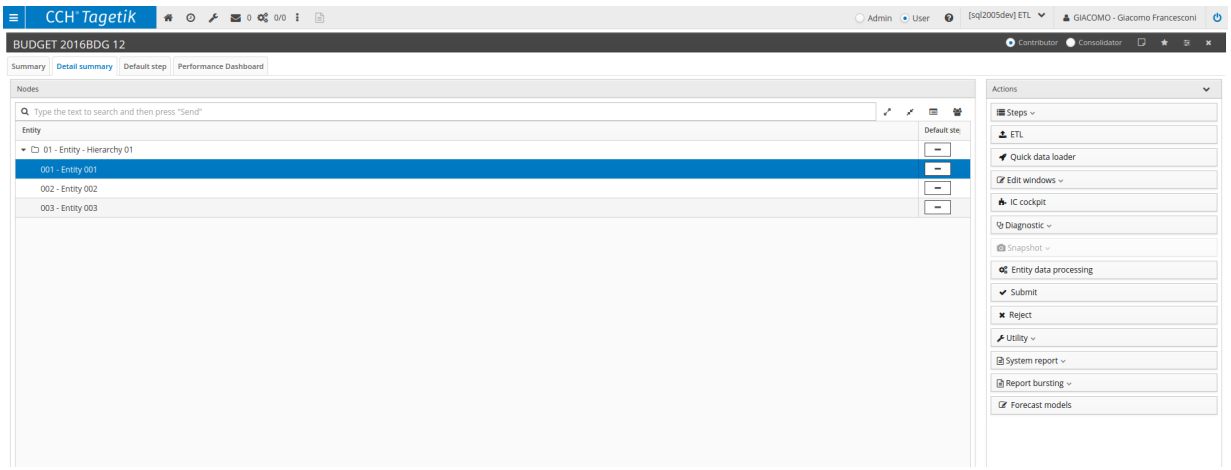
The ETL data processing can be accessed and run from several entry points

Process cockpit

From both the Contributor and Consolidator cockpits, you can access the ETL execution window both from the detail summary and from the individual step using the



button located in the **Action bar** on the right-hand side. In both cases, you must select a node of the hierarchy or the individual lowest level element. In the first case, the ETL will be run on all the entities belonging to that node.



Workflow

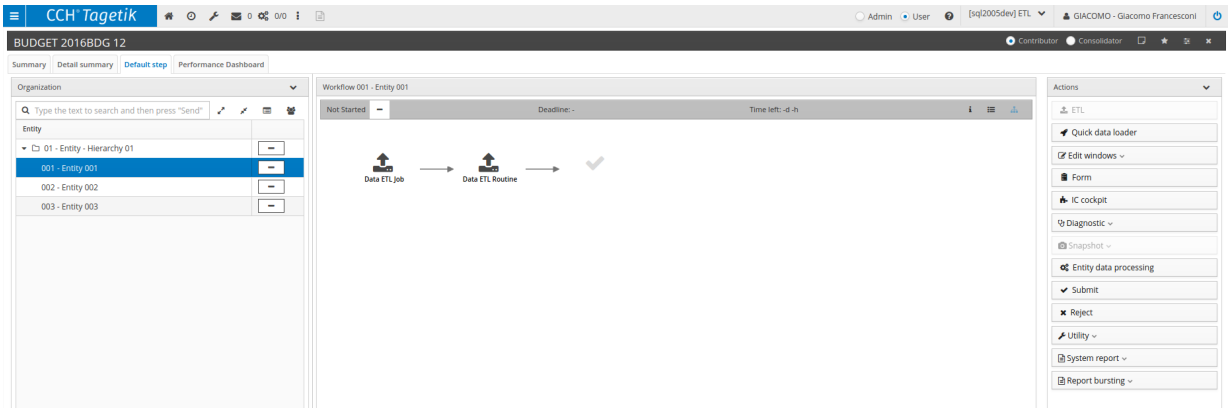
In the panel related to the individual step, if the node or entity has a workflow task containing one of the following types of task:

- ETL job for values
- ETL job for tables
- ETL routine for values
- ETL routine for tables

the button located in the **Action Bar** which allows the ETL to be executed will be disabled.

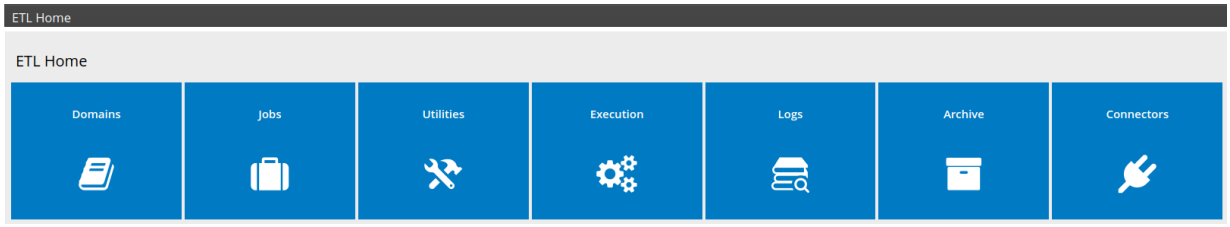
This is because if one of those tasks is present, the ETL can only be run in the defined execution order.

To run the ETL from a workflow task, you must click on the task and select the **Run** menu.



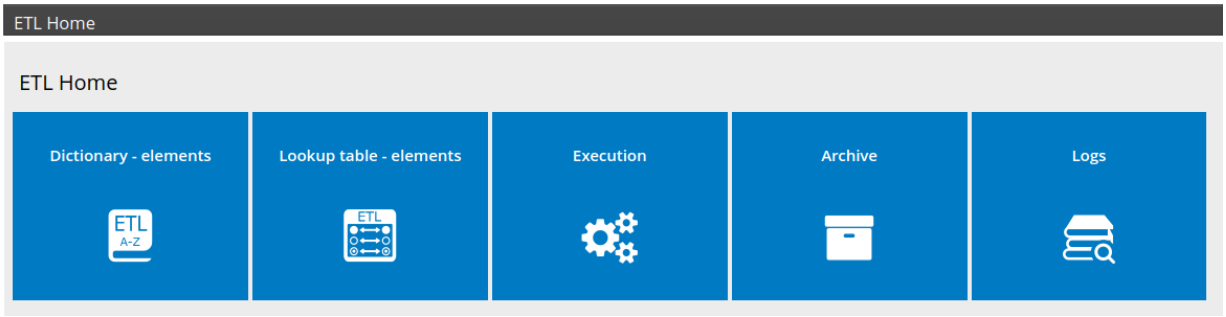
ETL - Admin user

From the ETL tile located in the Admin user's homepage From Navigation panel > ETL you can access the window by clicking on the **Execution** tile.



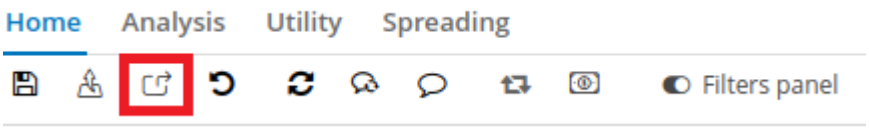
ETL - End user

From the ETL tile located in the End user's homepage From Navigation panel > ETL you can access the window by clicking on the **Execution** tile.



Web forms

From the button of the same name on the Home tab of the Reporting Data Entry WebForm.



IMPORTANT: The window that opens from each of these entry points will be editable for each user. It is possible:

- to select the ETL job or the individual routine to be run (except when accessed from a workflow task)
- if an ETL job has been selected, you can activate or deactivate one or several of its routines
- to activate or deactivate complete steps of a routine or its extraction, transformation and loading tasks

To make the window non-editable, you must enable function **0004108** (belonging to the ETL_ELAB_USER but disabled by default) in the user's role. By doing this, the window will restrict editing depending on the entry point from which it is accessed:

- Admin Home: always editable
- End User Home: always editable
- Process cockpit: the job or individual routine selection will be editable, but the steps selection and the selection and editing of extraction, transformation and loading tasks will not be editable
- Workflow: the job or individual routine selection will not be editable (values preset by the task), and the step selection and the selection and editing of extraction, transformation and loading tasks will not be editable
- Web form: the job or individual routine selection will be editable, but the steps selection and the selection and editing of extraction, transformation and loading tasks will not be editable

ETL execution window

Admin / <application database>

ETL > Execution

Processes > select process > Contributor > select Overview/step tab > select entity or node > (Actions section) > ETL

Processes > select process > Consolidator > select Overview/step tab > select entity or node > (Actions section) > ETL

Forms tile > click grouping of forms > (next to form)

User / <application database>

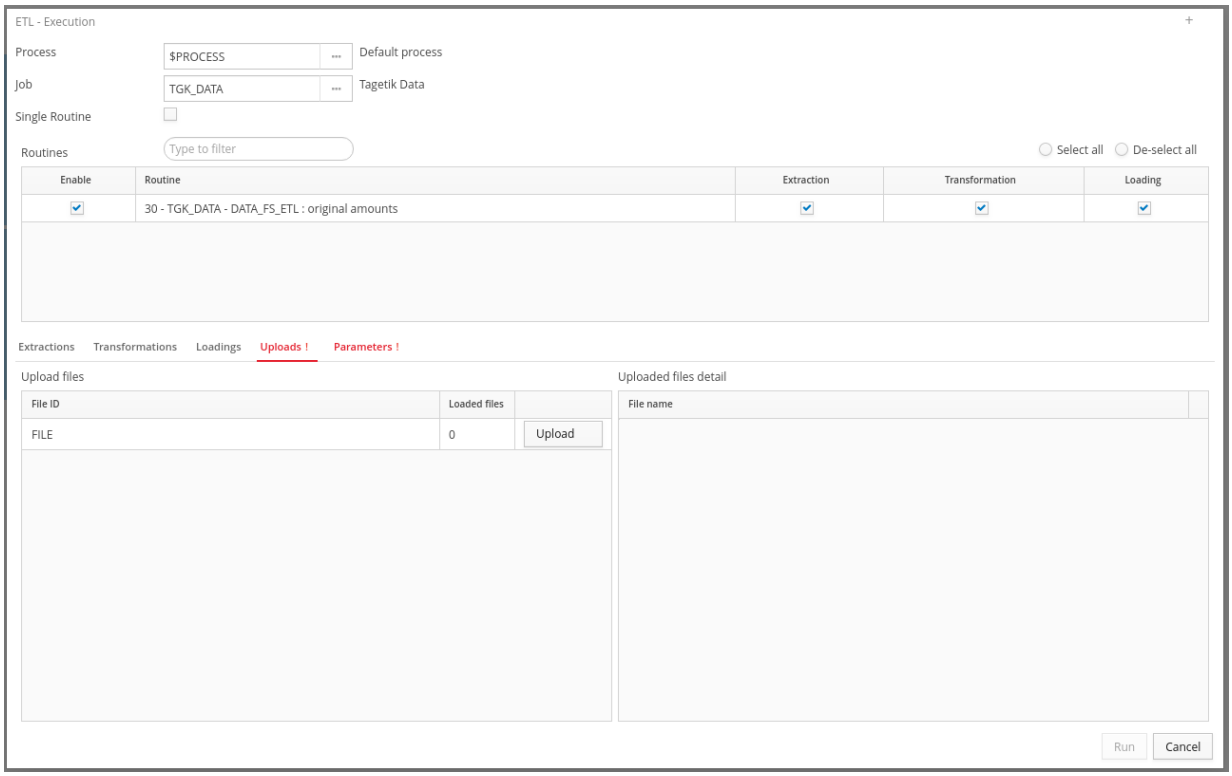
ETL > Execution

Purpose of the page

This window allows you to manage the setup of the ETL routines to be run. Here, you can edit the definition configured in the setup for the current execution, via the following sections:

- Process and Job/Routine selection
- Routines
- Extractions
- Transformations
- Loading
- Uploads
- Parameters

***IMPORTANT**: Any missing information that blocks the ETL execution is reported to the user by changing the colour of the information caption (to red).*



Process and Job/Routine selection section

Section	Description
Process	Process on which to run the ETL If the execution takes place from the process cockpit, the **Process** will be defined automatically as the cockpit process and will not be editable If the execution takes place from a workflow task, the **Process** will be defined automatically as the cockpit process and will not be editable
job	ETL job to be run The system shows the **Jobs** restricted to the current user as selectable. If the execution takes place from the process cockpit, the selectable **jobs** will be those present in the execution plan of that process for the selected entities. If the execution takes place from a workflow task: the **job** will not be selectable and will be defined automatically as the one set in the task definition.
Single Routine	The job's ETL routine to be run If the execution takes place from a workflow task. If the task is related to an ETL routine, the **Single Routine** option will be activated automatically.

Note: Once the **Job** has been selected or the **Single Routine** option has been activated, the system will update the contents of the window.

Routines section

This section can be populated using the following options:

- all **routines belonging to the Job** not restricted for the current user or for the Process execution plan, if a **Job** has been selected,
- all the **routines of all the ETL domains** visible by the user if the **Single Routine** option has been activated and the window was opened in one of the following ways:

- from the processes cockpit, we will only see the routines belonging to the execution plan of that process for the selected entities.
- from a task flowchart whose type is ETL routine, we will only see the routine set in the task definition.

Field	Description
Enable	indicates whether the routine must be run. If the **Single Routine** option is active, you will only be able to activate one routine. If the window was opened from a task flowchart whose type is ETL Routine, you will not be able to deactivate the option activated by default on the individual routine shown.
Routine	reports the order of the routine within the job (if a job has been selected), the ETL domain code, the routine code and its description.
Extraction	indicates whether the extraction step must be run for the routine. The option cannot be deactivated if extracted data storage is not active in the definition and it is therefore not possible to start with the transformation step.
Transformation	indicates whether the transformation step must be run for the routine. The option cannot be deactivated if transformed data storage is not active in the definition and it is therefore not possible to start with the loading step.
Loading	indicates whether the loading step must be run for the routine.

***IMPORTANT**: The activation/deactivation of options generates the update of the following sections...*

Extractions tab

This section is populated with all the extractions of all the routines active in the **Routines** section. If a **job** has been selected, only Routines defined in the extractions of the selected job will be show.

***IMPORTANT**: The activation/deactivation of tasks or the editing of attributes generates the update of the **Upload** and **Parameters** tabs.*

Field	Description
Enable	indicates whether the extraction task must be run.
Routine	routine to which it belongs.
Extraction task	sorting and description of the task, code and description of the data source.
Input type	allows you to select the input type from those configured in the task's data source. The default value is defined in the job's extractions.
Extraction type	allows you to select the extraction type from among: • *Application Server*: loading is carried out either from queries or from files present on directories visible by the application server indicated in the Permitted dictionaries. • *Client*: loading is carried out via the uploading of files by the client. • *Endpoint/Tagetik Data Loader*: loading is carried out using an **Endpoint** that also allows you to use the **Tagetik Data Loader**.
Endpoint/Tagetik Data Loader	allows you to select an endpoint that will be used if the **Extraction type** is *Endpoint/Tagetik Data Loader*.
Directory	allows you to define a directory that will be used if the **Input type** is from file and the **Extraction type** is not from *Client*.
File names	allows you to indicate a list of file names (comma separated) that will be used if the **Input type** is from file and the **Extraction type** is not from *Client*
File ID	allows you to indicate the ID of the file to be loaded via upload if the **Input type** is from file and the **Extraction type** is *Client*. This is used by the system to identify the extraction tasks that utilize the same data source in order to limit the number of uploads of files that contain data to extract. By labelling two or more tasks with the same **File ID** the system will require the upload of one single file, thus preventing it from requesting the uploading of superfluous files that have already been loaded.
File management type	allows you to select the action to be carried out on the loaded file, if the **Input type** is from file and the **Extraction type** is not from *Client*, choosing between: • *None*: no operation will be carried out. • *Delete*: the file will be deleted if loading is completed successfully. • *Store*: the file will be moved to an " archive file if loading is completed successfully.

***IMPORTANT**: If the configuration of an extraction task is not complete or if the same **File ID** has been defined for different **Data source types**, the "Extractions" label in the panel will turn red and it will not be possible to run the ETL.*

***Note**: If extraction tasks have been defined for the selected routines but none of them is active, the "Extractions" label in the panel will become yellow but it will be possible to run the ETL.*

Transformations tab

This panel is populated with all the transformations of all the routines that are active in the **Routines** panel.

Field	Description
Enable	indicates whether the transformation task must be run.
Routine	routine to which it belongs.
Transformation task	the sorting, type and description of the task.

***Note**: If transformation tasks have been defined for the selected routines but none of them is active, the "Transformations" label in the panel will become yellow but it will be possible to run the ETL.*

Loadings tab

This panel is populated with all the loadings of all the routines that are active in the **Routines** panel.

Please find below the most important fields.

Field	Description
Enable	indicates whether the loading task must be run.
Routine	the routine to which it belongs.
Loading task	the sorting, type and description of the task.
Detail task	description of the macro loading type and the descriptions of the detail tasks
Filter by origin	filter type to be applied to the origin in the deletion step.

***Note**: If loading tasks have been defined for the selected routines but none of them is active, the "Loadings" label in the panel will turn yellow but it will be possible to run the ETL.*

Upload tab

This panel is populated with all the loadings of all the routines that are active in the **Routines** panel.

It includes all the uploads to be carried out, identified with the **File type** and the **File ID** if tasks with **Input type** set to from file and **Extraction type** set to from *Client* are present in the Extractions panel.

The fields and buttons in the tab are shown below.

Field	Description
Upload files section	
File type	value of the **Data source type** field defined in the **Extractions** panel.
File ID	value of the field of the same name defined in the **Extractions** panel.
Loaded files	number of files currently loaded for that **File ID**.
Loaded detail files section	
File name	This includes the names of the files loaded for every record selected in this panel.
	allows you to remove the loading of the file

***IMPORTANT**: If the uploaded file includes a **Mime type** that is not the one required by the **File type**, the system will show the "Invalid file type" error and block the upload.*

Parameters tab

This panel is populated with all the parameters used in all the active tasks in the **Extractions**, **Transformations** and **Loadings** panels of all the routines that are active in the **Routines** panel.

Element	Description
Main section	
Description	This includes the ETL domain code, and the parameter code and description.
Additional loading parameters section	
Options on the parameters that are present only if defined for at least one routine that is active in the **Routines** panel.	
Step	filter on the steps belonging to the selected **process** so that only data related to accounts present in those steps are loaded. This option is only visible if the **Step** option has been activated for at least one active routine
Account hierarchy and Nodes of the account hierarchy	filter on the nodes of the account hierarchy so that only data related to accounts related to the indicated nodes are loaded. These options are only visible if the **Account hierarchy** option has been activated for at least one active routine. ***IMPORTANT**: If the account hierarchy has already been specified in the routine definition step, you need only specify the nodes to be used.*

***IMPORTANT**: If the execution takes place from the process cockpit or a flowchart task, any ETL parameters related to ETL dimension whose **Identity** is Scenario, Period or Entity and any additional parameter related to the steps will not be editable and will be defined automatically with the input parameters inherited from the process cockpit.*

The system will also show the following elements, based on the parameter type:

Parameter type	Element shown
Tagetik or From dimension dictionary	A **lookup** for selecting the value and a check box for activating the automatic definition option.
Free	a **text input** for manually defining the value.
From imported data	a check box for activating the automatic definition option.

Execution buttons

Element	Description
Run	allows the execution of ETL data processing. Only active in the following cases: • if at least one routine has been activated in the **Routines** panel • if at least one *Extraction*, *Transformation* or *Loading* step has been selected for the active routines • If at least one active routine requires the selection of the process in its definition, the **Process** filter must be defined. • if there are active tasks that must be configured consistently in the **Extractions** panel (*see fields description in the panel named* **Extractions**). • if some extractions require the loading of files via upload, these must be carried out in the **Uploads** panel (at least one for every **File ID**). • if parameters are present in the **Parameters** or **Additional loading parameters** panel, they must be defined. • if a **job** has been selected, the ETL data processing related to the job (ETL_ESECUZIONE_JOB) will be run. • If the **Single Routine** option has been activated, the ETL data processing related to the Routine (ETL_ESECUZIONE_ROUTINE) will be run.
continue	allows you to resume the data processing in case it was stopped, by applying the loading logics to the data not signalled by the editability check.

Element	Description
Skip	Only active if the post loading task stop has returned **warnings**. Allows you to move on to the next loading task. The following tasks can return warnings: • extraction tasks that do not load any record. • *Check* type transformation tasks. • loading tasks that apply editability checks. The ETL only stops if the option related to the stop in case of warnings has been defined on the task's routine.

***IMPORTANT**: ETL data processing can only be stopped to allow the user to perform the following operations:*

- select the values of the ETL parameters whose type is from *Imported Data* for which the *Define automatically* option has not been activated. The system will show a window containing as many records as there are parameters to be defined, which include the description of the parameter and a lookup allowing the value to be selected. The parameters must all be defined in order to continue with the ETL's execution.
- after a task that has returned warnings

Logs and audits

Contents

This section includes the following topics:

ETL Log window 77

ETL Log window

Admin | <application database>

ETL tile > LOGS

User | <application database>

ETL tile > LOGS

***IMPORTANT:** from this entry point, you cannot view logs related to executions carried out by users other than the current user.*

Purpose of window

This page allows you to consult the logs related to executions of ETL routines, both those run individually and those run within an ETL job.

Page sections

Element	Description
Events	Shows the list of events related to executions of ETL routines.
Event details	Displays in a tree, for each event, the details of the results generated by executing the routine's extraction, transformation and loading tasks (including micro tasks).

Events section

Element	Description
Result	Overall result of the execution: • ○: all tasks of the routine completed successfully • △: at least one task of the routine completed with warnings • ✖: at least one task of the routine completed with errors
User	Code and description of the user that ran the routine (only visible if the window was opened from an entry point that allows executions run by all users to be shown
ETL domain	Code and description of the ETL domain related to the routine run
Routine	Code and description of the routine run;
Start date	The timestamp related to the start of the routine's execution;
End date	The timestamp related to the end of the routine's execution. **Note**: this can be omitted if the execution was not completed due to a blocking error.

Event details section

Element	Description
Result	Task execution results: • ○: execution successful • △: execution completed with warnings • ✖: execution completed with errors • ⁄: not executed
Task description	Indicates a description of the data processing task, from the following: • the description of a step: *Extraction*, *Transformation*, *Loading*; • the description of an activity related to data storage: *Store data offline*, *Restore data online*; • the description of a task containing the sorting and description defined by the user.
Date	indicates the timestamp related to the start of the task's execution;
Record	indicates the number of records processed by the task.
Information	Only visible if the selected record has no other details;
Query	Only visible if the selected record is an extraction or transformation task or a loading micro task. Contains a list of the queries run by the task.

Actions in window

Element	Description
	allows you to export the content of every detail panel onto Excel
	**Delete Log**: this utility allows the advanced deletion of ETL logs. Logs can be selected based on the following options: Task execution **result**: • ○ execution successful • △ execution completed with warnings • ✖ execution completed with errors • ⁄ not executed **User**: code of one or several executing users **Routine**: code of one or several ETL routines **Date from/Date to**: fields for defining the time range in which the execution was run **Delete warnings only**: only deletes the warnings contained in the warnings tab of the *Executed with warnings* logs. ***IMPORTANT**: If the option is not active, the deletion will completely remove the selected logs.*

Logs deletion

Element	Description
Result	Task execution results: • ○ execution successful • △ execution completed with warnings • ✖ execution completed with errors • ⁄ not executed
User	code of one or several executing users
Routine	indicates the code of one or several ETL routines
Date from/Date to	fields for defining the time range in which the execution was run
Delete warnings only	If enabled, this option only deletes the warnings contained in the warnings tab of the *Executed with warnings* logs. If the option is not active, the deletion will completely remove the selected logs.

ETL execution data archive

Admin | <application database>

User | <application database>

ETL home > Staging Data Query

***IMPORTANT**: data stored are only available if the options that enable data archiving are active for an ETL routine.*

***Note**: from this entry point, you cannot view data stored or extracted by users other than the current user.*

Purpose of the page

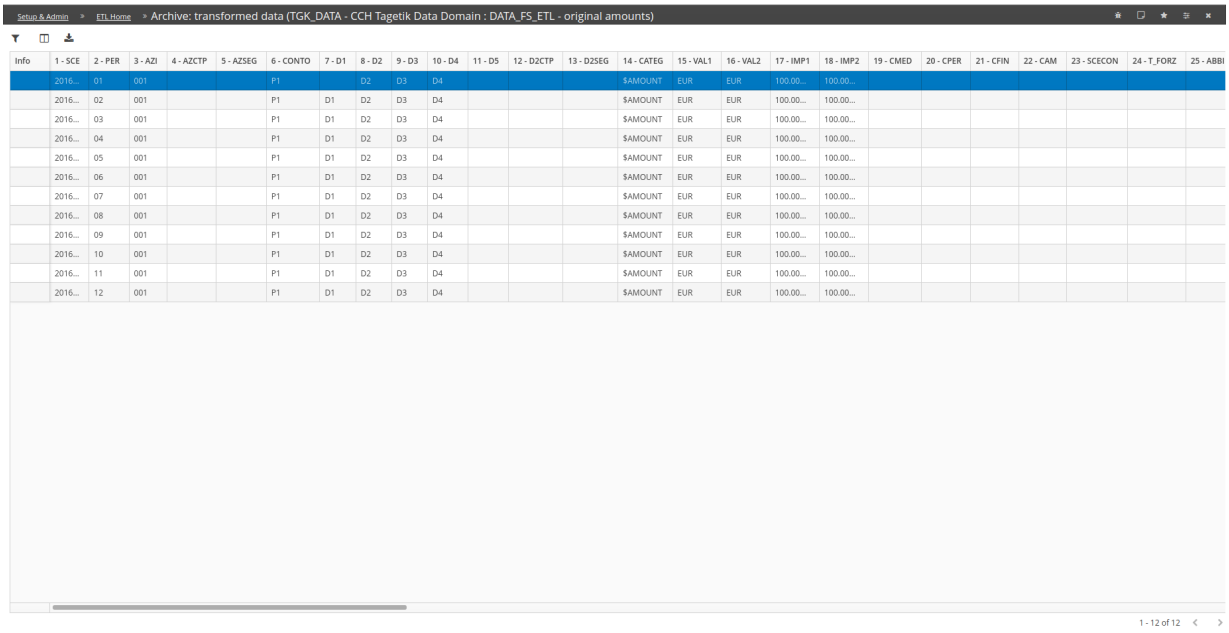
This page allows you to consult data stored in the extraction and transformation phase related to ETL routine executions run individually and run within an ETL job.

The window is made up of a read-only table containing only the columns related to the ETL Dimensions belonging to the ETL domain selected in the filter step, which shows the data stored by the extraction or transformation phase for the archive type and for the ETL routine (and potentially for the User).

ETL Archive tab

Field	Description
Archive type	Indicates the archive type: of the *Extracted data* or the *Transformed data*.
ETL domain / ETL routine	filters the archive for an individual ETL routine.
User	Only active for the **Admin user** entry point and for the **Archive type***Extracted data*: filters the archive for an individual user.

What it looks like



Element	Description
A	Archive command bar
B	List of data stored by the extraction or transformation phase

Buttons on the page

Element	Description
	This filters the data shown on the page.
	Selects the columns to be shown/hidden in the table and those to be locked when scrolling the table.
	Exports the content of every archive in Excel format.

Appendix

Contents

This section covers the following topics:

ETL via APIs 82
ETL syntax 84

ETL via APIs

Introduction

CCH Tagetik also allows you to run ETL job data processing via OPEN RESTful APIs.

STATUS	SWAGGER
ACTIVE	-
FORMAT	ROOT ENDPOINT
REST/OpenAPI	**<api root>/etl/job/v1**

Access conditions

To be able to use these specific APIs, the user must be authenticated (Basic Auth, OAuth2).

APIs must also be enabled in the Repository. For reverse proxies, further settings are necessary. They can be used with both *http* and *https* protocols.

For more details on authentication and enabling, see REST API Overview.

APIs for running ETL jobs

Method

POST

URL

```
<api-root>//etl/job/v1/database_id/etl_job_code
```

http://*server*:*port*/*tagetikcpm*/*api*/*etl*/*job*/*v1*/database_id**/*etl_job_code***

Parameters

database_id = ID of the database on which to run the ETL

Property

The following property can be included in JSON format in the body of the request

Property	Description	Options
process	If the ETL job requires it, this allows you to indicate the process with which it must be run.	-
etlParameters	If the ETL job uses ETL parameters in the ETL routine tasks, their values can be indicated by specifying the concatenation between *etl_domain_code*|*etl_parameter_code* as the parameter name.	Parameter types: • From CCH Tagetik, from Dimension dictionary, Free or from Imported Data: specifies a value or a list of values separated by **,** if the parameter accepts multiselection. • From Imported data intersection: specifies a value composed of the concatenation of the values of the tuple separated by the **;** character or a list of previously described values separated by **,** if the parameter accepts multiselection.

Example

```
{"process" : "ACTUAL","etlParameters" : [{"code" : "ED1|ENTITY","value": "001"},{"code" :
"ED2|SCENARIO_PERIOD","value": "2022ACT;01"}]}
```

Data returned

In the response, the following will be returned in JSON format:

- the following data processing ID:
- **dataprocessingId**
- an error will be returned if the parameter validation has failed

```
{ "dataprocessingId":"FF8081817A4264E5017A42D08E3C21EB" }
```

APIs for checking ETL job data processing status

```
<api-root>/etl/job/v1/dataprocessingId
```

Method

GET

URL

Data returned

The response returns the data processing status in JSON format:

status: RUNNING/FAILED/COMPLETED

Status	Description
RUNNING	Running
FAILED	Failed
COMPLETED	Completed successfully
WARNING	Completed with warnings

E.g.: {"requestDateTime": "2021-06-25T10:56:42.850+0000", "status": "COMPLETED"}

APIs for stopping ETL job data processing

Method

DELETE

URL

```
<api-root>/etl/job/v1/dataprocessingId
```

In the response, the following ID will be returned in JSON format:

message: "Stop on dataprocessingOid *dataprocessingId*" / "Notify on dataprocessingOid *dataprocessingId*"

Message	Description
Stop on dataprocessingOid	Data Processing was stopped
Notify on dataprocessingOid	Data processing with reported stop but currently still in progress

E.g.: {"message": "Stop on dataprocessingOid FF8081817A4264E5017A42D08E3C21EB"}

ETL syntax

Management of ETL filter or SQL query strings

For every ETL management window that includes an ETL filter or an SQL query where dynamic syntax is allowed, parameters can be defined to set the execution filters of a routine or an ETL job.

A link to this page will be shown for the management windows of an ETL where an ETL filter or query can be inserted in SQL language.

List of parametric elements

A list of the available parametric elements is provided below:

Parameter	Value returned in the execution step
{TABLE}	Staging table
{MAP}	ETL Domain code
{IMP}	Routine code
{JOB}	ETL job code
{USER}	Code of the user running the routine or ETL job
{PROCESS}	Data collection process code
{DATETIME}	Date and time when the routine or ETL job is run. Format: MMDDYYYYHHMMSSLLL (MM month, DD day, YYYY year, HH hours, MM minutes, SS seconds, LLL milliseconds)
{DATE}	Date when the routine or ETL job is run. Format: MMDDYYYY (MM month, DD day, YYYY year)
{TIME}	Time when the routine or ETL job is run. Format: HHMMSSLLL (HH hours, MM minutes, SS seconds, LLL milliseconds)
{DATE_YYYY}	The year (4 digits) when the routine or ETL job is run
{DATE_YY}	The year (2 digits) in which the Routine or Job ETL is run
{DATE_MM}	The month (2 digits) when the routine or ETL job is run
{DATE_DD}	The day (2 digits) when the routine or ETL job is run
{TIME_HH}	The time (2 digits) when the routine or ETL job is run
{TIME_MI}	The minute (2 digits) when the routine or ETL job is run
{TIME_SS}	The second (2 digits) when the routine or ETL job is run
{TIME_ML}	The millisecond (2 digits) when the routine or ETL job is run
{codice_parametro_ETl}	Parameter value. EXAMPLE: Suppose we have defined a parameter “PARAM_AZIENDA” whose type is “from imported data” which takes the value of the ETL Dimension and contains the entities for which the data are extracted (e.g. A01, A02, A03), you need only write COD_AZIENDA IN ({PARAM_AZIENDA}) because the system translates that instruction into COD_AZIENDA IN ('A01', 'A02', 'A03')

Parameter	Value returned in the execution step
{sigla_dimensione_ETL}	The system determines, from the ETL dimensions table, the type of dimension with the indicated acronym. The system can also automatically convert the value contained in the dimension element to the same dimension type. EXAMPLE: if we have a number type ETL Dimension with the acronym {IMP1}, supposing we need to multiply the value of that dimension by -1, and considering that the staging table has all string type columns, you need only write {IMP1}*-1, instead of TO_NUMBER(IMP1) *-1
{AW|[workspace code]|[dataset code]}	Used to recover the physical name of the specified Analytical Workspace dataset partition if the dataset is not partitioned or is partitioned by Business Cycle and the ETL is run from a single submission process.
{AW|[workspace code]|[dataset code]|[scenario code]|[period code]}	Used to recover the physical name of the specified Analytical Workspace dataset partition if the dataset is partitioned by Business Cycle and the ETL is run from a scenario/period submission process.
{AW|[workspace code]|[dataset code]|[entity code]}	Used to recover the physical name of the specified Analytical Workspace dataset partition if the dataset is partitioned by Business Cycle/Entity and the ETL is run from a single submission process.
{AW|[workspace code]|[dataset code]|[scenario code]|[period code]|[entity code]}	Used to recover the physical name of the specified Analytical Workspace dataset partition if the dataset is partitioned by Business Cycle/Entity and the ETL is run from a scenario/period submission process.
{AW|[workspace code]|[dataset code]|[typology code]|[process code]|[scenario code]|[period code]|[entity code]}	Used to recover the physical name of the specified Analytical Workspace dataset partition regardless of the dataset partition type or execution context. If one of the syntax parameters is not significant, it is necessary to set the **$** character, e.g.: • {AW\|001\|00001\|$|$\|$|$\|$} • {AW|001|00001|$\|BUDGET\|$|$\|$} • {AW|001|00001|$\|ACTUAL\|2020ACT\|12\|$} • {AW|001|00001|$\|ACTUAL\|2020ACT\|12\|001} • {AW\|001\|00001\|CALC\|BUDGET\|$|$\|001} • {AW\|001\|00001\|CALC\|ACTUAL\|2020ACT\|12\|$} • {AW|001|00001|CALC|ACTUAL|2020ACT|12|001}

***Note**: it is possible to use the syntax **#{}#** to prevent the system, when converting certain key words, from delimiting the value between single quotes.*

*For example, the syntax **{PROCESS}** is converted to 'BUGDET'; while the syntax **#{PROCESS}#** is converted to BUDGET.*

This is useful, for example, when you want to use a parametric syntax as part of a table name or as a parameter in a stored procedure.

***Note**: It is also possible to use Unified SQL Syntax.*

Warning

- Whenever an ETL Dimension needs to be referenced within an ETL filter or an SQL query, personal names must not be used. Instead, the field names of the staging table columns as defined in the Tagetik application database—such as CAMPO1, CAMPO2, etc.—must be used
- Whenever it is necessary to write a filter to apply to the staging table to refer to its fields, it is mandatory to use the names they have on the Tagetik application database, i.e. CAMPO1, CAMPO2 etc.
- Whenever it is necessary to write a filter to be applied to the table on which data is loaded to refer to its fields, it is mandatory to use the names they have on the database where the table is located
- It is not possible to use both characters '{' and '}' in a query in this order except to identify the parameters specified above.

ETL connectors

Introduction

Standard or wizard connectors are tools used to create an ETL domain which allows for the integration of data from different ERP solutions in Tagetik Software S.r.l., thus minimising the time that the client spends on data preparation.

ERP (Enterprise Resource Planning) solutions coordinate the flow of data between an entity's processes by helping to automate and manage the main business processes to obtain optimal performance.
In Tagetik Software S.r.l., there are the following connectors:

Element	Description
Dynamics AX	integrates the actual general accounting data present in the Dynamics AX ERP. Dynamics AX is an ERP solution for accounting, financial and decision-making management of medium-sized and large entities. Dynamics AX is part of the Microsoft Dynamics suite (http://www.microsoft.com/it-it/dynamics/default.aspx) Two versions of the AX connector are available in CCH Tagetik : • Dynamics AX 2012 • Dynamics AX 2012 R2. (Differentiated from the first by the **Partition** field. Dynamics AX
Dynamics Nav	integrates the actual general accounting data present in the Dynamics NAV ERP. Dynamics NAV is an ERP solution for the accounting and financial management of small and medium-sized companies. Dynamics NAV is part of the Microsoft Dynamics suite (https://www.microsoft.com/en-us/dynamics365/nav-overview). Dynamics Nav
SAP	integrates the tables and accounting data present in a database of the SAP© (management software for large companies) ERP SAP connector

Dynamics AX Connector window

Admin

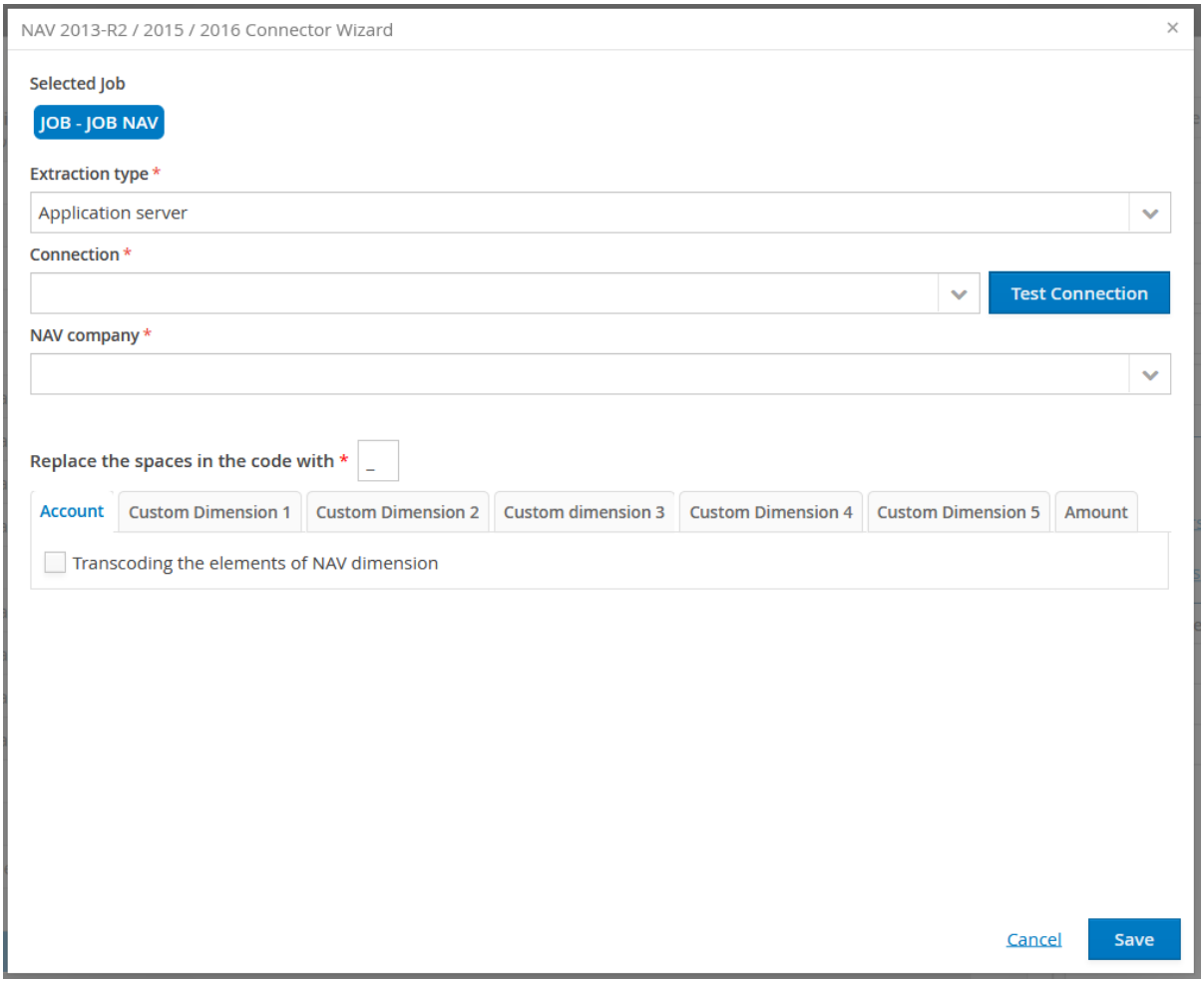
Setup & Admin > ETL > Jobs > AX 2012 Connector Wizard

Setup & Admin > ETL > Jobs > AX 2012 R2 Connector Wizard

Navigation panel > ETL > Jobs > AX 2012 Connector Wizard.

Navigation panel > ETL > Jobs > AX 2012 Connector Wizard.

***IMPORTANT**: the window can be accessed by editing a job or creating a new one.*



Purpose of the page

This window allows you to create an ETL domain to import data from Dynamics AX. The created ETL will be made up of two routines:

- Tables routine: allows you to populate both the account table and the Tagetik custom dimension tables
- General accounting data routines

The system will also generate any necessary lookup tables, based on the choices made by the user, who can run the generated ETL at any time using the usual Tagetik procedures.

Configuration fields

Field	Description
Extraction type	Defines the data source. The following options are available: • **Application server**: if the queries to be run are located on the CCH® Tagetik application server. • **Endpoint/Tagetik Data Loader**: if the queries to be run are located on an Endpoint or the Tagetik Software S.r.l.. **Note**: Any Endpoint/Tagetik Data Loader on which the data to be extracted is located must be specified in the field of the same name.
Connection	***IMPORTANT**: only visible and editable if the extraction type is from Application server.* Allows you to view all the secondary connections configured on the application server.
Partition	Active field only present for the Dynamics AX 2012 R version

Account tab

This tab allows you to define the relationship between the elements of the Tagetik Account dimension table and the Dynamics AX account dimension.

Element	Description
Chart of accounts	filter which allows you to include only the elements of the AX account dimension related to the applied filter. if the filter is not applied, the system will include the whole table of AX accounts.
Transcoding the elements of AX dimension	defines the type of relationship between the elements of the Company dimension defined in the AX data and the elements of Tagetik's Entity dimension. if selected: it automatically creates the lookup table between the elements of the Tagetik account dimension and the elements of the AX account dimension in the generated ETL domain. The user can then define the relationship between the elements. If not selected: it indicates that the elements of the Tagetik account dimension have the same identification code as the elements of the AX account dimension.

Custom dimensions tab

The system presents as many tabs as there are custom destinations managed in the Tagetik environment (and therefore a maximum of 5).
The names shown for these tabs will be those defined in Tagetik's general configuration for the managed custom dimensions.
The tabs are similar to one another and allow you to define the relationship between the elements of the Tagetik custom dimensions and the Dynamics AX dimensions.

Element	Description
Default element	in Dynamics AX, you can omit information for a certain dimension in registering accounting data. This means that, for a certain row of accounting records, there may be no information related to the dimension that you want to import into Tagetik. In Tagetik, this is not allowed: if a custom dimension is managed, that information will always be defined in the actual data. Therefore, you must define a default element for the custom dimension being considered, which will be used when the source element in the row originating from AX is NULL. This default element can be set as a fixed value, using the search function, or by selecting the Automatic by Entity option.
Automatic	Indicates that the default element must be calculated automatically by the system when data are imported from AX. In Tagetik, a default element for each managed custom dimension is defined for every element of the Entity dimension. This option tells the system to define the custom dimension as the Entity's default element, if the value is NULL in the AX data.
Related to AX dimension	by default, the custom dimension being considered has no relationship with Dynamics AX. In this case, in Tagetik the custom dimension will always be defined as the default element. However, if this option is selected, it indicates that the custom dimension being considered must be related to a Dynamics AX dimension. In this case, the user must define the following two fields
AX Dimension	the user must choose the AX dimension to which the custom dimension being considered corresponds. The system presents a list of the dimensions available in the client's Dynamics AX environment.
Transcoding the elements of AX dimension	if selected, it indicates that you must define a lookup table between the elements of the Tagetik custom dimension and the elements of the selected AX dimension. The system will automatically create the lookup table in the generated ETL domain. The user can then go on to define the relationship between the elements using Tagetik's normal ETL tools. Otherwise, if not selected, it indicates that the elements of the Tagetik custom dimension have the same identification code as the elements of the selected AX dimension.

Amount tab

Field	Description
Transcoding the elements of AX dimension	defines the type of relationship between the elements of the Company defined in the AX data and the elements of Tagetik's Entity dimension. • If selected, it indicates that a lookup table between the two dimensions is necessary. The system will automatically create that rule and the user must subsequently specify the existing relationships between the elements. • if not selected, the elements of the two dimensions in the two environments will have the same IDs. **Note**: this is common when the Entity table in Tagetik has been populated previously and its elements have the same codes as those present in AX.
Only standard allocations	if the option is selected, the system will only include data related to the standard AX allocations; otherwise, all the General Ledger data will be will be included.

Field	Description
Category	CCH Tagetik Category dimension code assigned to all rows imported from AX.
IC management	enables automatic management for IC transactions also. Dynamics AX: does not provide for the native management of InterCompany transactions. **Note**: it is best practice to define a custom AX dimension whose elements are used to specify the IC Companies: in this case, the connector can manage InterCompany transactions automatically.
AX Dimension	only enabled if the **IC Management** option is selected. This is the AX dimension adopted by the client for the management of IC transactions.
Transcoding the elements of AX dimension	only enabled if the **IC Management** option is selected. In Tagetik, InterCompany entities are the same as in the Entity table. Therefore, to avoid defining the same lookup tables twice (it is likely that the elements of the custom dimension dedicated to IC transactions have been defined with the same codes as used by the AX Companies), the system adopts the same rule defined for the AX Company. Select the option when you need to define a lookup table between the elements of the AX dimension specified by the user and Tagetik's IC elements.

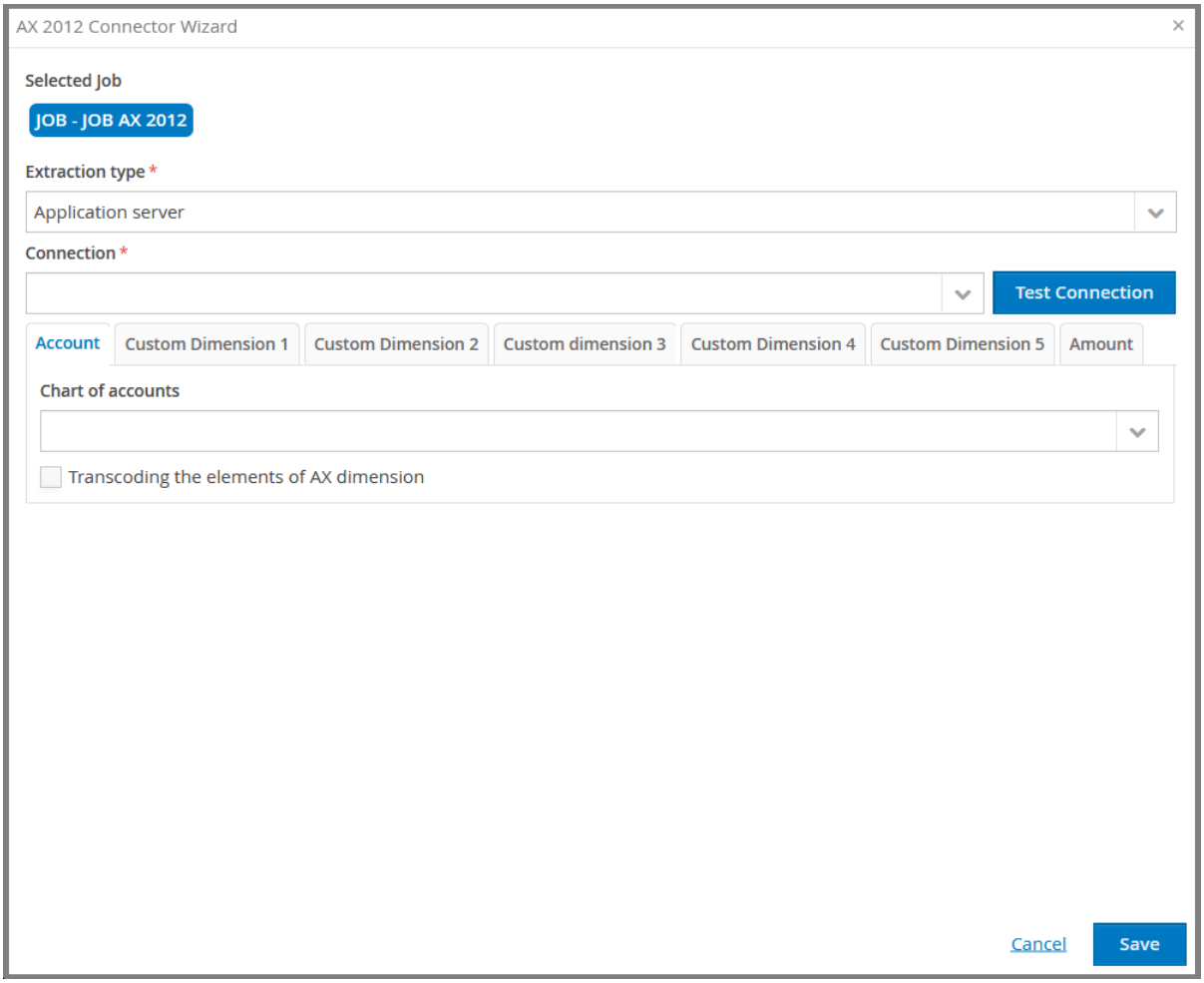
Dynamics NAV Connector window

Admin

Setup & Admin > ETL > Jobs > NAV 2013 R2/2015/2016 Connector Wizard

Navigation panel > ETL > Jobs > NAV 2013 R2/2015/2016 Connector Wizard

***IMPORTANT**: the window can be accessed by editing a job or creating a new one.*



Purpose of the page

This window allows you to create an ETL domain to import data from Dynamics NAV. The created ETL will be made up of two routines:

- Tables routine: allows you to populate both the account table and the Tagetik custom dimension tables
- General accounting data routines

The system will also generate any necessary lookup tables, based on the choices made by the user, who can run the generated ETL at any time using the usual Tagetik procedures.

Configuration fields

Field	Description
Extraction type	Defines the data source. The following options are available: • **Application server**: the queries to be run are located on the CCH® Tagetik application server. • **Endpoint/Tagetik Data Loader**: the queries to be run are located on an Endpoint or the Tagetik Software S.r.l.. ***Note**: Any Endpoint/Tagetik Data Loader on which the data to be extracted is located must be specified in the field of the same name.*
Connection	Only visible and editable if the extraction type is from Application server. ***IMPORTANT**: Allows you to view all the secondary connections configured on the application server.* ***IMPORTANT**: The NAV database must be configured as "Tagetik secondary database".* *Through **Test Connection**, the system runs a check on the status of the connection and, once established, it allows you to complete the setup of the **NAV entity** and **Replace spaces in the code with "_"** fields.*
Endpoint/Tagetik Data Loader	Only visible and editable if the extraction type is **Endpoint/Tagetik Data Loader**
NAV Entity	List of companies present in NAV
Replaces spaces in the code with "_"	Allows you to specify the character to be used to replace the spaces in the definition of the dimension elements (permitted in NAV, but not in CCH Tagetik)

Account tab

This tab allows you to define the relationship between the elements of the Tagetik Account dimension table and the Dynamics NAV account dimension.

Option	Description
Transcoding the elements of NAV dimension	Option selected: automatically creates the lookup table between the elements of the Tagetik account dimension and the elements of the NAV account dimension in the generated ETL domain. The user can then define the relationship between the elements. Option not selected: the elements of the Tagetik account dimension have the same identification code as the elements of the NAV account dimension.

Custom dimensions tab

In these tabs, you can define the relationship between the elements of the Tagetik custom dimensions and the Dynamics NAV dimensions. The number of tabs will depend on the number of custom destinations managed in the Tagetik environment (maximum 5).
The names of the tabs will correspond to those defined in the configuration of CCH Tagetik 's custom dimensions.

Element	Description
Default element	In Dynamics NAV, you can omit information for a certain dimension in the recording of accounting data. Given that this is not allowed in CCH Tagetik (if a custom dimension is managed, that information is always defined in the actual data), it is necessary to define a default element of the custom dimension being considered; this will be used when the source element in the row originating from NAV is NULL. The default element can be set as a fixed value: • by selecting a value using the search function • by using the Automatic from Entity option
Automatic	Option for automatically calculating the default element during the importing of data from NAV step. If selected, the system will define the custom dimension as the Entity's default element, if the value is NULL in the NAV data. In Tagetik, a default element for each managed custom dimension is defined for every element of the Entity dimension.
Related to NAV dimension	If selected, this relates the considered custom dimension to a Dynamics NAV dimension. If not selected in CCH Tagetik, the custom dimension will be defined as the default element of the same custom dimension.
NAV Dimension	The field is only active if the **Related to NAV dimension** option is enabled: allows you to define the NAV dimension to which the considered custom dimension corresponds, by selecting an element from the list of available dimensions in the client's Dynamics NAV environment.
Transcoding the elements of NAV dimension	If selected, this option automatically creates, in the generated ETL domain, a lookup table between the elements of the Tagetik account dimension and the elements of the NAV account dimension. The user can then define the relationship between the elements. If not selected: it indicates that the elements of the Tagetik account dimension have the same identification code as the elements of the selected NAV account dimension.

Amount tab

This tab allows you to define the rules for loading general accounting data from the NAV database.

Element	Description
Category	CCH Tagetik Category dimension code that will be assigned to all rows imported from NAV.
Include CTP	if selected, it will include information related to the counterparty.

SAP Connector Wizard window

Admin

Setup & Admin > ETL > Jobs > SAP© Connector Wizard

***IMPORTANT**: the window can be accessed by editing a job or creating a new one.*

Purpose of the page

This window allows you to create an ETL domain to import data from SAP©. The created ETL will be made up of two routines:

- Table routines:
  - relationship between the elements of the Entity table with the Companies defined in SAP©;
  - relationship between the elements of the Tagetik account dimension and those of the SAP© account dimension;
  - relationship between the Tagetik custom dimensions and the SAP© consolidation dimensions;
- General accounting data routines

Configuration fields

Field	Description
Extraction type	Defines the data source. The following options are available: • **Application server**: if the queries to be run are located on the CCH® Tagetik application server. • **Endpoint/Tagetik Data Loader**: if the queries to be run are located on an Endpoint or the Tagetik Software S.r.l.. **Note**: The Endpoint/Tagetik Data Loader on which the data to be extracted is located must be specified in the field of the same name.
Connection	***IMPORTANT**: only visible and editable if the extraction type is from Application server.* Allows you to view all the secondary connections configured on the application server. The extracted data are saved in a support database (staging database), without applying any transformation logic; these will then be applied during the import run by the ETL, by reading the data from this support database. In the “Connections” section, the user must provide all the necessary information for the connection to the staging database: ◦ *Host Database*: this is the ID of the server on which the staging database is located; ◦ *Database Catalog*: this is the name of the staging database ◦ *Linked Server*: used if Tagetik’s application database and the staging database are located on two different servers. Identifies the channel (Linked Server), present on Tagetik’s application database, through which the data present on the staging database can be accessed
Replace the spaces in the code with	in SAP©, the identification code of an element of a dimension can contain spaces; however, spaces are not permitted in Tagetik; therefore, in this field the user must specify a character to be used as a replacement for the spaces that the ETL will encounter during the import

Entity tab

This tab allows you to define the relationship between the elements of the Entity table in Tagetik and the Companies in SAP©. The setup will be used both during the Entity table population phase and in the load general accounting data phase.

Element	Description
Create filter for SAP© Company:	if selected, this allows you to select the SAP companies to be loaded in the ETL execution step.
Transcoding the elements of AX dimension	if selected: it automatically creates the lookup table between the elements of the two tables. The user can then define the relationship between the elements. If not selected: it indicates that the elements of the entity dimension in Tagetik have the same identification codes as the companies in SAP©.
Include CTP	if the option is selected, the system will set up the generated ETL so that it also imports information on Intercompany Companies from SAP©.

Account tab

This tab allows you to define the relationship between the elements of the Account table in Tagetik and the elements of the Account table in SAP©. As stated, the connector does not populate the Account table in Tagetik: therefore, the set options will only be used in the load general accounting data phase.

Option	Description
Manage SAP© transaction type:	if selected, this allows you to import the transaction type managed in SAP© into Tagetik; the generated ETL will automatically create a lookup table for the Account dimension, which will have two fields as its input, the SAP© Account/Entry type pair, while it will have the Tagetik account as the output. **Note:** default code, which will be used during the loading of data from SAP© for the rows in which the Entry Type field is NULL.
Associated with SAP© dimension	if selected, this indicates that there is a relationship between the Tagetik account and the SAP© account.

Custom dimensions tab

In these tabs, it is possible to define the relationship between the elements of the Tagetik custom dimensions and the SAP© dimensions.
The number of tabs will depend on the number of custom destinations managed in the Tagetik environment (maximum 5).
The names of the tabs will correspond to those defined in the configuration of CCH Tagetik 's custom dimensions.

Field	Description
Default element	identifies the element of the CCH Tagetik custom dimension that will be used by ETL on loading of General Accounting data, if the value on the source row for the selected SAP© custom dimension is NULL. The field will be disabled if the "Automatic" option is selected.
Automatic	option which allows you to automatically set the default element of the CCH Tagetik custom dimension. If selected, ETL will use the default element indicated for the entity used for every imported row.
Associated with SAP© dimension	if selected, this identifies a relationship between the custom dimension in Tagetik and a custom dimension in SAP.
SAP dimension	presents a list of custom dimensions in SAP© from which to chose the dimension to relate to the custom dimension in Tagetik. The field will be disabled if the **Associated with SAP dimension** option is not active.

Field	Description
Transcode the elements of the SAP dimension	if selected, this option allows you to create a lookup table between the elements of the two tables. Otherwise, the elements of the custom dimension in Tagetik will have the same identification codes as the elements in the selected SAP© custom dimension table.
Include CTP	Option only present in the custom dimension 2 tab in Tagetik. If selected, this indicates that the counterparty was also managed for the custom dimension in SAP©. Not all SAP© dimensions have a Counterparty: for this reason, the check box is only enabled if the user selects a SAP© custom dimension that has a counterparty.

Amount tab

Option	Description
Category	identifies the code of the CCH Tagetik Category dimension set for all the imported rows. Only includes “Amount” type categories.
Transcode the SAP© currency elements	if selected, this option defines a lookup table to create a relationship between the SAP© currencies and the Tagetik Software S.r.l. currencies, whose codes may differ.

Un

Customizations

ETL routines generated using the *“enabled for SAP®” Tagetik Standard Connector* wizard access data of the SAP ECC’s FI module which are relevant for the purposes of a typical consolidation process. As they are standard Tagetik routines, they can be extended or customised to also extract data from tables or extractors of other SAP ECC or SAL BW components (e.g. for amounts managed on Special Ledgers or data of other modules such as CO or SD).